

**FEDERAL UNIVERSITY OF RIO GRANDE DO SUL
ENGINEERING SCHOOL
GRADUATE PROGRAM IN INDUSTRIAL ENGINEERING**

LUCAS MIGUEL ALENCAR DE MORAIS CORREIA

**ORGANIZATIONAL STRUCTURING OF
HEALTHCARE DIGITAL TRANSFORMATION**

Porto Alegre

2026

Lucas Miguel Alencar de Moraes Correia

Organizational Structuring of Healthcare Digital Transformation

Tese submetida ao Programa de Pós-Graduação em Engenharia de Produção da Universidade Federal do Rio Grande do Sul como requisito parcial à obtenção do título de Doutor em Engenharia, na área de concentração em Sistemas de Produção.

Orientador: Alejandro Germán Frank, Dr.

Porto Alegre

2026

Lucas Miguel Alencar de Moraes Correia

Organizational Structuring of Healthcare Digital Transformation

This Ph.D. thesis was considered adequate to obtain the title of Ph.D. in Engineering and approved in its final form by the Advisor and by the Examining Board designated by the Graduate Program in Industrial Engineering at the Federal University of Rio Grande do Sul.

Prof. Alejandro Germán Frank, Ph.D.

Advisor PPGEP/UFRGS

Prof. Michel José Anzanello, Ph.D.

Coordinator PPGEP/UFRGS

Examination board:

Prof. Tarcisio Abreu Saurin, Ph.D.

Departamento de Engenharia de Produção e Transportes / UFRGS

Prof. Sílvio César Cazella, Ph.D.

Departamento de Ciências Sociais Exatas e Aplicadas / UFCSPA

Prof. Glauco Henrique de Sousa Mendes, Ph.D.

Departamento de Engenharia de Produção / UFSCar

Até aqui o Senhor nos ajudou.

1 Samuel 7:12

“Nada verdadeiramente valioso advém da ambição ou do mero senso do dever; é melhor que venha do amor e da devoção à humanidade e às coisas objetivas.”

Albert Einstein

Dedication

A minha mãe, Rosilda (in memoriam)

ACKNOWLEDGMENTS

First and foremost, I thank God for all the blessings granted throughout my life and for His constant faithfulness. Everything is owed to Him; without Him, I am nothing. I am deeply grateful to my parents, Rosilda (*in memoriam*) and Lucas, for their support throughout this entire journey. I thank my mother for her unconditional love, for facing countless challenges, and for never giving up on my education. She was present whenever I needed her, serving as my haven throughout my life. Wherever you may be, Mom, know that this achievement is ours. I love you forever.

I also sincerely thank my advisor, Prof. Alejandro G. Frank, who guided me in the development of this work and in every opportunity, he provided for me to become a better researcher and scholar, always encouraging me to aim higher. He was far more than an advisor; he became a true friend, present in both the good and the challenging moments of this journey.

I am grateful to my family and friends who, directly or indirectly, contributed along this path. I thank my beloved grandmother Maria (Dona Didi) and my sisters Hianne and Hivia for their constant support, encouragement, and affection. It is a privilege to share life with you, and I hope that in this new chapter we will be able to spend even more time together.

I would like to thank the examination board for their valuable comments and suggestions, which undoubtedly enriched this work and contributed to my development as a researcher. I also express my sincere gratitude to the professors at the Federal University of Rio Grande do Sul (UFRGS) for their teaching, knowledge exchange, and inspiration throughout my academic training.

I extend my gratitude to Professors Néstor Ayala, Camila Dutra, and Maria Auxiliadora Tinoco for their contributions to my academic and professional development. I also thank all the colleagues and friends I made at NEO (Organizational Engineering Group) for their companionship and support over these four years.

I also thank Prof. Ben Armstrong for his support, valuable contributions, and shared knowledge during my sandwich PhD period at the MIT Industrial Performance Center (MIT-IPC). I am equally grateful to Rafaela, Ana Victória, and Silvia for their partnership, friendship, and support during my stay in the United States, which made this experience lighter and more meaningful.

I gratefully acknowledge the financial support provided by CAPES and CNPq for the development of this research. I also express my gratitude to Hospital de Clínicas de Porto Alegre (HCPA), Hospital Mãe de Deus, and Grupo Hospitalar Conceição for supporting this study by providing valuable data and insights throughout the research period.

My sincere thanks to everyone involved in the PPGEP and to the UFRGS, that provided fundamental opportunities for my academic and professional journey.

Finally, I thank all those who, directly or indirectly, contributed to the completion of this work.

CREDITS

This thesis was developed in the NEO from the Graduate Program of Industrial Engineering at the UFRGS.

The development of this thesis was made possible by the doctoral scholarship granted to the candidate by the Coordination for the Improvement of Higher Education Personnel (Coordenação de Aperfeiçoamento de Pessoal de Nível Superior - CAPES) of the Brazilian Federal Government.

This thesis was also supported by resources from the research project “*Impacts of digital technologies on operations workers: adaptation processes and learning curves to enhance sociotechnical outcomes,*” funded under Call No. 14/2023 of the National Council for Scientific and Technological Development (Conselho Nacional de Desenvolvimento Científico e Tecnológico - CNPq), which supports international scientific, technological, and innovation research projects. The project involves a collaboration with the Massachusetts Institute of Technology (MIT), United States, a global leader in engineering.

RESUMO

A Transformação Digital Healthcare (HCDT) refere-se à integração de tecnologias digitais aplicadas tanto para os cuidados clínicos quanto aos processos administrativos nas operações de saúde. Essa transformação busca a personalização em tempo real dos serviços para pacientes e profissionais, facilitando a transição para um ambiente centrado no paciente. A literatura sobre HCDT tem demonstrado quais tecnologias digitais são adotadas e onde podem ser aplicadas. Além disso, estudos empíricos têm identificado associações positivas entre a adoção dessas tecnologias e a melhoria tanto nas melhorias dos pacientes quanto da eficiência operacional dos sistemas de saúde. No entanto, a HCDT tem sido majoritariamente abordada sob uma perspectiva tecnocêntrica, onde o foco recai predominantemente sobre a identificação das tecnologias em si e suas aplicações, enquanto a estruturação organizacional que antecede a implementação da HCDT tem sido negligenciada. Por outro lado, o sucesso da transformação digital nas organizações depende de uma estrutura organizacional sólida, que leve em consideração diferentes elementos organizacionais para apoiar o processo de implementação. Diante disso, esta tese tem como objetivo geral propor uma estrutura organizacional que integre estratégias digitais, recursos digitais, tecnologias digitais e os microfundamentos de rotinas operacionais – compreendidos como as interações entre indivíduos, estruturas hierárquicas e processos – para a implementação da HCDT. O estudo adotou uma abordagem mista, combinando métodos qualitativos e quantitativos, organizada em três artigos. Na vertente qualitativa, correspondente ao Artigo 1, foram analisados três estudos de caso em hospitais com diferentes tipos de propriedade, utilizando o framework TOE (Technology–Organization–Environment). Para isso, realizou-se a triangulação de múltiplas fontes de dados, incluindo entrevistas semiestruturadas, observação participante e análise documental. Os dados qualitativos foram analisados utilizando uma abordagem de codificação indutiva, partindo de conceitos de primeira ordem centrados no informante para temas teóricos de segunda ordem e dimensões agregadas. Os resultados revelam três configurações possíveis para estruturar organizacionalmente a HCDT, considerando as estratégias e diferentes modelos de gestão dos recursos digitais, cuja escolha depende do modelo de negócio da organização. Na vertente quantitativa, correspondente ao Artigo 2, a tese apresenta um estudo survey conduzido com 303 hospitais no Brasil. Os dados foram analisados por meio de técnicas de Análise Fatorial Exploratória e Confirmatória, regressões hierárquicas por Mínimos Quadrados Ordinários (OLS) e análises de mediação e moderação para o teste das hipóteses. Os resultados indicam que há uma associação positiva entre o alinhamento da estratégia digital e os recursos digitais

(seja gerenciados interna ou externamente) na implementação de tecnologias que apoiam os profissionais nos serviços de saúde, analisadas sob uma perspectiva social da visão sociotécnica. No entanto, não foram encontradas evidências significativas dessa associação no caso da adoção de tecnologias voltadas para a automação de processos, analisadas sob uma perspectiva técnica. Além disso, os resultados indicaram que a inserção em uma rede de atores da área da saúde, geograficamente próxima, modera positivamente a relação entre a estratégia digital e os recursos digitais externos, mas não os recursos internos. Adicionalmente, também vertente qualitativa, um estudo de caso qualitativo, correspondente ao Artigo 3, foi realizado para explorar como a HCDT molda os microfundamentos de rotinas operacionais. Para tanto, foram mobilizadas múltiplas fontes de evidência, incluindo entrevistas semiestruturadas, observação participante e análise de documentos institucionais. Os resultados indicam que a HCDT pode emergir a partir das interações entre os microfundamentos organizacionais, nas quais o conhecimento, as experiências e os processos de tentativa e erro vivenciados pelos profissionais de saúde orientam o alinhamento entre estratégias, recursos e tecnologias digitais. Além disso, os achados evidenciam que a gestão da HCDT não precisa ocorrer exclusivamente de forma top-down, como tradicionalmente observado, podendo também ser impulsionada a partir do nível operacional. Finalmente, uma estrutura organizacional é apresentada como principal produto resultante dos achados dos três artigos. Esta tese demonstra que a implementação da HCDT requer, de fato, uma estrutura organizacional que considere as estratégias digitais, a gestão dos recursos digitais e dos microfundamentos antecipadamente a implementação de tecnologias digitais. Sua principal contribuição é fornecer evidências empíricas que auxiliem sistemas de saúde a estabelecerem uma estruturação organizacional sólida para a implementação do HCDT, considerando não apenas aspectos técnicos, mas também ambientais, sociais e organizacionais.

Palavras Chaves: Transformação Digital, Healthcare, Estruturação Organizacional.

ABSTRACT

Healthcare Digital Transformation (HCDT) refers to the integration of digital technologies applied to both clinical care and administrative processes in healthcare operations. This transformation aims to enable real-time personalization of services for patients and professionals, facilitating the transition to a patient-centered environment. The literature on HCDT has demonstrated which digital technologies are adopted and where they can be applied. Moreover, empirical studies have identified positive associations between the adoption of these technologies and improvements in both patient outcomes and the operational efficiency of healthcare systems. However, HCDT has predominantly been approached from a technocentric perspective, focusing mainly on identifying the technology itself and its applications, while the organizational structuring that precedes HCDT implementation has been neglected. On the other hand, the success of digital transformation in organizations depends on a solid organizational structure that considers different organizational elements to support the implementation process. Considering this, the general objective of this thesis is to propose an organizational structure that integrates digital strategies, digital resource, digital technologies, and the microfoundations of operating routines—understood as the interactions among individuals, hierarchical structures, and processes—for the implementation of HCDT. The study adopted a mixed-methods approach, combining qualitative and quantitative methods, and was structured into three articles. The qualitative stream, corresponding to Article 1, analyzed three case studies of hospitals with different ownership types using the Technology–Organization–Environment (TOE) framework. Multiple data sources were triangulated, including semi-structured interviews, participant observation, and document analysis. The qualitative data were analyzed using an inductive coding approach, moving from first-order informant-centric concepts to second-order theoretical themes and aggregate dimensions. The findings reveal three possible configurations for the organizational structuring of HCDT, considering digital strategies and different models of digital resource management, with the choice of configuration depending on the organization’s business model. The quantitative stream, corresponding to Article 2, presents a survey study conducted with 303 hospitals in Brazil. The data were analyzed using Exploratory and Confirmatory Factor Analysis, hierarchical Ordinary Least Squares (OLS) regressions, and mediation and moderation analyses to test the proposed hypotheses. The results indicate a positive association between the alignment of digital strategy and digital resources (whether managed internally or externally) and the implementation of technologies that support healthcare workers, as analyzed from the

social perspective of the sociotechnical view. However, no significant evidence of this association was found for the adoption of technologies aimed at process automation, analyzed from a technical perspective. Additionally, the findings indicate that embeddedness in a geographically proximate healthcare network positively moderates the relationship between digital strategy and external digital resources, but not internal ones. Additionally, within the qualitative stream, a qualitative case study corresponding to Article 3 was conducted to explore how HCDDT influences the microfoundations of operating routines. Multiple sources of evidence were mobilized, including semi-structured interviews, participant observation, and the analysis of institutional documents. The results indicate that HCDDT can emerge from interactions among organizational microfoundations, in which healthcare professionals' knowledge, experiences, and trial-and-error learning processes guide the alignment between strategies, digital resources, and digital technologies. Moreover, the findings demonstrate that HCDDT management does not need to occur exclusively in a top-down manner, as traditionally observed, but can also be driven from the operational level. Finally, an organizational structure is presented as the main product resulting from the findings of the three articles. Overall, this thesis demonstrates that the implementation of HCDDT requires an organizational structure that anticipates the definition of digital strategies, the management of digital resources, and the configuration of microfoundations prior to the implementation of digital technologies. Its main contribution lies in providing empirical evidence to support healthcare systems in establishing robust organizational structuring for HCDDT implementation, considering not only technical aspects but also environmental, social and organizational aspects.

Keywords: Digital Transformation, Healthcare, Organizational Structuring.

LIST OF FIGURES

Figure 1. Research methodology	26
Figure 2. Conceptual models developed throughout the research phases (Source: prepared by the author)	29
Figure 3. Analytical framework of digital resource management strategies for HCDT	47
Figure 4. Data structure of the qualitative analysis	53
Figure 5. Configuration of internal management of digital resources for HCDT within the public health system.....	55
Figure 6. Configuration of external management of digital resources for HCDT within the private health system.....	59
Figure 7. Analytical framework for digital transformation in the education and research healthcare system.....	64
Figure 8. Conceptual research model	98
Figure 9. Slope for the moderating role of healthcare network embeddedness (EMBEDDEDNESS) on STRATEGY vs EXTERNAL.	106
Figure 10. Theoretical model	132
Figure 11. Bodies of information and communication technology, innovation, and data.....	137
Figure 12. Data collection	139
Figure 13. Coding process and data structure	144
Figure 14. Empirical model of how HCDT as a sociotechnical system shapes the interactions among the microfoundations of operating routines.....	162
Figure 15. Articles relationship	176
Figure 16. Organizational Structure for the Healthcare Digital Transformation	177

LIST OF TABLES

Table 1. Article's scope.....	30
Table 2. Hospitals' characteristics and interviewees' profile.....	49
Table 3. Sources of primary and secondary data used.....	51
Table 4. Summary of the main findings from the three case studies.....	68
Table 5. Sample composition (n=303 hospitals)	100
Table 6. Exploratory Factor Analysis for digital technologies.....	102
Table 7. Results of confirmatory factor analysis	103
Table 8. Bivariate correlations.....	103
Table 9. Regression results for OLS direct effects (a).....	107
Table 10. Bootstrapping results of the indirect effects	108
Table 11. Digital technologies used by the hospital as mentioned in the interviews	150

TABLE OF CONTENTS

1. INTRODUCTION.....	18
1.1 DISSERTATION TOPIC AND OBJECTIVES	23
1.2 JUSTIFICATION OF THE RESEARCH PROBLEM.....	23
1.3 STUDY DESIGN	25
1.3.1. Research method	25
1.3.2. Research Design.....	27
1.4 STUDY DEFINITION	32
1.5 STRUCTURE OF THE DISSERTATION.....	33
1.6 REFERENCES	33
2. STRUCTURING RESOURCES IN HEALTHCARE DIGITAL TRANSFORMATION: A COMPARISON ACROSS PUBLIC, PRIVATE AND RESEARCH HOSPITALS.....	38
2.1 INTRODUCTION	38
2.2 THEORETICAL BACKGROUND.....	42
2.2.1 Managing DT in healthcare organizations	42
2.2.2 Research opportunities on digital resource management for HCDDT – The TOE framework	45
2.3 RESEARCH METHOD	47
2.3.1 Selection criteria for the cases and the participants.....	48
2.3.2 Data collection procedures	49
2.3.3 Data Analysis – validity, reliability and interpretation	51
2.3.4 External validation	54
2.4 RESULTS	54
2.4.1 Case study 1: Public hospital (hospital A)	54
2.4.2 Case study 2: Private hospital (hospital B)	59
2.4.3 Case study 3: Research-oriented hospital (Hospital C).....	63
2.4.4 Cross-case comparative summary	68
2.4.5 Comparison with secondary data for external validation	69
2.5 DISCUSSION.....	69
2.6 CONCLUSION.....	72
2.6.1 Theoretical implications.....	72
2.6.2 Managerial implications.....	73
2.6.3 Limitations and future research.....	74
2.7 REFERENCES	75
Appendix A - Representative studies in top-tier journals	82
Appendix B - Interview guideline	84

3. ALIGNING DIGITAL STRATEGY AND RESOURCES TO CREATE HEALTHCARE 4.0 ORGANIZATIONS: IMPLICATIONS FOR SMART OPERATIONS AND SMART WORKING	85
3.1 INTRODUCTION.....	85
3.2 THEORETICAL BACKGROUND	89
3.2.1 Healthcare 4.0 and the strategic alignment of sociotechnical systems	89
3.2.2 Network embeddedness theory	91
3.3 Hypotheses development.....	93
3.3.1. Healthcare digital strategies and the development of Healthcare 4.0 dimensions	93
3.3.2. The alignment of healthcare digital strategy and resources and the role of network embeddedness	94
3.3.3. Contribution of digital resources for the development of Healthcare 4.0 dimensions	97
3.4 RESEARCH METHOD	99
3.4.1 Sampling.....	99
3.4.2 Survey instrumentation	100
3.4.3 Constructs definition and validation.....	101
3.4.4 Response bias and common method variance	103
3.4.5 Data analysis	104
3.5 RESULTS	104
3.5.1 Power analysis.....	108
3.5.2 Robustness checks.....	109
3.6 DISCUSSIONS AND CONCLUSIONS	109
3.6.1 Theoretical implications.....	113
3.6.2 Managerial implications.....	115
3.6.3 Limitations and future research.....	116
Acknowledgements	117
3.7 REFERENCES.....	117
Appendix A: Questionnaire structure.....	123
4. HEALTHCARE DIGITAL TRANSFORMATION AS A SOCIOTECHNICAL SYSTEM: SHAPING THE INTERACTIONS BETWEEN INDIVIDUALS, STRUCTURES, AND PROCESSES	125
4.1 INTRODUCTION	125
4.2 THEORETICAL BACKGROUND.....	129
4.2.1 Sociotechnical system	129
4.2.2 Organizational design for the HCDDT from a sociotechnical perspective	129
4.2.3 The microfoundations of operating routines	132
4.3 RESEARCH DESIGN	135
4.3.1. Case description	135
4.3.2. Data collection procedure.....	138

4.3.3.	Data analysis	142
4.4	RESULTS	144
4.4.1.	HCDT as a sociotechnical system – organizational aspects.....	145
4.4.2.	HCDT as a sociotechnical system – technical aspects	149
4.4.3.	HCDT as a socio-technical system – social aspects.....	151
4.4.4.	New digital operating routines in healthcare.....	159
4.5	DISCUSSIONS.....	161
4.6	CONCLUSIONS	166
4.7	REFERENCES	168
	Appendix A	173
	Appendix B: Interview guideline	175
5.	CONCLUSIONS	176
5.1	CONTRIBUTIONS.....	180
5.2	LIMITATIONS AND FUTURE RESEARCH	184

1. INTRODUCTION

The demands for improvements in healthcare systems have grown significantly in recent decades, requiring new perspectives for delivering healthcare (Deloitte, 2020). This issue has become more relevant due to increasing life expectancy and the aging population (WHO, 2019), reinforcing the need for the development of disruptive approaches (Wang et al., 2018; Kraus et al., 2021). In this context, digital technologies have played a key role in healthcare systems by breaking paradigms in management and medical care (Ford et al., 2017; Marbough et al., 2023). Since the 1960s, emerging technologies have traditionally been integrated into medical treatments and administrative processes in healthcare settings (Baytechit, 2018). However, it was only after the 1990s, with the advancements of the Internet, that the importance of digital technologies was significantly boosted, giving rise to the terms "e-health" and "smart health" (Aceto et al., 2018).

With the advent of the Industry 4.0 movement, healthcare systems have advanced toward the modernization of operations and the implementation of smart, connected medical care (Tortorella et al., 2020a). This revolution in healthcare is commonly referred to as Healthcare 4.0 (Thuemmler & Bai, 2017; Kumari et al., 2018) or Healthcare Digital Transformation (HCDT) (Kraus et al., 2021). HCDT represents a digital transformation process in healthcare environments, adapting the principles and applications of Industry 4.0 to the sector (Thuemmler & Bai, 2017). This transformation occurs through the incorporation of advanced technologies, such as the Internet of Things (IoT), cloud computing, big data & analytics, artificial intelligence, wearable devices, telemedicine, collaborative robots, augmented and virtual reality, electronic health records, blockchain, among others, applied to both administrative processes and patient care (Tortorella et al., 2022a; Saha et al., 2022; Raimo et al., 2023; Saha & Rathore, 2024). This enables the real-time personalization of services for patients and professionals, facilitating the transition to a patient-centered environment, which is the focus of Healthcare 4.0 (Alloghani et al., 2018; Wang et al., 2018; Tortorella et al., 2022a).

Studies in the field of HCDT have identified positive associations between the adoption of digital technologies and improvements in performance, both in terms of patient care (e.g., Wani & Malhotra, 2018; Queenan et al., 2019; Mukherjee & Sinha, 2020) and operational performance (e.g., Devaraj et al., 2013; Sharma et al., 2016; Mukherjee & Sinha, 2020; Tushe et al., 2025). However, although there is an understanding in the literature about HCDT, when it comes to its structuring, the role of organizational management that influences digitalization

is often overlooked (Kraus et al., 2021). This occurs because most studies have emphasized a technocentric view, focusing on identifying technology and its application (Tortorella et al., 2020a; Tortorella et al., 2022a; Bardhan et al., 2025). In contrast, the organizational structuring, which determines how digital transformation should occur, is not addressed with the necessary depth (Mauro et al., 2024; Bendoly et al., 2025).

In this context, it is essential to understand how to organizationally structure to facilitate the adoption of digital technologies (Kraus et al., 2021; Verhoef et al., 2021; Oludapo et al., 2024). Digital transformation should not focus solely on technical aspects (technology), but also consider other intrinsic organizational factors, such as organizational, social (human factors) and environmental aspects (Marcon et al., 2022; Marcon et al., 2025). Otherwise, the adoption of technologies may be unsuccessful (Zimmer et al., 2023), and barriers may arise that hinder the digitalization journey, especially in complex environments such as the healthcare sector (Tortorella et al., 2021; Iyanna et al., 2022; Mauro et al., 2024).

Organizational structuring for digital transformation refers to the process of planning, defining, and designing a structure that includes key organizational elements aimed at facilitating the introduction and integration of digital technologies (Kraus et al., 2021; Verhoef et al., 2021). Among these organizational elements, digital strategies, digital resources, digital technologies, and microfoundations of operating routines stand out (Chanias et al., 2019; Roscoe et al., 2019; Verhoef et al., 2021). This is because HCDDT is neither purely technological nor purely strategic, but rather a complex sociotechnical process that only materializes when these four elements are aligned and integrated. These four elements are interdependent: digital strategy guides which technologies should be adopted; digital resources determine whether the strategy is feasible; digital technologies generate value only when embedded in daily work practices; and microfoundations, in turn, feed back into the strategy through organizational learning arising from interactions with these technologies.

More specifically, digital strategy guides the objectives of digital transformation and establishes organizational priorities; without strategic clarity, the adoption of technologies tends to be fragmented, reactive, and technocentric (Chanias et al., 2020). Resources—also referred to as digital resources—represent the organizational means that enable the operationalization of the strategy and generally constitute prerequisites for digital transformation. These resources are often scarce or distributed across different actors, which makes their management particularly relevant (Verhoef et al., 2021). They include, for example, technical resources such

as hardware, software, and technological infrastructure that support digital technologies, as well as financial investments, among others. They also encompass social resources, such as human resources (e.g., information technology teams), accumulated knowledge, information assets, and digital culture. Such resources can be managed through different strategies: they may be developed internally by the organization itself (in-house), acquired from external partners (outsourcing), or governed through hybrid models that combine both approaches (Franco et al., 2024).

Digital technologies, in turn, constitute the most visible facet of digital transformation. They refer to computing-based tools and systems that use digital data for various purposes, such as task execution, information processing, and facilitating communication across areas and departments. These technologies materialize the strategy and depend on digital resources for their implementation; however, they generate value only when effectively integrated into work processes and practices. A purely technocentric perspective tends to overlook the fact that digital transformation does not automatically reshape organizations but instead requires its incorporation into the organizational and social context (Bendoly et al., 2025). In this regard, it becomes essential to consider operating routines, as it is within these routines that healthcare professionals interact with digital technologies in their daily patient care. Conceptually, operating routines refer to the execution of known procedures aimed at generating value for the organization, which can be modified through the adoption of digital technologies (Roscoe et al., 2019). These routines are composed of microfoundations, which relate to the effects of interactions among individuals, structures, and processes, influencing outcomes at the macro-organizational level. The development, combination, and implementation of operating routines form the basis of operational capabilities that, over time, become consolidated within the organization (Roscoe et al., 2019).

Organizational structuring aims to align these four elements (strategy, resources, digital technologies and operating routines) within an organizational structure so that they function harmoniously, moving from a macro management perspective to a micro-operational perspective. Models derived from well-established theories in literature, such as the Resource-Based View (RBV) (Barney, 1991), Sociotechnical Dynamics (Marcon et al., 2022), and the Microfoundations of Operating Routines (Felin et al., 2012), for example, may serve as pathways for this structuring. However, each model presents particularities and some limitations. For example, the RBV does not thoroughly explain how resources can be orchestrated (Sirmon et al., 2007); sociotechnical dynamics tend not to comprehensively

address other aspects, such as organizational and environmental elements (Marcon et al., 2022); and the microfoundations have been scarcely explored within the healthcare context (Roscoe et al., 2019). Therefore, although these models can be considered a relevant theoretical background, they require critical analysis and potential reformulation when applied to the context of HCDT, particularly because healthcare is recognized as a complex system composed of multiple interrelated and interacting elements.

Other studies in literature have proposed organizational structuring models for digital transformation. Verhoef et al. (2021) presents three stages of digital transformation: digitization, digitalization, and digital transformation. The study identifies and delineates growth strategies for digital firms, as well as the assets, resources, and capabilities required for a successful digital transformation. The authors postulate that digital transformation requires specific organizational structures and entails implications for the performance metrics used to assess outcomes. However, the study does not address the different possible governance arrangements and strategies for these organizational structures, nor does it examine the specific role of external partners in digital transformation.

Kraus et al. (2021) developed an organizational model explaining how different forms of technology implementation in healthcare can generate operational efficiency gains associated with HCDT. The model encompasses five interconnected dimensions: operational efficiency, patient-centered approaches, organizational factors and managerial implications, workforce practices, and socioeconomic aspects. However, it does not specify the organizational resources required, the strategies for managing these resources, or how the workforce can systematically contribute to achieving HCDT. Chaudhuri et al. (2023) proposed a framework of resources and capabilities required for co-creation in implementing 3D printing services in healthcare. However, although the model addresses technical, organizational, and environmental aspects with support from external partners, it focuses on a single technology rather than the broader scope of HCDT. Moreover, it does not consider how resources should be managed or which strategies may be adopted across different healthcare system settings.

Technological and organizational maturity models have also been proposed in the literature, such as the Digital Readiness Assessment Maturity (DREAMY) model developed by De Carolis et al. (2025), which provides a framework for assessing firms' digital maturity, identifying strengths and weaknesses, and guiding their transition toward Industry 4.0 and 5.0. Nevertheless, this model primarily focuses on maturity assessment and does not specify in detail

the strategies, resources, or organizational changes required for digital transformation, particularly when applied to sectors beyond manufacturing.

Tortorella et al. (2022a) proposed a model comprising four digital technology packages focused on healthcare supply chains, diagnosis, treatment, and patient monitoring. Nonetheless, the model adopts a predominantly technocentric perspective, emphasizing which technologies can be used and how they can be applied, while overlooking the organizational elements that support their implementation. Mishra and Sharma (2022) developed the Digital Transformation Evaluation Model (DTEM), which uses three dimensions—convergence, maturity, and adoption—to assess telehealth providers. The results of its application suggest that DTEM is an effective tool for analyzing these providers and identifying areas for improvement. Nevertheless, although the model was applied in empirical cases, the study highlights a lack of standardization in the telehealth field, emphasizing the need for organizations to revisit and better structure their organizational arrangements to support telemedicine implementation.

However, it can be observed that, among the organizational structuring or digital transformation maturity models cited, both present limitations in that they fail to provide a holistic and integrated approach encompassing digital strategies, digital resources, technologies, and the microfoundations of operating routines. Moreover, they do not explore how these elements can be aligned and examined from technological, organizational, environmental, and social perspectives, which reveals an opportunity to investigate how such structuring can be carried out to support the broad implementation of technologies.

Studies have emphasized the need to organizationally structure HCDT and to identify high-impact digital technologies, as well as the factors that hinder, constrain, or influence its implementation, particularly considering the dimensions technology, organizational and environmental (Mauro et al., 2024; Duus et al., 2026). In addition, Ahmadi et al. (2018), Mauro et al. (2024) and Mao et al. (2024) highlight the importance of considering inter- and intra-organizational factors for the implementation of digital transformation — or even HCDT — from a sociotechnical perspective (Kronlid et al., 2024; Green et al., 2026). Kraus et al. (2021) argue that improvements in the quality of medical care are directly linked to digital transformation and require organizational structures to support it. Therefore, we decided to study the organizational structuring of HCDT and propose an organizational structure based on the four main elements previously mentioned: strategies, resource management, digital

technologies, and changes in operating routines, which will be explored throughout the three articles of this thesis.

Based on these considerations, the following research question is proposed: *How can healthcare organizations be organizationally structured to implement Healthcare Digital Transformation?*

1.1 DISSERTATION TOPIC AND OBJECTIVES

The present study focuses on Operations and Technology Management. The general objective of this dissertation is *to propose an organizational structure that integrates strategies, resources, digital technologies, and the microfoundations of operating routines for the implementation of HCDDT*.

To achieve this goal, the following specific objectives were proposed:

- (i) To propose the main strategies for managing digital resources aimed at the adoption and development of HCDDT.
- (ii) To analyze the effects of aligning digital strategy with digital resources, as well as the moderating effect of network embeddedness, to support the implementation of digital technology packages.
- (iii) To identify how HCDDT shapes the interaction effects among the microfoundations of operating routines and create new healthcare digital operating routines.

Through this proposal, the aim is to develop an organizational understanding that can assist healthcare managers and decision-makers in implementing HCDDT in healthcare systems. The goal is to make medical operations and care more efficient, improving outcomes for both patients and professionals, as well as operational performance.

1.2 JUSTIFICATION OF THE RESEARCH PROBLEM

This thesis is justified based on identified theoretical and practical gaps. Regarding theoretical aspects, it is significant because HCDDT represents a new paradigm in the healthcare field, bringing forth various issues to be researched. Although the topic has garnered interest

from researchers worldwide, leading to an increase in academic studies, there are still gaps in understanding how to manage and develop healthcare systems (Kraus et al., 2021; Tortorella et al., 2022a; 2022b; Mauro et al., 2024). Specifically, this study addresses the management of the adoption of digital technologies that support the principle of digital transformation to achieve the concept of Healthcare 4.0.

HCDT is seen as a strategic approach by organizations seeking a multilevel transformation that goes beyond basic technological advancements. This includes changes in the medical model, shifting toward personalized services where the focus is not only on treating diseases but also on preventive healthcare for patients, as well as transformations in the routines of healthcare professionals. For this reason, HCDT is widely investigated, with various solutions proposed to analyze in detail the new technologies that fall within its scope. However, the concept of HCDT is still considered complex to implement, primarily due to the intrinsic nature of healthcare systems, which involve different areas and departments with non-standardized processes and high uncertainties (Iyanna et al., 2022; Bendoly et al., 2025). Additionally, unlike manufacturing, where products can be reworked, in healthcare, the poor implementation of digital technologies can result in the loss of patients' lives. Most studies in the literature involving HCDT focus only on the technocentric view with a patient-centered approach (e.g., Tortorella et al., 2022a; Raimo et al., 2023). However, for HCDT to be achieved, organizational structuring is necessary, ensuring that organizational and social elements are engaged with digitalization. In this sense, several authors argue that there is a need to structure HCDT to support decision-making regarding the implementation of technologies as a gateway and management tool for digitalization (e.g., Kraus et al., 2021; Iyanna et al., 2022; Dal Mas et al., 2023; Marbough et al., 2023; Mauro et al., 2024).

From a practical perspective, although numerous studies claim that the implementation of digital technologies promotes improvements for both operations and patients, this paradigm still seems distant from the reality of many healthcare organizations (Tortorella et al., 2020b). The HCDT journey requires significant organizational effort on the part of healthcare systems (Dal Mas et al., 2023; Mauro et al., 2024). This may occur because healthcare systems often lack sufficient knowledge about which elements to consider and which paths to follow to organizationally structure HCDT (Kraus et al., 2021). Another possibility is that healthcare organizations underestimate digital transformation, lacking clarity on possible strategies, considering it a costly process with difficulties in accessing the necessary digital resources to manage it and understanding how it changes operations (Iyanna et al., 2022). Therefore,

understanding the role of organizationally structuring HCDT, as proposed in this thesis, aims to reduce these conflicts and assist managers in advancing the digitalization of operations and patient care.

1.3 STUDY DESIGN

This section provides a comprehensive overview of the methodologies and research approaches used to achieve the general and specific objectives within the thesis theme.

1.3.1. Research method

This thesis adopts a research method based on the development of three sequential scientific articles, which are articulated to propose an organizational structure for HCDT. These articles employ qualitative and quantitative methods in a complementary manner to achieve the research objectives; therefore, the thesis can be classified as a mixed-methods study (Dalfovo et al., 2008). Figure 1 presents the research methodology.

The research began with a literature review of the main scientific articles on HCDT. For this purpose, three search criteria were adopted. First, studies published after 2011 correspond to the emergence of the Industry 4.0 concept; second, articles published in top-tier journals ranked ABS 3 or 4; and third, journals in the field of technology and operations management, as this thesis is situated within this research domain. The results of this literature review are presented in the Appendix of chapter 2 corresponding to Article 1. Subsequently, based on this review, Article 1 was developed as a qualitative study, employing a multiple case study approach and the use of semi-structured interviews.

Building on the insights derived from Article 1, Article 2 was developed as a quantitative study aimed at extending and validating the findings of the first article. In this second article, statistical analyses of primary data collected through a survey.

Articles 1 and 2 focused on the macro and meso levels of healthcare organizations and revealed a gap in understanding at the micro-organizational level. Article 3 was developed with the objective of completing the analysis across the different organizational levels. In this third article, a qualitative approach through a case study was also employed to finalize the analysis. Finally, drawing on the integrated findings of the three articles, an organizational structure for the implementation of HCDT is developed and presented, which is discussed in the final chapter of the dissertation.

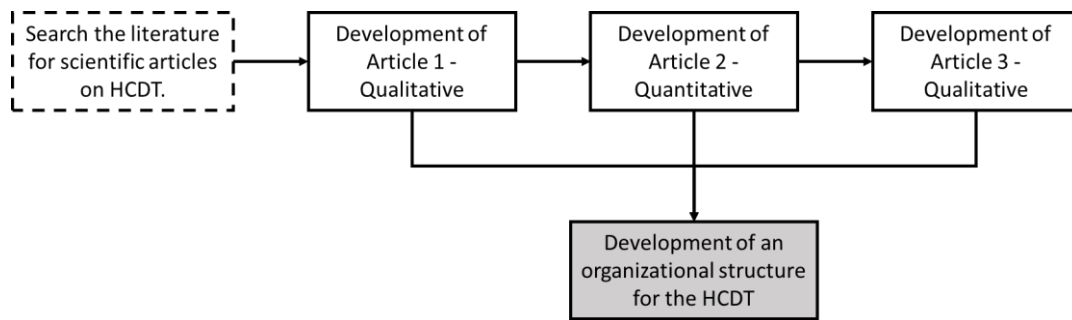


Figure 1. Research methodology

The research approach for Article 1 can be classified as exploratory, as it aims to analyze and deepen the understanding of the relationship between the technological, organizational, and environmental contexts for structuring the management of digital resources for HCDT, based on a multiple case study (Marconi and Lakatos, 2010). The exploratory approach allows for the analysis of a research problem whose mechanisms are not yet fully understood and require further investigation to implement Healthcare 4.0. Additionally, it examines existing knowledge and empirical reality (Gil, 2008). Therefore, this article adopts an inductive approach and uses three different cases to achieve its objective.

On the other hand, for Article 2, a deductive and descriptive approach is employed, as it starts from a theoretical proposition about empirically studied relationships to provide insights into how organizational aspects (the alignment between digital strategy and digital resources and the role of network embeddedness) contribute to the adoption levels of Healthcare 4.0. Subsequently, hypotheses are formulated and tested using statistical tools. This article relies on primary quantitative data collected through questionnaires to achieve its objective.

Finally, like Article 1, the research approach of Article 3 can also be classified as exploratory, as it seeks to analyze and deepen the understanding of how HCDT, as a sociotechnical system, shapes microfoundations — involving the interaction effects between individuals, structures, and processes — in hospitals because of HCDT. This process may enable the emergence of new digital operating routines (Roscoe et al., 2019), which we aim to demystify through a case study in the healthcare sector. Therefore, this article also adopts an inductive approach.

However, by employing a mixed-methods approach and a combination of inductive and deductive reasoning, this thesis aims to provide a comprehensive understanding of the role of organizational structuring in HCDT, with the goal of contributing to theoretical development and offering practical insights for healthcare operations management.

1.3.2. Research Design

The research method developed in this thesis follows the theory-building approach proposed by Van de Ven (2007). According to this theory, the process involves the interaction between theory and practice to generate conceptual models. Achieving this requires the following four key stages: problem definition, theory construction, development of an explanatory model, and proposition of solutions based on the findings (Van de Ven, 2007). The problem definition was addressed in the introduction section, while the remaining stages are described in the subsequent chapters.

This thesis aims to bridge the research gap by analyzing its implications and enhancing the understanding of organizational structuring for HCDT. Therefore, the study is conducted in three principal phases (Figure 1), presented in three scientific articles, each with a specific objective that contributes to achieving the overall goal. Figure 2 illustrates the conceptual models developed throughout the research phases, while Table 1 provides an overview, including research questions, objectives, and methods.

Phase 1 aims to explore and understand how hospitals with different ownership structures organize themselves to manage digital resources for HCDT, using multiple case studies and in-depth interviews. The studies were conducted in three large hospitals located in Southern Brazil: Grupo Hospitalar Conceição, Hospital Mãe de Deus, and Hospital de Clínicas de Porto Alegre (HCPA), representing, respectively, public, private, and teaching and research management models. To achieve this, a framework is considered (Model 1 in Figure 2), which evaluates three configurations of digital resource management across three organizational dimensions for technological innovation: technological, organizational, and environmental.

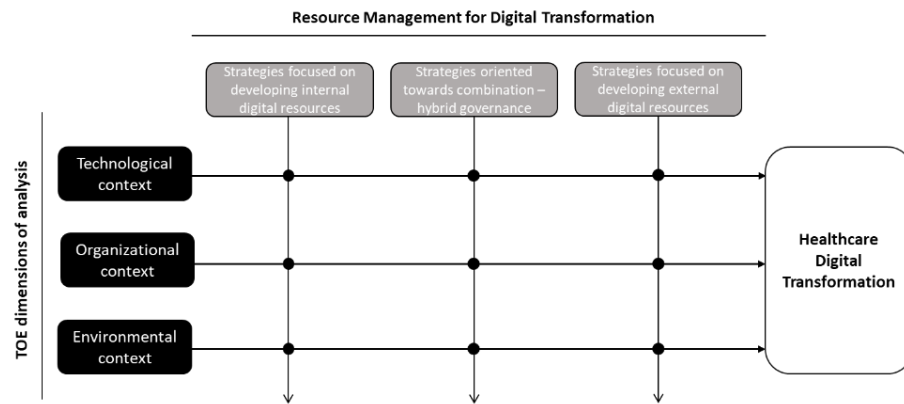
Building on the insights from Phase 1, Phase 2 aims to understand, through a quantitative study conducted in Brazilian hospitals with different ownership structures (public, private, and teaching/research hospitals), the dynamics of aligning digital strategy with digital resources (whether managed internally or externally) as an organizational approach to implementing digital technologies as a sociotechnical system. This implementation is analyzed in two key applications: process automation (smart operations, technical aspect) and support

for healthcare workers (smart working, social aspect) (Model 2 in Figure 2). The study is conducted using primary data collected through a survey carried out by the National Association of Private Hospitals, Fundação Dom Cabral, and the Organizational Engineering Group (NEO-UFRGS).

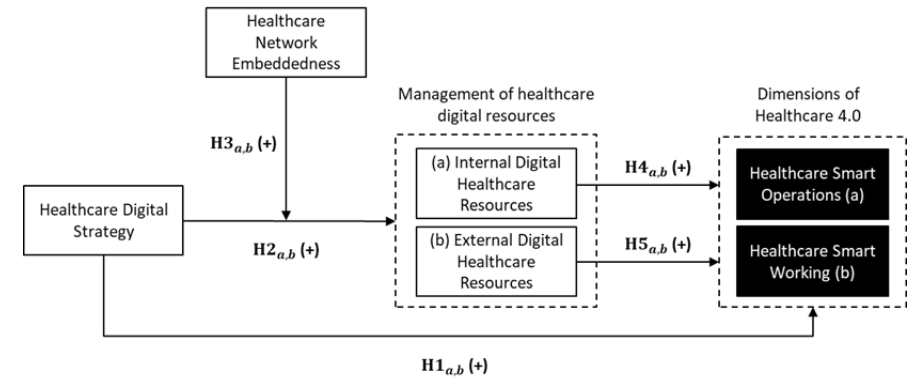
Finally, based on the findings from Phase 2, it became evident that a detailed micro-level analysis of the organization was necessary, as this is where digital technologies are operationalized and integrated into the organizational structuring for HCDT. In this context, Phase 3 aims to explore, through a qualitative case study, how HCDT as a sociotechnical system shapes the microfoundations of operating routines. To achieve this, a framework is adopted (Model 3 in Figure 2), which examines how digital strategy, digital resources and digital technologies influences the interaction effects of microfoundations—individuals, hierarchical structures, and processes—potentially leading to the creation of new healthcare digital operating routines. This case study was conducted at the HCPA, recognized as one of the leading public hospitals in Latin America. The institution holds several national and international certifications and is widely acknowledged for its initiatives in digital transformation. This study was made possible through a collaborative research project carried out between 2024 and 2025 involving the NEO-UFRGS and HCPA.

In this way, it is possible to develop a comprehensive understanding of the phenomenon of organizational structuring in HCDT.

Phase 1 – (Model 1) Configurations for managing digital resources



Phase 2 – (Model 2) Alignment of digital strategy with digital resources for the adoption of Healthcare 4.0 dimensions and the moderating role of healthcare network embeddedness



Phase 3 – (Model 3) HCDDT as a sociotechnical system: shaping the interactions between individuals, structures, and processes

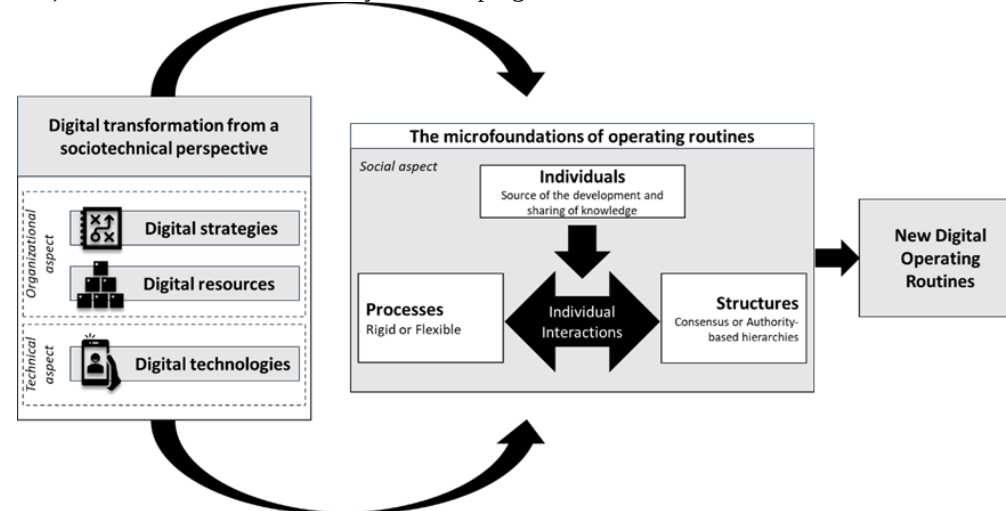


Figure 2. Conceptual models developed throughout the research phases (Source: prepared by the author)

Table 1. Article's scope

	Research question	Objective	Background theoretical	Research method
Article 1	How do hospitals deal with digital resource management strategies to adopt and develop HCDT?	This study aims to define configurations of digital resource management for HCDT in hospitals and analyze how they are implemented. To this aim, we employ the Technological, Organizational, and Environmental (TOE) framework to analyze these three forms of digital resource management in hospitals from a holistic perspective.	1. Managing DT in Healthcare organizations. 2. An analytical framework to understand digital resource management strategies for HCDT.	Qualitative research 1. Multiple case study in hospitals 2. Semi-structured interviews 3. Analysis of the interviews grouped by content and methodology Gioia
Article 2	What is the role of digital strategy and digital resources – as organizational strategic alignment – in the development of Healthcare 4.0 sociotechnical dimensions, namely healthcare smart operations and healthcare smart working? And what is the effect of healthcare network embeddedness in this context?	This study aims to analyze the contribution of digital strategies and resources for the development of a sociotechnical view of Healthcare 4.0, considering smart operations and smart working as the main dimensions. Furthermore, analyze the moderating role of healthcare network embeddedness in the strategy-resources relationship.	1. Healthcare 4.0 and the strategic alignment of sociotechnical systems. 2. Network embeddedness theory 3. Hypotheses development.	Quantitative research 1. Survey with hospitals 2. Exploratory factor analysis (EFA) 3. Confirmatory factor analysis (CFA) 4. Linear regression 5. Bootstrapping analyses
Article 3	How does HCDT, as a sociotechnical system, shape microfoundations and generate new digital operating routines in healthcare?	This study aims to examine how HCDT as a sociotechnical system shapes the microfoundations involving individuals, structures, and processes—viewed as social aspects.	1. Sociotechnical system 2. Organizational design for the HCDT from a sociotechnical perspective 3. The microfoundations of operating routines	Qualitative research 1. Case study in hospitals 2. Semi-structured interviews 3. Participant observation 4. Analysis of the interviews grouped by content and methodology Gioia

- (a) Article 1 published in the International Journal of Technological Innovation, Entrepreneurship and Technology Management – Technovation (Qualis Capes A1);
- (b) Article 2 published in the Technological Forecasting and Social Change (Qualis Capes A1);
- (c) Article 3 to be submitted for publication in the International Journal of Operations & Production Management (Qualis Capes A1).

Article 1 - “Structuring resources in healthcare digital transformation: a comparison across public, private and research hospitals”.

The objective of this phase was to identify how different hospital ownership structures handle various digital resource management strategies to adopt and develop HCDT. Additionally, it aimed to identify the key factors that lead these hospitals to follow specific

management models. This phase had a qualitative approach, with eight interviews conducted with managers involved in healthcare digital transformation. The analyzed hospitals are implementing new digital technologies to digitize operations and medical care. As key findings, the article presents three different organizational structuring strategy frameworks for resource management in HCDDT. These frameworks vary according to the hospital ownership structure and illustrate the dynamic involvement of the technological, organizational, and environmental contexts in fostering innovation.

Article 2 – *“Aligning digital strategy and resources to create Healthcare 4.0 organizations: implications for smart operations and smart working”*.

The objective of this phase was to validate the previous stage and identify the existence of an association between the alignment of digital strategy and digital resources for the implementation of Healthcare 4.0 as a sociotechnical system, and the role of the network embeddedness in the alignment. This phase had a quantitative approach, conducted through a survey involving 303 hospitals engaged in digital transformation. The key findings indicate that a strong healthcare digital strategy positively influences the adoption of technologies aimed at process automation (smart operations, technical aspect) and support for workers' activities (smart working, social aspect). On the other hand, digital resources, whether managed internally or acquired externally for Healthcare 4.0, positively mediate only the relationship between digital strategy and smart working. However, no evidence was found to support the same mediation effect for smart operations. Additionally, healthcare network embeddedness positively moderates only the relationship between healthcare digital strategy and external digital resources. In contrast, there was no evidence that network embeddedness enhances the alignment between strategy and internal digital resources.

Article 3 – *“Healthcare digital transformation as a sociotechnical system: shaping the interactions between individuals, structures, and processes”*.

In this final phase, the objective is to deepen the understanding developed in the previous study by analyzing how HCDDT, as a sociotechnical system, shapes the microfoundations of operating routines. These microfoundations involve interactions among individuals, hierarchical structures (based on consensus or authority), and processes (rigid or flexible), which are understood as the social aspect of the organization. This analysis is particularly relevant because it focuses on the micro-organizational level, where digital technologies are effectively operationalized and where direct interactions between people and

technologies take place. As a result, the consolidation of these interactions over time leads to the emergence of new operating routines. This phase adopts a qualitative approach, involving semi-structured interviews, focus groups with healthcare professionals, participant observation across different areas and departments, as well as the analysis of documents and reports from a public teaching and research hospital in southern Brazil that is in the process of implementing digital technologies in its operations. The findings indicate that interactions among individuals, structures, and processes enable a customized digital transformation aligned with professionals' needs, as they mobilize their knowledge and experience to define the alignment between strategies, resources, and digital technologies. Moreover, the results show that HCDDT does not necessarily emerge in a top-down manner, as traditionally observed in healthcare systems, but can also arise from the micro level of the organization, contributing to the reduction of potential human-machine conflicts and enabling the emergence of new healthcare digital operating routines.

1.4 STUDY DEFINITION

For the development of this research, the practical context of Brazil was considered, as developing countries often face difficulties and barriers in the process of organizational digital transformation, particularly in the healthcare sector and even more so within public healthcare systems. Thus, the structured implementation of HCDDT in hospitals can contribute to accelerating national development, fostering digital transformation, and improving population health outcomes. However, developed countries present different institutional contexts and levels of technological maturity. Therefore, the territorial scope represents a limitation of this study. To partially mitigate this limitation, a sandwich PhD research period lasting nine months was conducted at the Massachusetts Institute of Technology (MIT), at the Industrial Performance Center (MIT-IPC), in the United States. This experience enabled technical visits to leading hospitals and interviews with managers, thereby complementing the understanding of the organizational structuring of HCDDT in a developed-country context.

Moreover, this study does not aim to conduct comparisons between developed and developing countries, but rather to analyze environmental, organizational, technological, and social aspects — including elements such as digital strategies, digital resources, digital technologies, and operating routines — that can be applied in practice to structure HCDDT across different economic contexts. Nevertheless, it is acknowledged that hospitals may require

different management models over time, because of institutional, technological, and organizational changes.

1.5 STRUCTURE OF THE DISSERTATION

This thesis proposal is structured into five chapters. The first chapter provides a detailed discussion of the research problem, objectives, and corresponding justifications. Additionally, it outlines the research methodology, study structure, and scope. Building on this foundation, Chapters 2, 3 and 4 present the core articles developed thus far, each addressing one of the study's specific objectives, as illustrated in Figure 2. These chapters offer a comprehensive analysis of the articles, including their methodologies, key findings, and theoretical and practical contributions, thereby enriching our understanding of the phenomenon.

The fifth and final chapter of this thesis proposal is dedicated to the conclusions and contributions of this dissertation. It focuses on the overall objective of the research, providing an integrative overview of the findings identified in the previous chapters. In this way, the fifth chapter contributes to the cohesion and advancement of the research, bringing to light opportunities that follow from this thesis.

1.6 REFERENCES

- Aceto, G., Persico, V., Pescapé, A. (2018). The role of Information and Communication Technologies in healthcare: taxonomies, perspectives, and challenges. *Journal of Network and Computer Applications*, 107, 125–154.
- Ahmadi, H., Nilashi, M., Shahmoradi, L., Ibrahim, O., Sadoughi, F., Alizadeh, M., & Alizadeh, A. (2018). The moderating effect of hospital size on inter and intra-organizational factors of Hospital Information System adoption. *Technological Forecasting and Social Change*, 134, 124–149.
- Alloghani, M. et al. (2018). Healthcare services innovations based on the state of the art technology trend industry 4.0. 2018 11th International Conference on Developments in eSystems Engineering (DeSE). *Anais...IEEE*.
- Baytechit. (2018). The History of Healthcare Technology and the Evolution of HER. Accessed Juno 15, 2024. <https://www.baytechit.com/history-healthcare-technology/>
- Baptista, J. et al. Digital work and organisational transformation: Emergent Digital/Human work configurations in modern organisations. *Journal of Strategic Information Systems*, 29(2), 101618.

- Bardhan, I. Kohli, R. Oborn, E. Mishra, A. Tan, C. H. Tremblay, M. C. Sarker, S. (2025). Humancentric information systems research on the digital future of healthcare. *Information Systems Research*. Articles in Advance, 1–20.
- Bendoly, E. Li, M. Smith, J.; Pavlou, P. A. (2025). Operational Process and Agency Considerations in Managing Emerging Technologies in Healthcare [Editorial]. *Journal of Operations Management*, 1-12, 2025.
- Brunzini, A., Peruzzini, M., Barbadoro, P. (2023). Human-centred data-driven redesign of simulation-based training: a qualitative study applied on two use cases of the healthcare and industrial domains. *Journal of Industrial Information Integration*, 35(100505), 100505.
- Chanias, S., Myers, M. D., Hess, T. (2019). Digital transformation strategy making in pre-digital organizations: The case of a financial services provider. *Journal of Strategic Information Systems*, 28(1), 17–33.
- Chaudhuri, A., Naseraldin, H., & Narayanamurthy, G. (2023). Healthcare 3D printing service innovation: Resources and capabilities for value Co-creation. *Technovation*, 121, 102596
- Dal Mas, F. et al. The challenges of digital transformation in healthcare: An interdisciplinary literature review, framework, and future research agenda. *Technovation*, v. 123, n. 102716, p. 102716, 2023.
- Dalfovo, M. S., Lana, R. A., Silveira, A. (2008). Métodos quantitativos e qualitativos: um resgate teórico. *Revista Interdisciplinar Científica Aplicada*, 2(3), 1-13.
- De Carolis, A., Sassanelli, C., Acerbi, F., Macchi, M., Terzi, S., & Taisch, M. (2025). The Digital REadiness Assessment MaturitY (DREAMY) framework to guide manufacturing companies towards a digitalisation roadmap. *International Journal of Production Research*, 63(15), 5555–5581.
- DELOITTE. Digital Transformation: Shaping the Future of European Healthcare 2020. Available at: <https://www2.deloitte.com/content/dam/Deloitte/uk/Documents/life-sciences-health-care/deloitte-uk-shaping-the-future-of-european-healthcare.pdf> (Accessed Juno 15, 2024).
- Devaraj, S., Ow, T. T., Kohli, R. (2013). Examining the impact of information technology and patient flow on healthcare performance: A Theory of Swift and Even Flow (TSEF) perspective. *Journal of Operations Management*, 31(4), 181–192.
- Duus, R., Cooray, M., & Lilley, S. (2026). AI adoption in healthcare organizations: Spheres of development and the virtue of visible value. *Technological Forecasting and Social Change*, 226(124541), 124541.
- Elhoseny, M. et al. (2018). A hybrid model of Internet of Things and cloud computing to manage big data in health services applications. *Future Generations Computer Systems*, 86, 1383–1394.
- Felin, T., Foss, N. J., Heimeriks, K. H., & Madsen, T. L. (2012). Microfoundations of routines and capabilities: Individuals, processes, and structure. *Journal of Management Studies*, 49, 1351–1374.
- Ford, G. et al. (2017). The role of digital disruption in healthcare service innovation. In: *Service Business Model Innovation in Healthcare and Hospital Management*. Cham: Springer International Publishing, 57–70.

- Franco, C. W., Benitez, G. B., Sousa, P. R., Neto, F. J. K., Frank, A. G. (2024). Managing resources for digital transformation in supply chain integration: The role of hybrid governance structures. *International Journal of Production Economics*, 278(109428), 109428.
- Gil, A. C. (2008). Métodos e técnicas de pesquisa social / Antonio Carlos Gil. - 6. ed. - São Paulo: Atlas.
- Green, N., Tappin, D., & Bentley, T. (2025). The impact of telehealth videoconferencing on the healthcare work system: designing for sustained practice. *Applied Ergonomics*, 130(104641), 104641.
- Grover, P.; Kar, A. K.; Davies, G. (2018). “Technology enabled Health” – Insights from twitter analytics with a socio-technical perspective. *International Journal of Information Management*, 43, 85–97.
- Iyanna, S. et al. (2022). Digital transformation of healthcare sector. What is impeding adoption and continued usage of technology-driven innovations by end-users? *Journal of Business Research*, 153, 150–161.
- Kronlid, C., Brantnell, A., Elf, M., Borg, J., & Palm, K. (2024). Sociotechnical analysis of factors influencing IoT adoption in healthcare: A systematic review. *Technology in Society*, 78(102675), 102675.
- Kraus, S. et al. (2021). Digital transformation in healthcare: Analyzing the current state-of-research. *Journal of Business Research*, 123, 557–567.
- Kumari, A. et al. (2018). Fog computing for Healthcare 4.0 environment: Opportunities and challenges. *Computers & Electrical Engineering: an International Journal*, 72, 1–13.
- Mao, H., Liu, S., & Gong, Y. (2024). Balancing structural IT capabilities for organizational agility in digital transformation: a resource orchestration view. *International Journal of Operations & Production Management*, 44(1), 315–344.
- Marbough, D. et al. (2023). Healthcare 4.0 digital technologies impact on quality of care: a systematic literature review. *Total Quality Management & Business Excellence*, 34(15–16), 2157–2182.
- Marcon, É. et al. (2025). Combining Organizational and Social Factors to Support Industry 4.0 Implementation: A Sociotechnical and Configurational Analysis of Technology Adopters. *IEEE Transactions on Engineering Management*, 72, 146-160.
- Marcon, É. et al. (2022). Sociotechnical factors and Industry 4.0: an integrative perspective for the adoption of smart manufacturing technologies. *Journal of Manufacturing Technology Management*, 33(2), 259–286.
- Marconi, M. D. A.; Lakatos, E. M. (2010). Fundamentos de metodologia científica. Atlas.
- Mauro, M., Noto, G., Prenestini, A., Sarto, F. (2024). Digital transformation in healthcare: Assessing the role of digital technologies for managerial support processes. *Technological Forecasting and Social Change*, 209, 123781.
- Mishra, V., & Sharma, M. G. (2022). Digital transformation evaluation of telehealth using convergence, maturity, and adoption. *Health Policy and Technology*, 11(4), 100684.

- Mukherjee, U. K., Sinha, K. K. (2020). Robot-assisted surgical care delivery at a hospital: Policies for maximizing clinical outcome benefits and minimizing costs. *Journal of Operations Management*, 66(1–2), 227–256.
- Oludapo, S., Carroll, N., Helfert, M. (2024). Why do so many digital transformations fail? A bibliometric analysis and future research agenda. *Journal of Business Research*, 174, 114528.
- Queenan, C., Cameron, K., Snell, A., Smalley, J., Joglekar, N. (2019). Patient heal thyself: Reducing hospital readmissions with technology-enabled continuity of care and patient activation. *Production and Operations Management*, 28(11), 2841–2853.
- Roscoe, S., Cousins, P. D., Handfield, R. (2019). The microfoundations of an operational capability in digital manufacturing. *Journal of Operations Management*, 65(8), 774–793.
- Saha, E., Rathore, P. (2024). The impact of healthcare 4.0 technologies on healthcare supply chain performance: Extending the organizational information processing theory. *Technological Forecasting and Social Change*, 201, 123256.
- Saha, E. et al. (2022). The interplay of emerging technologies in pharmaceutical supply chain performance: An empirical investigation for the rise of Pharma 4.0. *Technological Forecasting and Social Change*, 181, 121768.
- Sharma, L. Chandrasekaran, A. Boyer, K. K. Mcdermott, C. M. (2016). The impact of Health Information Technology bundles on Hospital performance: An econometric study. *Journal of Operations Management*, 41(1), 25–41.
- Sirmon, D. G., Hitt, M. A., Ireland, R. D. (2007). Managing firm resources in dynamic environments to create value: looking inside the black box. *Academy of Management Review*, 32(1), 273–292.
- Thuemmler, C., Bai, C. Health 4.0: Application of industry 4.0 design principles in future asthma management. Em: Health 4.0: How Virtualization and Big Data are Revolutionizing Healthcare. Cham: Springer International Publishing, 2017. 23–37.
- Tortorella, G. L. et al. (2022a). Healthcare 4.0 digital applications: An empirical study on measures, bundles and patient-centered performance. *Technological Forecasting and Social Change*, 181, 121780.
- Tortorella, G. L. et al. (2022b). Measuring the effect of Healthcare 4.0 implementation on hospitals' performance. *Production Planning & Control*, 33(4), 386–401.
- Tortorella, G. L. et al. (2021). Impacts of Healthcare 4.0 digital technologies on the resilience of hospitals. *Technological Forecasting and Social Change*, 166, 120666.
- Tortorella, G. L. et al. (2020a). Healthcare 4.0: trends, challenges and research directions. *Production Planning & Control*, 31(15), 1245–1260.
- Tortorella, G. L. et al. (2020b). Effects of contingencies on healthcare 4.0 technologies adoption and barriers in emerging economies. *Technological Forecasting and Social Change*, 156, 120048.

- Tushe, S. Kc, D. S. Ding, H. Yeung, H. (2025). Multichannel healthcare: Impact of asynchronous telemedicine adoption on patient flow. *Manufacturing & Service Operations Management*, 27(1), 59–74.
- Van De Ven, Andrew H. (2007). *Engaged scholarship: a guide for organizational and social research: a guide for organizational and social research*. OUP Oxford.
- Verhoef, P. C. et al. (2021). Digital transformation: A multidisciplinary reflection and research agenda. *Journal of Business Research*, 122, 889–901.
- Wang, Y. et al. (2018). An integrated big data analytics-enabled transformation model: Application to health care. *Information & Management*, 55(1), 64–79.
- Wani, D.; Malhotra, M. (2018). Does the meaningful use of electronic health records improve patient outcomes? *Journal of Operations Management*, 60(1), 1–18.
- World Health Organization (WHO) (2019). Life expectancy. Available at: <https://www.who.int/data/gho/data/themes/mortality-and-global-health-estimates> (Accessed on June 15, 2024).
- Zimmer, M. P. et al. (2023). Digital workplace transformation: Subtraction logic as deinstitutionalising the taken-for-granted. *The Journal of Strategic Information Systems*, 32(1), 101757.

2. STRUCTURING RESOURCES IN HEALTHCARE DIGITAL TRANSFORMATION: A COMPARISON ACROSS PUBLIC, PRIVATE AND RESEARCH HOSPITALS

Lucas Miguel Alencar de Moraes Correia, Alejandro Germán Frank

This article has been published in Technovation (JCR: 10,9)

<https://doi.org/10.1016/j.technovation.2025.103320>

Abstract

Healthcare organizations have witnessed major advances in the use of digital technologies. While prior research on Healthcare Digital Transformation (HCDT) has focused on the impact of digital technologies on healthcare operations, understanding how to manage the digital resources necessary for HCDT implementation remains unclear. We address this gap by examining how hospitals manage digital resources to foster HCDT. Drawing on the Technological, Organizational, and Environmental (TOE) framework and supported by the Resource-Based View, Relational View, and hybrid governance theories, we conduct a qualitative multi-case study of three large hospitals: one public, one private, and one research-oriented. Data collection includes semi-structured interviews, document analysis, and on-site visits. Our study reveals that HCDT strategies are shaped by hospital business models and financial structures, leading to distinct digital resource governance approaches: internal development in public hospitals, external sourcing in private hospitals, and hybrid governance in research hospitals. We propose a typology aligned with strategic logics and clarify how each model engages with digital ecosystems—ranging from internal coordination to external orchestration and collaborative innovation. Despite their dominant strategies, all hospitals maintain complementary resource bases to ensure resilience. Additionally, we contribute to theory by conceptualizing hybrid governance in research hospitals as a funding-driven mechanism, offering a new rationale for its adoption. These insights advance the understanding of digital resource management in healthcare and provide practical guidance for designing context-sensitive HCDT strategies.

Keywords: Healthcare; Digital transformation; Digital technologies; Digital resources; Business models.

2.1 INTRODUCTION

Digital Transformation (DT) has been developed in different sectors, including healthcare organizations. In this case, it is defined as Healthcare Digital Transformation (HCDT) (Kraus et al., 2021) or Healthcare 4.0 (H4.0) (Thuemmler & Bai, 2017). HCDT considers several digital technologies that have been applied and integrated to enhance and modernize healthcare systems, aiming for efficiency in processes and patient well-being (Raimo et al., 2023). HCDT has been analyzed from the organizational perspective with the use of

digital technologies to support the administration of hospitals (Cerchione et al., 2023) and from the operational perspective to execute routines like surgeries, emergencies, and related activities (Beane, 2019; Wang et al., 2018).

Several studies have considered the specificities of technologies that can support HCDT (e.g., Tortorella et al., 2021; Tortorella et al., 2022a). These studies are devoted to discussing how digital technologies impact healthcare operations. However, a crucial factor for the implementation and success of HCDT in healthcare environments, and which is usually dismissed, is the management of organizational resources related to digital technologies, referred to as *digital resources* (Piccoli et al., 2024; Verhoef et al., 2021). The view of digital resources considers what hospitals must have to effectively implement digital technologies and transform their process from analog to digital (Elia et al., 2021). It acknowledges that digital transformation is not just about technology but also about how different resources are combined to digitalize the processes (Chaudhuri et al., 2024; Franco et al., 2024). In this sense, hospitals can adopt different strategies to manage the required digital resources for developing the HCDT.

Hospitals may *internally* develop digital resources to enhance healthcare and administrative processes through HCDT (Chaudhuri et al., 2023). This strategy aims to create digital capabilities within the hospital that can be used to improve operational efficiency and enhance healthcare professionals' skills (Oliveira et al., 2024). On the other hand, hospitals can focus on *outsourcing* HCDT to external partners, forming a digital healthcare ecosystem. Outsourcing can be used to mitigate risks and reduce costs (Ayala et al., 2017). This strategy enables the formation of external alliances for acquiring knowledge and technology from partners with expertise in digital solutions (Dalenogare et al., 2023). The disadvantage of this approach is that it can create a high dependence on a potential strategic area that hospitals can develop (Dalenogare et al., 2022). Both strategies require different management profiles for HCDT, as organizational structures have peculiarities reflecting different forms of associated benefits and risks (Franco et al., 2024). A third way is that hospitals can also manage digital resources by developing *hybrid governance* to achieve better operational performance (Mura et al., 2021). This strategy aims to find a balance in managing and gaining synergies between *exploitation* and *exploration* activities (March, 1991). For example, the complementarity between internal and external digital resources helps the hospital exploit internal and external digital resources (Franco et al., 2024). However, in such cases, the cost of maintaining two

different forms of digital resources can become higher, and other benefits must be higher to justify this combination (Parmigiani, 2007).

Although there is a wide understanding of resource management to gain a competitive advantage, how to manage such resources for implementing HCDT remains diffuse (Kraus et al., 2021). The literature has addressed digital resource management to comprehend the co-creation process with specific technologies (Chaudhuri et al., 2023). There is also an understanding regarding developing capabilities from digital resources for HCDT (Oliveira et al., 2024). Nonetheless, the literature lacks evidence on how these three different resource strategies are adopted and their modus operandi in HCDT. In this sense, it is still necessary to understand how internal, external, and hybrid governance digital resource management strategies are organizationally structured in hospitals to promote HCDT. Therefore, we propose the following research question: *How do hospitals deal with digital resource management strategies to adopt and develop HCDT?*

This study aims to define configurations of digital resource management for HCDT in hospitals and analyze how they are implemented. To this aim, we employ the Technological, Organizational, and Environmental (TOE) framework to analyze these three forms of digital resource management in hospitals from a holistic perspective. First, we propose a theoretical framework for models of digital resource management, considering three governance strategies (Franco et al., 2024): strategies focused on developing internal digital resources, strategies focused on developing external digital resources, and strategies oriented towards a combination of both through hybrid governance. We utilize the TOE framework in a qualitative multi-case study research approach to evaluate these strategies. The research is based on semi-structured interviews guided by an open script conducted in three hospitals of different natures: private, public, and research-oriented. We combine the Resource-Based View (RBV), Relational View (RV), and hybrid governance to investigate how digital resource management is developed in hospitals to implement HCDT.

Our research clarifies how digital resource management in hospitals is structured by describing how hospitals manage and utilize digital resources for HCDT. Our final frameworks reveal that the management of digital resources for HCDT in hospitals varies according to the business profile of each institution. Each analyzed hospital presents a distinct business model that defines how digital resources are accessed to promote technological innovations. We also demonstrate how the elements of the TOE framework support the structuring of digital resource

management for HCDT. In the case of the public hospital, it adopts a complete internal model, focusing on developing its own resources. This strategy is driven by the hospital's reliance on bidding processes for acquiring digital resources, which often require greater financial investments and significantly extend the timeline for digital transformation with high transactional costs to deal with technology providers. On the other hand, private hospitals opt to acquire external resources, outsourcing technologies, and services through direct contracts while establishing intra-organizational relationships to create an innovative ecosystem in the healthcare sector based on their power structure that allows them to manage contracts with technology providers based on performance. Meanwhile, the research hospital prioritizes a hybrid governance model, combining the development of internal resources with an emphasis on internal R&D with external collaborations. In this case, they aim to create internal platforms to follow external dissemination and expansion through research collaboration. The TOE framework clarifies how hospitals organize their digital resources in each model described. Therefore, we contribute to the theory by clarifying these different digital resource structures, while our findings also help managers build their digitalization strategies in HCDT.

Our study offers several novel contributions to the understanding of HCDT. First, we demonstrate that HCDT strategies are highly sensitive to the hospital's business model, challenging the prevailing notion of a uniform digital transformation path in healthcare. Second, we introduce a typology of digital resource governance—internal, external, and hybrid—each aligned with specific strategic logics (resource-based view, relational view, and hybrid governance, respectively) and shaped by the institution's financial structure and objectives. Third, our findings uncover distinct models of ecosystem orchestration: while private hospitals act as central coordinators of external providers, research hospitals engage in horizontal, collaborative innovation networks, and public hospitals rely on internal coordination mechanisms. Fourth, we show that despite the predominance of a specific governance model in each hospital type, all organizations must maintain complementary resource bases to ensure resilience and continuity, thereby advancing a more nuanced understanding of the internal–external resource dynamic. Finally, we contribute to the conceptual development of hybrid governance by highlighting its use in research hospitals as a funding-driven strategy—distinct from the balancing rationale commonly found in the literature—thereby revealing a new explanatory mechanism for its adoption in digital transformation initiatives.

The next sections of this paper are structured as follows. In Section 2, we provide the theoretical background for managing digital transformation in healthcare organizations, as well

as an analytical framework for understanding strategies for managing digital resources for HCDT. Section 3 presents the research method, discussing our qualitative approach to studying hospital cases. Section 4 presents the results, followed by Section 5, which presents the discussion and conclusion.

2.2 THEORETICAL BACKGROUD

2.2.1 Managing DT in healthcare organizations

HCDT is a term used in literature to describe the adoption and implementation of digital technologies in the healthcare sector (Kraus et al., 2021). It involves technologies such as the Internet of Things (IoT), cloud computing, Big Data & Analytics (Wang et al., 2018), and Artificial Intelligence (AI) (Dicuonzo et al., 2023). This enables real-time data collection, storage, and analysis of healthcare operations, facilitating the transition to a patient-centered environment (Tortorella et al., 2022). Other digital technologies have been used to support healthcare operations, including wearable devices and augmented/virtual reality and simulation (Raimo et al., 2023), telemedicine (Baudier et al., 2023), 3D printing (Chaudhuri et al., 2023), and collaborative robots (Beane, 2019). Furthermore, ERPs, electronic health records (Raimo et al., 2023), and blockchain (Massaro, 2023) are also used in the HCDT journey, specifically in administrative processes and the supply chain.

Managing HCDT involves dealing with various digital resources. First, technological infrastructure is a key resource to enable the use of digital technologies, including the hardware, software, and network structure necessary to run digital solutions (Verhoef et al., 2021). Second, human capital is also an important part of digital resources, which includes having a team of skilled workers to implement digital technologies, IT workers, and programmers (Fernandez-Vidal et al., 2022). The models for managing digital resources for HCDT in hospitals can follow different approaches (Parmigiani, 2007): developing digital resources *internally (in-house)*, managing the acquisition of external digital resources (*outsourcing*), and adopting a hybrid governance that combines both. These three perspectives are grounded on the competence view of the firms, which considers the choice of whom will be the most competent part for executing an activity (Ketokivi & Mahoney, 2020).

The *internal digital resources'* perspective is rooted in the Resource-Based View (RBV), which examines the relationship between internal characteristics and performance, arguing that organizations can define, own, and accumulate specific resources and capabilities

that are valuable, rare, and more difficult to imitate or substitute (Barney, 1991). These resources can be tangible, such as assets, technologies, people, and infrastructure, or intangible, such as “know-how” and reputation. The main characteristic of this approach is the total control that the organization has over its internal resources (Parmigiani, 2007). This is because the organization owns its resources, allowing it to determine how it will enhance operations (Barney, 1991). Additionally, it contributes to building knowledge and information through the internal development of digital resources and enhancing the capabilities and skills of human resources, qualifying healthcare professionals (Dal Mas et al., 2023; Piccoli et al., 2024). However, in this governance model, exploring physical resources involves using complex social resources, such as social relationships, culture, traditions, etc. This social complexity is a challenge in managing internal digital resources because it involves intangible elements that the organization often cannot control (Elia et al., 2021). Furthermore, it may entail high costs to maintain the operation of resources (e.g., IT teams to develop digital technologies), which can be a limiting factor for the hospital.

The *external digital resources*' perspective is rooted in the Relational View (RV), which argues that organizations can find digital resources and/or create value through their relationship with external parties (Dyer & Singh, 1998). The RV suggests that inter-organizational alliances allow organizations to overcome their limits and access critical routines, resources, and capabilities to become more innovative. Two main characteristics encompassing this approach are digital resource outsourcing and forming an ecosystem (Kamalaldin et al., 2020). Resource outsourcing is when an organization hires services or acquires goods from suppliers instead of conducting these activities internally (Gambal et al., 2022). Furthermore, it involves the transfer of specific responsibilities and functions, enabling the organization to focus on its core strategic activities, mitigating risks, and reducing costs (Kamalaldin et al., 2020). On the other hand, forming a partner ecosystem involves creating an arena of external alliances in which organizations engage in activities and share a common goal (Aloini et al., 2023; Schiavone et al., 2021). The innovation literature defines ecosystems as “*the alignment structure of the multilateral set of partners that need to interact in order for a focal value proposition to materialize*” (Adner, 2017, p. 40). This implies that, unlike simple outsourcing, when ecosystems are considered part of external digital resources, the alignment of external actors comes to the forefront. Such alignment may occur either around a network of companies with diverse technological capabilities or through technology platforms that integrate these actors (Gawer and Cusumano, 2014). The digital transformation literature has

emphasized the role of digital ecosystems in delivering integrated solutions for companies, recognizing that no single firm can independently manage the full spectrum of technical skills and competencies required (Benitez et al., 2020). Accordingly, an orchestrated combination of capabilities offers a more effective approach (Dalenogare et al., 2023). Studies on healthcare ecosystems have highlighted several benefits of creating an external network of actors, including the increase of knowledge transfer flow (Secundo et al., 2019) and value co-creation (Schiavone et al., 2021). However, although this approach offers benefits such as resource savings and a focus on essential competencies, it also poses challenges, including the organization's high dependence on external suppliers, hold-up problems, quality control issues, and potential opportunities in the strategic area of internal development that may go unrealized (Gambal et al., 2022).

A third option for managing digital resources is combining internal and external resources into a single management approach (Franco et al., 2024). This approach is called the '*hybrid governance*' structure (Blomqvist et al., 2002). Hybrid governance refers to adopting a strategy that involves both elements of internal and external governance, aiming to optimize efficiency in specific organizational processes (Blomqvist et al., 2002; Parmigiani, 2007). It has the advantage of considering the best of both approaches, as it keeps the strategic digital resources internally but also takes advantage of using strategic partnerships with specialized partners (Franco et al., 2024). For example, in situations where internal hospital teams are unable to develop or operationalize a specific digital technology, the hospital may seek to complement it by acquiring it externally or obtaining knowledge and information from partners but avoiding total dependence on external suppliers (Parmigiani, 2007; Steinhäuser et al., 2020). Through hybrid governance, hospitals can also adapt to constantly changing conditions, as evidenced by the COVID-19 pandemic and transaction-specific factors such as uncertainty and asset specificity (Blomqvist et al., 2002).

However, although this approach may offer various benefits, such as increased flexibility and exploration of external incentives, it also presents some complexities compared to the other two approaches (Franco et al., 2024). First, the cost of maintaining two different forms of digital resources can become higher for the organization because unexpected expenses and silent risks may arise due to the hospital's ability to deal, for example, with contracts and bidding processes (Parmigiani, 2007). Second, there is the complexity of its organization, dealing with ambivalences, especially regarding responsibilities when an issue arises concerning the resources and redundancies of information in the DT process (Blomqvist et al.,

2002). Therefore, as demonstrated by Franco et al. (2024), hybrid governance can be a viable structure, but under specific conditions when the complexity of combining internal and external resources will pay off with the value gained through the performance benefits that can be achieved.

2.2.2 Research opportunities on digital resource management for HCDDT – The TOE framework

Digital transformation literature has previously addressed the role of developing specific resources devoted to the digitalization process in companies. From an organizational perspective, acquiring or developing digital resources has been seen as a strategic imperative of digital transformation that is deeply connected with the *firm's organizational* structure and digital strategy (Verhoef et al., 2021). Moreover, digital transformation has been analyzed from the strategic alignment perspective between the necessary resources and the *environmental* demands (Liu et al., 2011). It has been an important factor in defining the *technological* aspects of the readiness for digital transformation (Chwiłkowska-Kubala et al., 2023). Other streams of the literature have recently addressed the challenging issue of how to manage such digital resources, including the internal development of organizational digital resources, the external acquisition through collaboration (Dalenogare et al., 2023), and the development of a hybrid structure (Franco et al., 2024). Franco et al. (2024) were the first to investigate these three options for resource development (internal, external, and hybrid) in digital transformation, considering the supply chain context.

However, when addressing the HCDDT literature, little is known about this issue. As shown in Appendix A, most existing studies on HCDDT focus on the implications of technology implementation in healthcare. In other words, they primarily examine how digital technologies influence and affect the healthcare domain but not what is required to implement these technologies. Few studies have considered a resource-based view of HCDDT (Chaudhuri et al., 2023; Dicuonzo et al., 2023; Mauro et al., 2024) or a relational view of HCDDT (Malik et al., 2023). These studies mainly use such theoretical perspectives to explore the adoption or impact of digital technologies in healthcare systems. However, none specifically address how these two types of resources are implemented or the necessary structures for doing so. Furthermore, the HCDDT literature lacks discussion on hybrid governance models, which have been explored in the supply chain management literature (Franco et al., 2024). Such models could offer

valuable insights, as digital technologies often operate in complex environments that demand higher expertise in technology implementation.

We aim to address this gap regarding the necessary resources for HCDT, first from a holistic perspective that integrates technological aspects with organizational and environmental factors. Additionally, we seek to explore the various approaches to resource development, considering, as Franco et al. (2024) suggest, that resource management may have different approaches for its development—whether through internal, external, or hybrid digital resource management models. Following this reasoning, the Technology-Organization-Environment (TOE) framework (Rocco DePietro et al., 1990) can provide insight into the structure of HCDT in different organizational dimensions (Kinkel et al., 2022; Tabim et al., 2021). The TOE framework explains that adopting and implementing technological innovations is influenced by three elements of the organization's context: technological context, organizational context, and environmental context (Rocco DePietro et al., 1990). The *technological context* includes technologies relevant to an organization's business ecosystem, including current internal practices and equipment within the organization, as well as the set of technologies available externally to the organization. The *organizational context* refers to various descriptive characteristics of how the organization works in terms of design, processes, and responsibilities. Additionally, informal relationships between employees, hierarchy, and the procedures they operate include decision-making and internal communication. Finally, the *environmental context* is where an organization conducts its business – i.e., the industry context, competitors, and its relationship with other institutions, such as customers, suppliers, and third parties (Marcon et al., 2025).

Integrating the different factors of the TOE framework allows for understanding the broader scenario in which the healthcare organization operates and the influence of these factors on the adoption of new technologies, as well as in supporting the management of digital resources to deal with HCDT (Rocco DePietro et al., 1990). We use the TOE dimensions to define the types of digital resources to be managed according to the analytical framework presented in Figure 3. The analytical framework is based on three digital resource management strategies described in Section 2.1, which are related to the three dimensions of the TOE framework. We aim to understand how these strategies are organized and operate within the TOE dimensions for HCDT. In this sense, the conceptual framework guides us to discover if and how the elements of RBV, RV, and hybrid governance are present, supporting managing digital resources for HCDT.

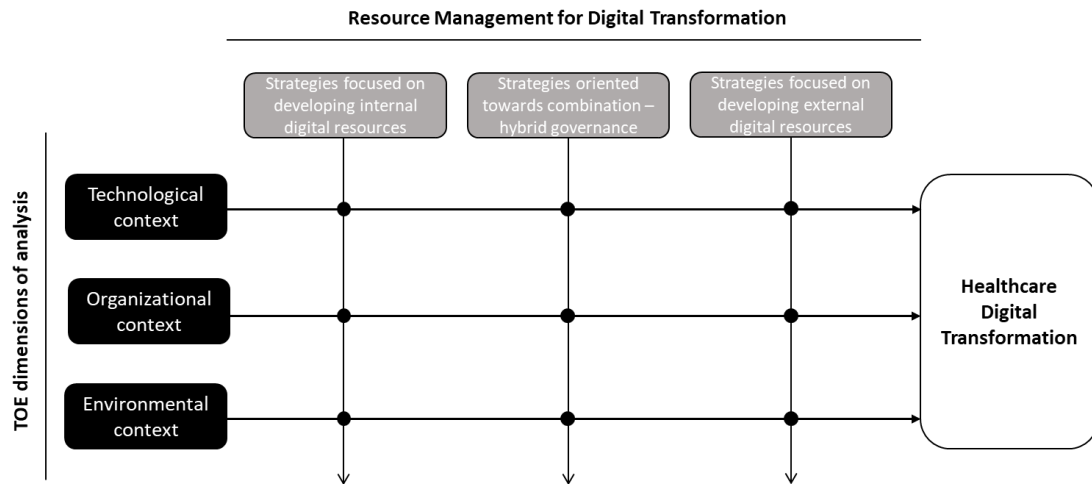


Figure 3. Analytical framework of digital resource management strategies for HCDT

2.3 RESEARCH METHOD

The literature on HCDT has predominantly focused on the application and impact of digital technologies, yet limited attention has been given to how digital resources are managed throughout the transformation process. This study seeks to deepen the understanding of how hospitals develop and structure these resources, offering insights into the “how” of digital resource configuration. Given this objective, a qualitative research approach is particularly appropriate, as it allows for in-depth exploration of complex processes and serves as a starting point for advancing this research agenda. However, such an approach entails limitations in terms of generalizability, which should be addressed in future studies using complementary research methods for validation (Voss et al., 2002).

In this context, we adopted a multiple-case study design, guided by the analytical framework presented in Figure 3. This method is especially effective for investigating contemporary and emerging phenomena within real-world settings (Barratt et al., 2011). Following the recommendations of Ketokivi and Choi (2014), we employed a priori theorization to structure our research framework. While the findings derived from this study are not statistically generalizable – as would be possible through quantitative methods – the case study approach offers a valuable foundation for generating novel insights and developing theoretical contributions (Yin, 2012). Accordingly, our investigation enables a nuanced analysis of distinct strategies for managing digital resources in HCDT, thereby contributing to the advancement of theory in this field. The operationalization of our qualitative research followed

the methodological guidelines proposed by Voss et al. (2002), as detailed in the subsequent subsections.

2.3.1 Selection criteria for the cases and the participants

A selection criteria was established to ensure that the cases selected were relevant to the study's purpose. First, the hospitals had to be from the same region, as we wanted to control the socio-economic context. Different regions or nationalities can present significant differences in the management practices adopted between organizations, especially in the healthcare field (Kaplan et al., 2010). Second, hospitals must have reasonable experience implementing HCDT (Tortorella et al., 2020) and integrating technological resources into their administrative processes and patient healthcare. As a third and final criterion, the study included organizations with different management structures: public, private, and research hospitals. This diversity allows for identifying the scope of the effects of HCDT on the operational performance and flexibility of healthcare organizations with different organizational structures and capital expenditure capacities.

The following criteria were established to select the interviewees representing each case. First, they should work as hospital managers, assuming a leadership role in the organization. This would ensure they have a broader view of the hospital and its processes, legitimizing their perceptions (Tortorella et al., 2022). According to Huy (2001), leaders are crucial in operationalizing radical changes, such as HCDT. Second, the interviewers should have a minimum experience time, which we set as at least five years of work experience. The interviewees' experience is crucial, as it provides practical knowledge and a deep understanding of challenges, patterns, and nuances in the operational environment. Finally, the interviewees should be involved in the HCDT initiatives.

As a result, we selected three large-sized Brazilian hospitals (i.e., with more than 150 inpatient beds – Brazilian Ministry of Health) that have been operating for over forty years: one public (Hospital A), one private (Hospital B) and one research (Hospital C), summarized in Table 2. We selected for the interviews a variety of representatives as described in Table 2. We selected representatives leading the HCDT initiatives, which resulted in variations in the number of interviewees per hospital. Hospital A had only two leaders for this initiative—the IT manager and the innovation coordinator—who provided all the necessary information. In contrast, Hospital C has three leaders overseeing the HCDT initiative: the HCDT coordinator,

who works closely with the IT coordinator and the Data Science coordinator. Additionally, during on-site visits, two other colleagues—one from quality management and a division chief—became involved after the formal interviews. Their input helped us better understand the environment, but they are not cited here as they were not the primary HCDDT representatives interviewed during the formal data collection process. Hospital B, on the other hand, was represented by two technology sector leaders and one business coordinator. In this case, the hospital includes an administration representative in its HCDDT initiative leadership. These differences in the number of interviewees reflect the distinct ways in which each institution structures its HCDDT activities. Table 2 also provides further details, such as the years of the institution, the number of beds that represent its operating capacity, and details of the interviewees.

Table 2. Hospitals' characteristics and interviewees' profile

Hospital	Ownership	Hospital age (years)	Number of beds	Interviewee	Position	Experience
A	Public	63	784	1A	IT Manager	40 years
				2A	Innovation Coordinator	11 years
B	Private	45	312	1B	IT Coordinator	8 years
				2B	Systems and data Coordinator	8 years
				3B	Business Coordinator	7 years
C	Research	53	836	1C	HCDDT Coordinator	6 years
				2C	IT Coordinator	14 years
				3C	Data science Coordinator	13 years

Note: IT = Information Technology.

2.3.2 Data collection procedures

Given the nature of multiple case studies like this research, Galletta (2013) suggests using semi-structured interview techniques as a more appropriate approach because the researcher can get closer to the interviewees, allowing for the identification and detailed capture of their behavioral aspects. Online semi-structured interviews were conducted with the managers of the hospitals (Table 2) using a video conference platform. The data were stored from audio and image recordings and later transcribed, providing greater security in storing interview information. It is important to note that prior contacts and scheduling were made with the interviewees to ensure data quality and participant comfort.

The interviews were conducted between October and November 2023, and coding, cross-interview analysis, and report verification were completed during the second half of November 2023. In this sense, the dimensions to develop HCDT (Figure 3) were presented at the beginning of each interview to establish a common understanding of the research. In addition, the anonymity of each interviewee was guaranteed to have reliable responses. The interviews were conducted following a semi-structured interview guideline (Appendix B). The interviews lasted between 60 and 80 minutes and were conducted by at least two researchers. According to Dubé & Paré (2003), the presence of more than one researcher increases the ability to process contextual information with a higher level of confidence. The interviews were recorded and transcribed for analysis and, besides, interviews' notes were taken by the researchers. Finally, at the end of the interviews, a summary of the answers provided by the interviewees was made to confirm the contributions and eliminate scattered comments that were out of the topic.

Furthermore, we combined the information obtained from the interviews with other data sources. Voss et al. (2002) and Yin (2012) recommend data triangulation from multiple sources of evidence, as it allows different data collection methods to contribute to the understanding of a specific phenomenon and strengthen construct validity (Carter et al., 2014). In this sense, our data triangulation approach considered three other different sources as a complement to individual interviews. First, we mapped hospitals' resources and relationships with their partners. Second, we analyzed documents and information available on the websites of the hospitals in this study to identify historical, technological, and organizational aspects, as well as the evolution of HCDT. Third, on-site technical visits were conducted at the locations, allowing for the observation of operations and services. We describe in Table 3 the different sources of data used for the three cases with additional details of the data collection process.

Table 3. Sources of primary and secondary data used

Hospital	Primary data			Secondary data
	Interviews	Transcription	Participant observation	Files
Hospital A	1A (58 minutes)	26 pages	Observation of administrative and care processes (1 visits – 5 hours)	Internal Bidding and Contracts Regulation (65pages)
	2A (56 minutes)	28 pages		Strategic Information Technology Plan & Information Technology Master Plan (46pages)
				Strategic and Communication Planning (25pages)
				Visits to the hospital websites
Hospital B	1B (1:12 hours)	31 pages	Observation of administrative and care processes (2 visits – 4 hours)	Hospital partners and outsourced solutions folder (2pages)
	2B (1:03 hours)	30 pages		Review of the hospital website
	3B (53 minutes)	23 pages		
Hospital C	1C (1:14 hours)	33 pages	Observation of administrative and care processes (3 visits – 6 hours)	Information and Communication Technology Strategic Plan (37pages)
	2C (1:05 hours)	31 pages		Information and Communication Technology Master Plan (91pages)
	3C (1:10 hours)	30 pages		Integrated Management report (218pages)
				Review of the hospital website

2.3.3 Data Analysis – validity, reliability and interpretation

For the qualitative data analysis, we employed an interactive, multi-stage process (Gioia et al., 2013; Merriam & Tisdell, 2015). The primary goal was to answer the research question by identifying relevant categories or emerging themes (Merriam & Tisdell, 2015). Accordingly, based on transcriptions of audio-recorded interviews, we analyzed the data following the steps of the Gioia methodology (Gioia et al., 2013). This methodology offers a systematic approach to theory development, ensuring qualitative rigor in the execution and presentation of case studies (Gioia et al., 2013). Previous studies have used the Gioia methodology to strengthen the robustness of qualitative research analyses (Schiavone et al., 2021; Zhang et al., 2023).

In the first step of the Gioia methodology, we organized and categorized raw data into first-order empirical concepts. Initially, we explored a wide range of terms from informants (e.g., *"The IT Director of Hospital C and the Hospital Management System Coordinator of Hospital C are the same person"*). During this stage, we aimed to adhere faithfully to the language used by the informants while making only minimal attempts to extract categories to prevent an unmanageable proliferation of categories early in the analysis. As the process progressed, we identified similarities and differences among these categories, allowing us to

refine and consolidate them into a more manageable structure. These categories were then labeled with concise descriptions (e.g., *"Public and private actors play active and distinct roles in digital transformation"*).

The second step involved identifying second-order themes. At this stage, we systematically analyzed the relationships between first-order empirical concepts, using axial coding (Corbin & Strauss, 2014) to group and consolidate them into broader second-order themes (Gioia et al., 2013). For example, from the first-order empirical concepts *"The hospital's business model impacts the definition of the digital transformation strategy"* and *"Each hospital has distinct models that influence different paths for HCDDT,"* we developed the second-order theme "Business model."

Once we identified a solid set of themes and concepts, achieving theoretical saturation, we proceeded to the third step, organizing second-order themes into aggregated or theoretical dimensions. At this stage, we aligned the emerging themes with the dimensions of the theoretical framework presented in Figure 3: (1) Technology, (2) Organization, and (3) Environment—ensuring consistency with the research objectives. Figure 4 provides a graphical representation of our data coding and structuring process, illustrating examples of first-order empirical concepts, second-order themes, and theoretical dimensions.

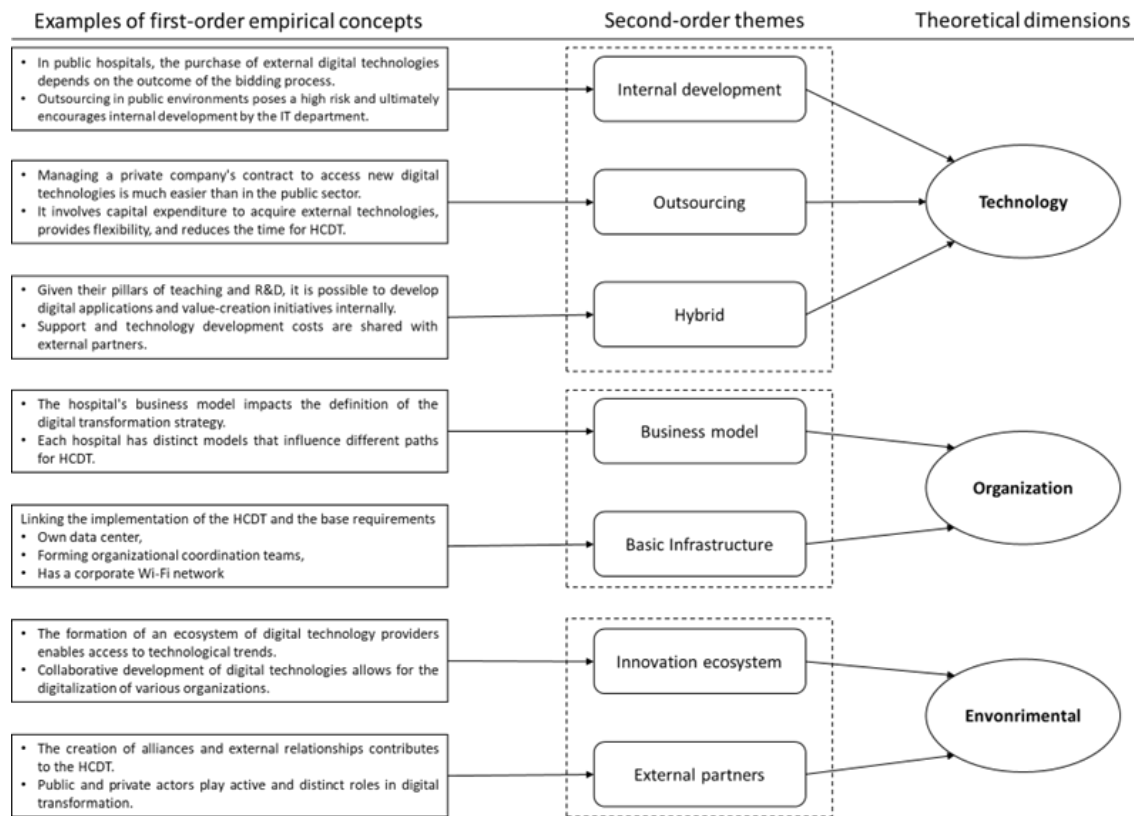


Figure 4. Data structure of the qualitative analysis

For validation, we cross-referenced data across the three theoretical dimensions for each type of management strategy, analyzing the most frequently occurring second-order themes and identifying the most relevant ones. To ensure coding consistency and establish inter-rater reliability, a second member of the research team independently repeated the coding process for both the first-order empirical concepts and the second-order themes (Gioia et al., 2013). This approach reinforced the reliability of our analysis of the semi-structured interviews, as all participants responded to the same structured questions in the same order. Furthermore, the independent analysis conducted by the second researcher resulted in a high level of agreement on the key themes, minimizing biases and enhancing the rigor of the qualitative analysis process. Following Gioia et al. (2013), we also made a conscious effort at this stage to avoid an in-depth review of the literature. Engaging too early with existing research could introduce confirmation bias and restrict our perspective. Finally, the use of multiple data collection methods further strengthened the reliability and validity of our study.

For the presentation, interpretation, and discussion of the cases (Goffin et al., 2019), we used the three dimensions of TOE as guidelines for our narratives in each case study. For each

dimension of TOE, we present a narrative that interrelates its highlighted elements and those considered relevant for data analysis. We focus on interpreting these dimensions using the perspectives of RBV, RV, and the hybrid governance between them. We discuss how the elements of these theoretical lenses provide support and are present in managing digital resources in TDHC. The narratives mentioned before were reviewed to establish data documentation, considering all data sources, reflections, ideas, and insights. Finally, as recommended by Ketokivi and Choi (2014), the authors considered the idiosyncrasy and transparency of reasoning to conduct multiple case studies rigorously. Figures 5, 6, and 7 synthesize the data analysis conducted, including the narrative of the case studies, designed through a final framework.

2.3.4 External validation

Since the exploratory case study investigates three hospital types within a single region to control socio-economic context, we conducted an external validation of our findings after the initial analysis. To this end, we used secondary data from business reports focused on healthcare and digital transformation. Specifically, we examined reports published by globally recognized consulting firms and institutions. These reports included case studies of digital transformation initiatives in public, private, and research/teaching hospitals, which we compared with our own findings. While this comparison is inherently subjective and has its limitations, it provided insights into the extent to which our results might be generalizable beyond the original context. We discuss the generalizability of our findings in Section 4.5 and address the study's limitations in the Conclusion (Section 5.3).

2.4 RESULTS

2.4.1 Case study 1: Public hospital (hospital A)

Hospital A is the largest unit of a group of public hospitals in the southern region of Brazil, providing high-complexity care in its outpatient clinic, emergency department, and inpatient services through the Unified Health System (Sistema Único de Saúde - SUS). This hospital offers 784 beds, representing about 55% of the group's total capacity, and manages approximately 25,000 patients per year. Recently, the hospital invested at least \$3 million in expanding its Intensive Care Unit (ICU) sector, aiming to further enhance its facilities by

increasing the number of beds from 40 to 59, thus becoming one of the largest ICUs of the Unified Health System in Brazil. Additionally, it features state-of-the-art digital equipment and a monitoring center that enables remote patient monitoring. We summarize in Figure 5 the findings regarding the different dimensions of TOE, which will be explained next.

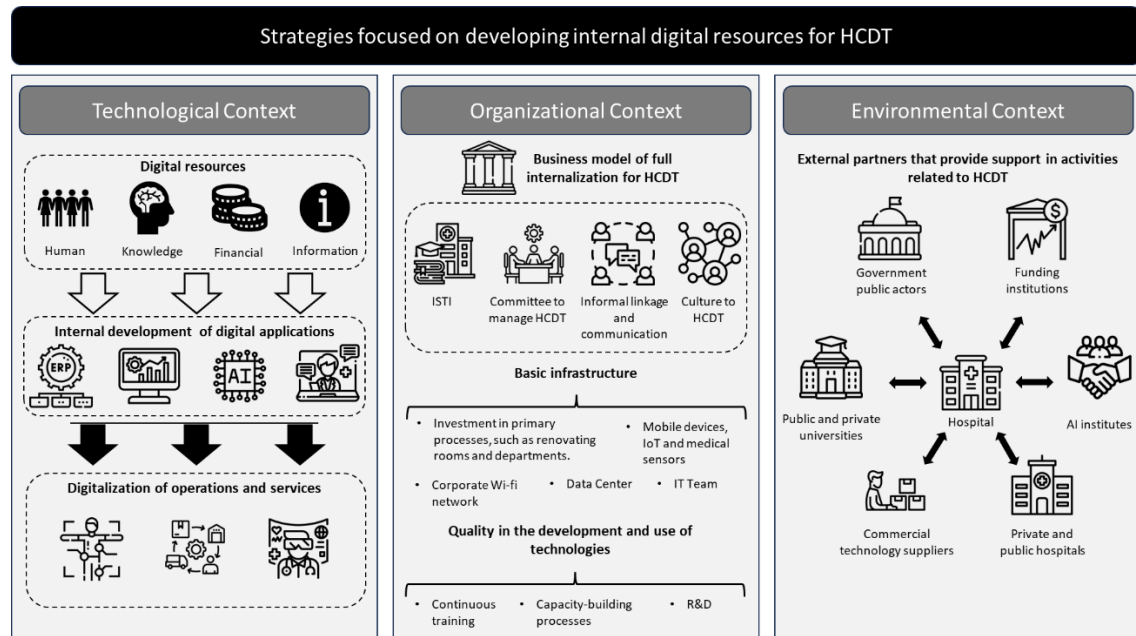


Figure 5. Configuration of internal management of digital resources for HCDT within the public health system

2.4.1.1 Technological context

Regarding the technological context (Figure 5), if Hospital A wishes to implement a digital technology resource available in the market, it must issue public notice and conduct a bidding process, seeking the most suitable market price. Since this makes the digitalization process very slow compared to technological evolution and market dynamics, the management of Hospital A decided to internalize the technological development, maintaining absolute control over the development of digital technology resources through its IT team.

When a technological project is approved, and the hospital has the necessary resources available, whether in terms of knowledge, information, human, technological, or financial resources, the IT team takes responsibility for the operationalization of the project, using their expertise and skills to develop and implement the required technologies. The IT team develops corporate management systems for department integration (e.g. Enterprise Resources Planning - ERP), electronic medical records, data mining systems (used, for instance, in cancer care),

KPI monitoring, mobile applications, telemedicine solutions and prediction and forecasting systems based on artificial intelligence.

The strategy of developing technological resources internally was also motivated by high costs, bidding process and time. As a public entity, most of the Hospital A's revenue is funded by government institutions, resulting in limited capital expenditure. This restricts Hospital A from making significant investments in digital transformation. Outsourcing to external suppliers can involve high costs, as it not only requires investments in the digital resources themselves but also in associated support services. Consequently, the hospital's budget could be affected, as unexpected expenses may arise from digitalization, such as software updates or equipment maintenance, compromising other priority expenditures necessary for operations. Furthermore, if support services are not provided, digital resources may become outdated or lack necessary maintenance, negatively impacting the hospital's efficiency. By acquiring a digital resource externally rather than developing it internally, Hospital A will depend on the bidding process to form contracts with suppliers. In this context, contract management can be challenging and bureaucratic. Hospital A must prove and ensure the transparency of the invested funds' destination to various governmental and public instances. In some situations, suppliers may not fully meet the established requirements in the contract, causing delays in delivering the digital resource or service. As a result, Hospital A may become dependent on the supplier, without the possibility of terminating the contract, since the financial resources have already been transferred and processed through transparency channels. Another limiting factor is the response time to meet internal demands of improving their operation. This is because Hospital A needs to conduct a thorough technical analysis of the environment that will receive the technology, considering requirements, adaptations, infrastructure, and presidential approval, among other factors that are to provide transparency of the public funds. This, coupled with the procedures of bidding processes (in the case of outsourcing), can take a long time, delaying improvements and even compromising patients' lives.

In sum, this hospital prioritizes internal resource development, emphasizing self-sufficiency due to the high transactional costs of dealing with technology providers. The internal team has a detailed understanding of the integration among processes, people, and technologies, facilitating the customization of digital transformation in the hospital. Manager 1A of Public Hospital A, for example, comments on the adoption of the strategy to customize their own management system that integrates all information and data. According to him, “*the*

volatility of software companies and the processes involved, especially for public environments, as well as the exchange and/or the creation of new companies, can affect the service provided by them, leading to delays or contract breaches.” Additionally, he also explained: “The response time for customization demands is much faster than dealing, for example, with the contracting of outsourced software, which is prevalent in private hospitals, managing their healthcare services”.

2.4.1.2 Organizational context

In relation to the organizational context of managing internal digital resources for HCDDT (Figure 5), Hospital A adopts the business model of full internalization, which is characterized by complete centralization of the development and implementation organizational structure. In terms of governance structure, Hospital A consists of various departments, including education and research, and information technology. Each of these departments reports to specific directorates, which, in turn, are under the oversight of the executive board. During the interview with the IT coordinator, it was emphasized that Hospital A started by establishing an internal culture focused on DT, with awareness programs for professionals and top management, aiming to use internal digital resources to develop HCDDT and address operational issues. For this purpose, a specific committee was formed to manage HCDDT, aligned with the hospital's strategies. Additionally, the executive leadership has established a Technology Incorporation Core, which can decide whether a technology project should or should not be integrated into the hospital. The HCDDT process enables professionals, departments, and management at Hospital A to establish informal linkage and communication through suggestions, visits, and meetings with teams, such as doctors, nurses, administrative staff, and supervisors, to identify issues and opportunities related to the development and implementation of digital technologies. According to Interviewee 1A and reinforced by Interviewee 2A, both these informal linkages and the strategies established in the hospital's strategic planning are two of the main sources of origin for innovation in operations.

In terms of basic infrastructure resources for HCDDT, initially, Hospital A decided to make a more focused investment in primary processes, such as renovating rooms and departments and implementing internal corporate Wi-Fi network access points that enable the connectivity of mobile devices, IoT and medical sensors, equipment, and systems to integrate processes and patients in real-time. According to the IT manager, this is one of the fundamental first steps to initiate DT, as it serves as the foundation for data collection and monitoring, as well as enabling interoperability within the hospital. All collected data (e.g. electronic medical

records, patient treatments, the number of consultations) is stored in a *Data Center* and will be used in future technological applications. Digital resources are developed by the internal IT team, consisting of 34 professionals, with 12 dedicated to development and 22 responsible for Level I support, basic support, and microinformatics support, all with higher education. These professionals possess specialized knowledge and are aligned with the hospital's organizational culture, allowing them to drive the HCDDT in operations through the valuable and rare digital resources developed.

Finally, to maintain the quality in the development and use of technologies, hospital employees undergo continuous training and capacity-building processes. Specific courses are offered by teaching and research management, allowing employees to learn about the processes and use of technologies. This practice is essential to ensure that the team is up-to-date and possesses the necessary skills to handle evolving technologies. Additionally, the hospital emphasizes the importance of intensity in research and development (R&D) with universities through applied projects.

2.4.1.3 Environmental context

In relation to the environmental context of managing internal digital resources for HCDDT (Figure 5), although Hospital A initially established a culture of complete internalization of digital resources, the hospital also relies on external partners that provide support in activities related to HCDDT. According to interviewees A1 and A2, there are six main types of partners that support Hospital A in managing digital resources internally for HCDDT.

The government provides necessary financial resources for the hospital's operation, maintenance, and technological investments. Additionally, they are related to compliance with regulations and guidelines imposed by the Unified Health System (SUS). Hospital A also establishes partnerships with funding institutions through participation in calls for financial resources to support the development of open-source software. Universities are essential partners, establishing cooperation agreements with the hospital for the development of non-profit research and development projects. These projects aim to advance the development of technological resources, using data and information from the hospital as a foundation, as well as gaining access to complementary information and knowledge. There are agreements established with AI and Analytics Institutes, where cooperation agreements are signed between the entities for project development. These agreements aim to leverage the expertise of these institutes in the use of AI and data analysis to drive the hospital's technological advancement.

Finally, commercial technology suppliers are included as the hospital establishes collaborative alliances through bidding processes with specialized technology companies or solution providers, although this is specialized for only specific cases.

2.4.2 Case study 2: Private hospital (hospital B)

Hospital B is a private institution that offers comprehensive healthcare solutions in various areas, including diagnosis, treatment, and monitoring, with a focus on providing humanized, safe, and patient-centered care. It is part of a group of healthcare business units, serving as the main business of the group. It comprises four large units and three major colleges, employing over 4,200 staff members. The hospital provides 312 active inpatient beds. It also houses over 2,000 advanced technology equipment and has a team of professionals from various fields, including more than 2,300 accredited physicians. This hospital stands out as an organization with multiple accreditations and awards related to healthcare digital transformation and was recognized as one of the top ten hospitals in Brazil in 2024 and holds the Gold Seal of Accreditation from the American College of Radiology (ACR). Figure 6 summarizes the TOE structure of this case which will be described next.

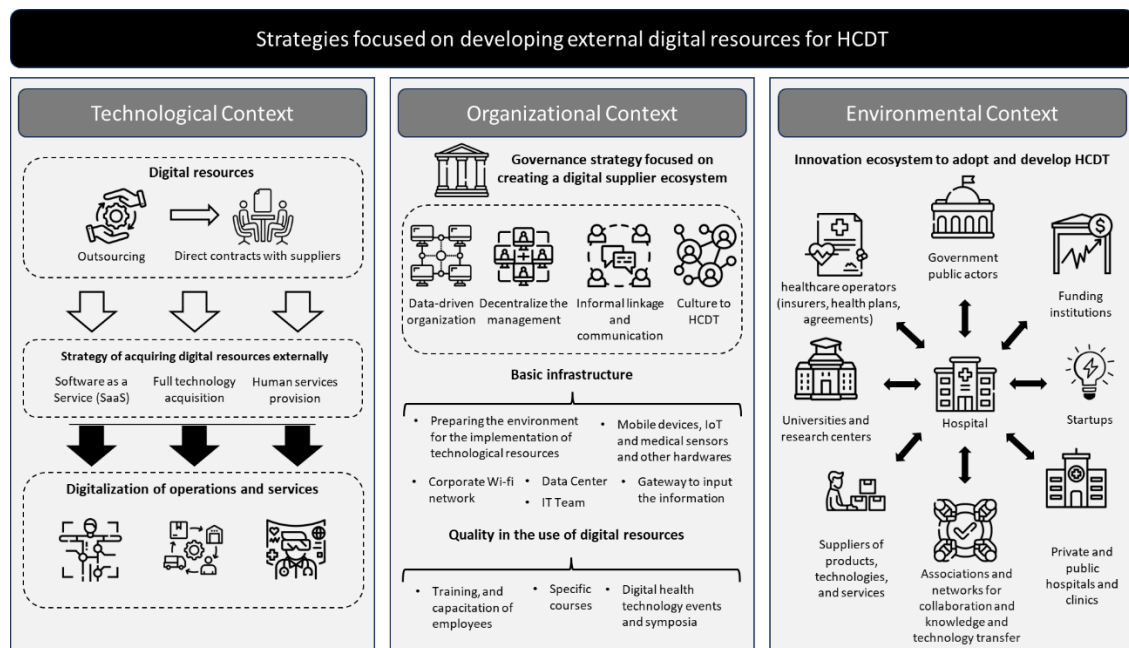


Figure 6. Configuration of external management of digital resources for HCDT within the private health system

2.4.2.1. Technological context

In relation to the technological context for HCDT (Figure 6), Hospital B has adopted the strategy of acquiring digital resources externally instead of developing them in house, establishing strong outsourcing partnerships with commercial suppliers in the sector. This strategy has allowed the hospital to implement cutting-edge digital technologies, adopt innovative management practices, and deliver services, thus creating a healthcare innovation ecosystem.

Interviewees 1B and 2B explained that the culture of the relationship with external partners enables the hospital to access technological resources in a practical and fast manner. According to the business coordinator (Interviewee 3B), Hospital B manages external digital resources through direct contracts with suppliers involving three modalities: Software as a Service (SaaS), full technology acquisition, and human services provision. This approach combines different components to drive Hospital B's DT, enabling better integration of technology and services to achieve its goals. Some of the technologies provided by the ecosystem are IoT and biomedical sensors for connectivity, wearable devices, cloud-based applications and systems, collaborative robots, big data and analytics, artificial intelligence algorithms, enterprise resource planning (ERP), patient management systems, telemedicine, and augmented and virtual reality. Additionally, the hospital utilizes technologies for process simulation, 3D printing, and room service systems, among others. According to Interviewee 1B, *"Together, they [these technologies] allow for vertical and horizontal integration of the hospital, providing visibility into all operations and automation of repetitive human and manual activities"*.

According to the IT Manager for digital transformation, the strategy of managing external digital resources to drive HCDT was motivated by four main aspects: availability and flexibility (changes and supplier replacement at any time) of technological options in the market to reduce potential risks; from an economic standpoint, it is more viable for the hospital to outsource rather than develop and manage internally, because Hospital B does not have costs of suit as high as in the public case; availability of financial resources for outsourcing; and streamlined bureaucratic processes for hiring, reducing the time required to implement new technologies.

2.4.2.2. Organizational context

Regarding the organizational context (Figure 6), over more than ten years, Hospital B has become a data-driven organization, focused on collecting data from both managerial and clinical aspects. Therefore, the strategies and investments for HCDDT are defined considering this scenario. Hospital B has a high level of maturity in terms of visibility and transparency of its operations, as emphasized by the IT Manager: *“Being a data-driven organization, we are aware of the main pain points and bottlenecks in processes and we know how DT can leverage improvements, whether in administrative, financial, accounting, procurement, pharmacy, surgical block, intensive care unit (ICU), hospitalization unit, among others”*. This allows top management to communicate with operational teams, understand their needs, and integrate information to support the decision-making process in technology-related investments.

In terms of organizational management, Hospital B is comprised of directorates that collaborate to oversee and implement digital transformation strategies within the hospital. According to the Business Coordinator (Interviewee 3B), the directorates have chosen to decentralize the management of some of their operations to third parties, such as equipment, systems, and technology maintenance, allowing the institution to focus on the creation of solutions and innovation. This results in cost reduction and a decreased workload, as well as production optimization.

When a digital resource project is implemented by an ecosystem partner, the IT team takes responsibility for monitoring and overseeing its functionality for six months. The IT team receives instructions and utilizes their knowledge and skills to ensure that the digital resources function efficiently and meet the hospital's internal needs. The team is composed of 50 specialized professionals in DT and is aligned with the organizational culture of staying up to date with market technological trends to drive digitalization in the hospital. This team strongly focuses on orchestrating the ecosystem based on the hospital's power structure to purchase advanced technology without major acquisition processes. In other words, they have financial resources and freedom to ensure operations are based on the best technology available in the market. The hospital has a service request to acquire technologies with well-defined rules. In this system, professionals can file complaints, requests, or suggestions for solutions to the problems they or their departments face. This process creates a record that the IT team will analyze, potentially generating knowledge and leading to technological innovations within the hospital. Thus, technologies are defined from a bottom-up approach based on the operational needs of the professionals.

In terms of basic infrastructure resources for technological innovation, the three interviewees highlighted that Hospital B prepares the environment for the implementation of technological resources, encompassing everything from civil construction to the installation of support equipment. According to Interviewer 1B, *“You can't make any modifications without a good infrastructure. This was a problem we had previously at the hospital, hindering the implementation of technologies”*. Like the public hospital, Hospital B also has an internal corporate network that enables connectivity for mobile devices (e.g. smartwatches, smartphones, tablets), IoT sensors, hardware, and medical equipment, integrating all departments in real-time. Interviewee 2B also emphasized that this is one of the fundamental elements to starting DT in a hospital as it serves as a foundation for data collection and transmission, enabling digitized processes and interoperability throughout the institution. He also pointed out that, for the implementation of digital technological resources, it is necessary to have a gateway to input the information, allowing partners to quickly connect their solutions.

Another important factor highlighted to increase the hospital's digitalization is the continuous investment in the adequacy, training, and capacitation of employees and monitoring of new technologies. Specific courses are conducted for employees to learn about the use of emerging technologies and services to develop new skills. Furthermore, participating in digital health technology events and symposia allows professionals from Hospital B to conduct benchmarking and exchange experiences with other hospitals regarding HCDDT.

2.4.2.3. Environmental context

In relation to the environmental context of managing external digital resources for HCDDT (Figure 6), Hospital B has established a culture of relationship with external partners to manage digital resources. In this way, it has created an ecosystem to adopt and develop HCDDT. This approach implies that the hospital had to create an orchestration capability to coordinate the ecosystem of partners. The ecosystem is composed of eight different types of external partners that develop innovative solutions for the healthcare sector.

Hospital B collaborates with public actors (e.g., government, regulatory or oversight agencies, such as the Health National Agency - ONS) to comply with norms and guidelines established for HCDDT. The hospital identifies and selects suppliers that align with its institutional guidelines and can contribute to the excellence of the provided services, establishing collaborative relationships through contracts. This enables them to obtain relevant certifications like the *Joint Commission International*. These suppliers are both national and

international, with some being located close to their facilities. Hospital B also develops and implements technological projects in its operations in partnership with startups. This allows to test proof of concepts within the hospital. Other hospitals and partner clinics are also engaged in such an ecosystem to provide knowledge on how to use new technologies. This is coordinated through the engagement with the National Association of Private Hospitals in Brazil (Associação Nacional de Hospitais Privados - ANAHP). Furthermore, insurers, health plans, and covenants, are also partners of the ecosystem, and they provide data and information to improve the joint digital services. The hospital also focuses on the innovativeness of the healthcare system by developing collaboration with universities and research centers. From a geographical perspective, Interviewee 1B emphasized that the proximity to the mentioned partners favors communication and the relationship between the parties. In this sense, the location of this hospital in a healthcare industry cluster is a keystone of this business model.

2.4.3 Case study 3: Research-oriented hospital (Hospital C)

Hospital C is a public research hospital focused on high complexity healthcare services in a wide range of specialties through the Unified Health System (SUS). It is academically affiliated to the largest public university of the region. The hospital offers approximately 836 inpatient beds and has 6.733 employees. In the last two years, the hospital has conducted over 60,279 hospitalizations and 883.507 consultations. The research conducted by this hospital, in turn, generates new knowledge, techniques, and technological advancements. During the Covid-19 pandemic, this hospital innovated by offering new technological services, such as telemedicine and telehealth, as well as distance health education for remote communities. Additionally, technologies were implemented for the protection of patients and staff, management of patients with respiratory problems, and team training as contingency measures to mitigate the effects of the pandemic. Figure 7 summarizes the TOE structure of this hospital, which will be described next.

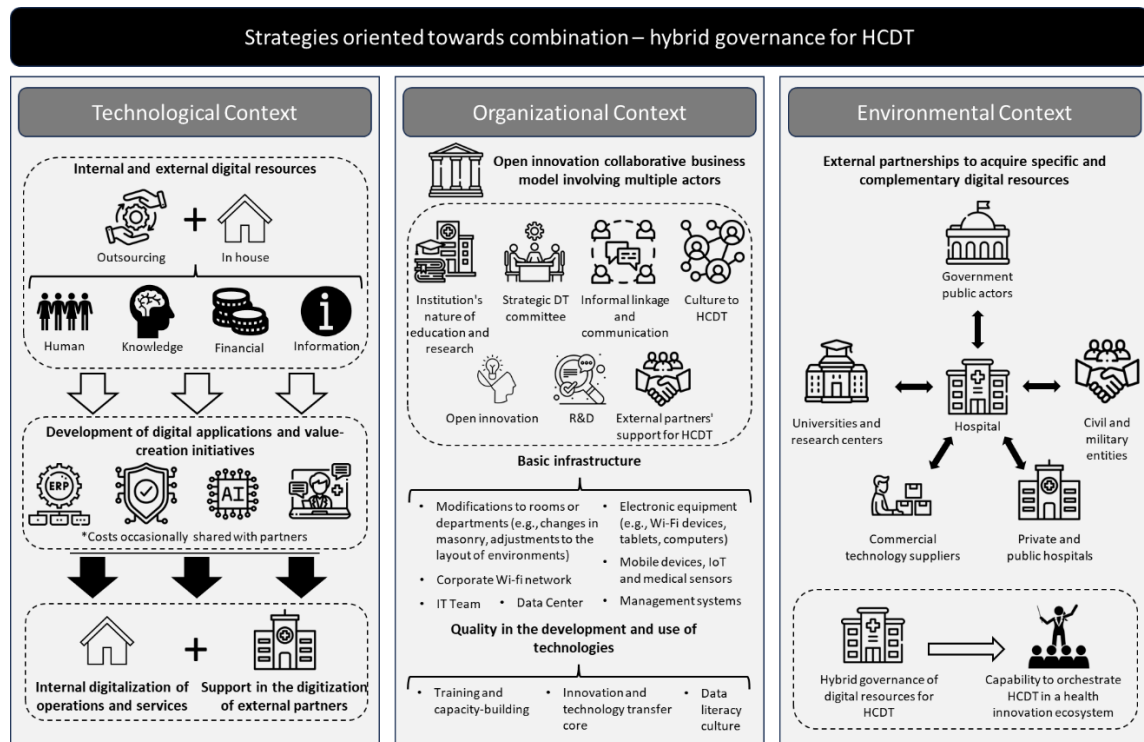


Figure 7. Analytical framework for digital transformation in the education and research healthcare system

2.4.3.1. Technological context

In relation to the technological context for HCDDT (Figure 7), Hospital C has established a hybrid governance strategy to manage digital resources, combining the internal development of digital resources with the establishment of collaborative alliances to acquire external digital resources. Internally, Interviewee 1C explained that Hospital C uses its own available resources, such as human, knowledge, and information, to develop digital technologies for HCDDT. For instance, the internal team develops IoT connectivity for equipment and patient monitoring, cloud-based computing applications, corporate management systems for department integration (e.g., ERP), electronic health records, e-prescriptions, telemedicine, and other solutions based on big data analytics, and AI-based prediction solutions. A *data science center* is also under development to store data and expand the hospital's DT capabilities.

However, according to Interviewer 2C, “*In some situations, there are limitations to internally develop technological projects for HCDDT, often requiring external digital resources that are scarce for the hospital or that it lacks the conditions to develop*”. To overcome these limitations, Interviewees 1C, 2C, and 3C explained that the institution maintains relationships with external partners to support and accelerate HCDDT at the hospital. This culture of partnerships enables Hospital C to access complementary digital resources and capabilities (e.g.

financial resources, costs shared, human resources, new skills, hardwares and softwares), whether for internal development or complete acquisition of the digital resources.

According to all interviewees from Hospital C, the strategy of managing digital resources through hybrid governance for HCDDT is primarily motivated by three reasons. First, as a research hospital, its essence lies in using teaching and research to develop technological projects. Second, there is a large IT team available with expertise in the subject and flexibility for updates. Third, the hospital aims to overcome the limitations and challenges of internal digital resources by leveraging the support of external digital resources through collaborative alliances. This is because, being a public entity, it shares the same configurations as a public organization; for example, outsourcing alone may involve high costs and bureaucratic contract management (see section 4.1.1). External partnerships are necessary to *complement* the internal resources and capabilities of the hospital, but not to substitute them. In addition to these three reasons, Interviewee 1C highlighted that Hospital C, being a teaching and research hospital, was motivated by two additional objectives: the replication of the technological resources developed and the organizational management model for HCDDT in other hospitals (public or private), and the acquisition of complementary resources to meet the requirements for HIMSS certification (Healthcare Information and Management Systems Society). For this, the hospital hired specialized consulting services to provide support in the development of the necessary actions.

2.4.3.2. Organizational context

Regarding the organizational context of hybrid governance of digital resources for HCDDT (Figure 7), Hospital C has adopted an open innovation model, involving multiple actors. This model is particularly applied due to the institution's nature of education and research, which receives resources from partners to drive R&D activities. Additionally, it plays a transformative role, supporting the technological development of other healthcare entities. Therefore, governance is focused on both internal and external resources. Strategies for HCDDT are defined considering this scenario, with the aim of fostering joint efforts for technological development and innovation both within and outside the organization. This approach has enabled Hospital C to stand out at the national level, being recognized as a benchmark for more than 84 hospitals and healthcare institutions across the country.

The internal IT team is a large group with over 90 skilled professionals, including statisticians and experts in BI and machine learning, responsible for the development,

operationalization, and maintenance of most technological applications used in both internal and external operations. These professionals are aligned with the organizational culture of research and development, enabling them to drive innovation both within and outside the hospital. In terms of organizational management, Interviewee 2C explained that Hospital C has a strategic HCDT committee (comprising representatives from all hospital directorates), responsible for coordinating the entire institution's HCDT strategy. In this committee, projects are defined and prioritized for technology adoption. The needs and demands of different departments are brought to the HCDT committee through formal and informal channels. As a result, problems and opportunities within the sector are meticulously assessed. This approach allows top management to establish informal links and communicate with departments at the operational level, understand their needs, and integrate the information to structure the decision-making process regarding technological innovations.

Regarding the basic infrastructure for technological innovation, the five interviewees emphasized that Hospital C invested in electronic equipment and an internal network that enables connectivity for IoT sensors and mobile devices, as well as management systems that allow real-time data and information transmission. Interviewees also highlighted that modifications were made to rooms or departments for the adoption of digital resources. Moreover, in terms of healthcare service quality and patient safety, Hospital C's employees also undergo training and capacity-building to enhance patient satisfaction and operational performance, making the most of available technologies. As explained by the ICT Manager, *“The hospital C invests in the training and continuous learning of professionals to achieve better results in the development of specialized technologies”*. Moreover, it is supported by an innovation and technology transfer board composed of members from the IT department, aiming to facilitate and enable innovation and technology transfer initiatives to society. This allows for the creation of research projects to develop innovative technological processes and resources. Additionally, the implementation of a data literacy culture within the institution is underway, enabling employees to act as multipliers of the use of these technological resources.

2.4.3.3. Environmental context

From the environmental context point of view (Figure 7), Hospital C establishes external partnerships to acquire specific and complementary digital resources. As observed in Hospitals A and B, these external partnerships have also allowed Hospital C to access knowledge sharing. However, as a research-oriented healthcare institution, its aim with such partnership is also to share good practices and help to build together with the external partners an HCDT vision for

the ecosystem, which includes even the technology transfer of the digital solutions developed internally.

The hospital's environmental context comprises five main actors. First, government actors operate as providers of the financial resources necessary for the investments in technological development. On the other hand, the Hospital C, as an educational and research institution, collaborates with the government in the development of public policies for the expansion of HCDDT in the Brazilian public healthcare system. The second partner are the Universities and research centers, that act as collaborators in the development of R&D projects for HCDDT, sharing such kind of costs through cooperation agreements. Third, public and private hospitals establish collaborative alliances with Hospital C to expand and adopt the technological resources developed by Hospital C in their operations, promoting the sharing of knowledge and information. Fourth, civil and military entities use their knowledge, information, and human resources to update or develop new modules and complementary applications for the hospital management system of Hospital C, especially when the hospital cannot do so on its own or when these entities design technological solutions that can be utilized by Hospital C. This collaboration is made possible through a cooperation agreement, in which Hospital C provides its internal hospital management system for these entities to use in their operations. In return, the military entities commit to supporting the management and continuous improvement of Hospital C's hospital management system. Finally, Hospital C maintains collaborative relationships with specialized technology companies or solution providers, which are contracted through bidding processes for the supply of specialized technological solutions that are beyond the skills of the internal IT team.

Another important element highlighted in the interviews was the capability of Hospital C to orchestrate HCDDT in a healthcare innovation ecosystem. Hospital C acts as an orchestrator of the digital health ecosystem, harmoniously coordinating various resources, stakeholders, and digital technologies to achieve digitization goals and enhance the overall performance of its operations and partners. This role was emphasized by Interviewee 2C, who stated, *"Hospital C has the capacity to orchestrate HCDDT, as it is an educational and research entity. Managing digital resources in collaboration with other stakeholders provides autonomy and leadership for innovation projects"*. He further adds, *"Coordinating technological projects fosters a culture of building alliances with partners, as well as developing high-quality and suitable solutions to meet our needs or certification requirements"*. Compared to Hospital B (Section 4.2), an important difference in such orchestration is that Hospital C aims to create a knowledge

flow among the partners to create new knowledge and technological progress. On the other hand, since Hospital B is a private profit-oriented institution, its goal is different, as it aims to orchestrate the ecosystem to obtain integrated solutions for its operations.

2.4.4 Cross-case comparative summary

We summarize the main insights obtained from the three case studies also in the comparison presented in Table 4. While in Figures 5 to 7 we described the detailed elements identified in the three cases, here we describe the overall patterns that help to compare between the cases. In this sense, our findings show that there is a clear difference in HCDT based on the business model followed by each of these hospitals. The financial structure and the main goal the institution has will shape a very different orientation of the digital resource management approach, as detailed in this table.

Table 4. Summary of the main findings from the three case studies

	Hospital A (Public hospital)	Hospital B (Private hospital)	Hospital C (Research hospital)
Model of the digital resources	Focus on internal digital resources	Focus on external digital resources	Focus on hybrid digital resources
Reasons for the model adopted	Financial resources for HCDT are from public sources	Financial resources for HCDT are from private sources	Financial resources for HCDT are obtained through R&D grants
	Acquiring digital resources demands long bidding processes with high switching costs	Acquiring digital resources can be flexible based on the best option available independently of cost requirements	Acquiring digital resources needs to be complementary to internal resources to justify the innovation projects
	To reduce switching costs and its associated risks, internal development is adopted.	To increase efficiency and use the best competitive option, outsourcing highly specialized and skilled digital resources is adopted	To increase the innovative results of the solutions and create an open innovation environment, a hybrid model is adopted
HCDT Structure	Technology	HCDT is more conservative and driven by organizational needs	HCDT is more aggressive and competitive based on performance
	Organization	HCDT is more innovative and experimental (adopting high risks)	Structure focused on providing through HCDT the most usual healthcare problems of population in general
	Environment	Structure focused on achieving through HCDT innovative solutions that can be shared with other institutions (open innovation)	Internal top-down centralized structure that coordinates departmental needs
		Orchestration of the technology providers to integrate the solutions for the hospital requirements.	Orchestration of a horizontal collaboration of different institutions and stakeholders.

2.4.5 Comparison with secondary data for external validation

To validate our findings beyond the original context, we examined secondary data from KPMG's (2024) business report on healthcare transformation, which presents international case studies on how digital transformation is reshaping hospitals. One such example is Nova Scotia Health in Canada (KPMG, 2024, pp. 26–30), a public initiative aimed at transforming the healthcare system. A notable convergence with our analysis is that their transformation was guided by the principles of "people, process, and technology," which aligns closely with our use of the TOE framework. Moreover, like our public hospital case, Nova Scotia Health implemented a centralized coordination center to internalize data-driven decision-making. This indicates that, even in other countries, public healthcare institutions tend to internalize key digital resources.

The KPMG report also highlights research hospitals, notably the case of the Digital Health Village developed by Helsinki University Hospital in Finland. This initiative established a hybrid ecosystem using a customizable digital platform—an approach that mirrors the configuration we observed in our research/teaching hospital case. According to the report, *“The Digital Health Village was built in such a way that healthcare professionals could take ownership of the design of the digital health pathways relevant to their specialty areas. That meant that the Digital Health Village could be rapidly scaled up as healthcare professionals worked in parallel on their own pathways”* (KPMG, 2024, p. 29). These pathways are intended not to replace, but to enhance, traditional care through digital tools such as guided self-care, remote monitoring, and automated reminders. The project was designed with scalability in mind, allowing internally proven solutions to be extended externally with appropriate support.

With respect to private hospitals, our findings are consistent with trends reported by consultancy sources such as S&S Insider (2025), which emphasize the growing outsourcing of IT and digital activities by private healthcare providers. This outsourcing is increasingly adopted as a strategy to enhance operational efficiency and reduce costs, particularly in markets like the United States, where private healthcare providers dominate (S&S Insider, 2022). In this sense, private hospitals tend to build digital ecosystems as those investigated in our results.

2.5 DISCUSSION

Our results provide a new perspective on the study of digital transformation in healthcare systems. The findings challenge the existing literature, as we demonstrate that there

is no single way to approach the management of HCDT. Prior studies simplify this aspect by considering general elements involved in HCDT management without differentiating between business model profiles (Tortorella et al., 2020; Tortorella et al., 2021; Raimo et al., 2023; Mauro et al., 2024).

In this regard, our results show that the management system can follow different orientations based on RBV, RV, or hybrid governance, depending on the type of hospital, its objectives with digital transformation, and the financial sources supporting technological changes. Prior literature typically considers HCDT either from the perspective of developing competitive internal resources (Chaudhuri et al., 2023; Dicuonzo et al., 2023) or building external relationships (Schiavone et al., 2021; Aranyossy & Halmosi, 2024; Malik et al., 2024). Our findings indicate that these two perspectives are linked to specific contexts: the first (internal resources) is more relevant to public hospital systems, whereas the latter (external resources) is more prominent in private hospital systems. Of course, a private hospital may seek to internalize digital resources for various reasons, just as public hospitals may opt for outsourcing. However, we identified an interrelated mechanism that generates greater advantage when combining public hospitals with internal resources and private hospitals with external resources.

These interrelated mechanisms are summarized in the key drivers of adoption, which we present in Table 4. They are rooted in the reduction of switching costs resulting from public healthcare bidding processes and in the pursuit of efficiency and specialization in private healthcare organizations. In sum, while specific circumstances may justify the opposite choices for public and private hospitals, the most expected approach can be clearly explained by our findings.

Furthermore, hospitals must consider what Ayala et al. (2017) highlighted in another research domain: although organizations may choose to externalize or internalize digital resources, it is always necessary to maintain the alternative as a complementary resource base. When HCDT relies on external digital resources, it is important to establish an internal foundation for long-term sustainability, particularly when digital transformation becomes a central competitive element. In such cases, private hospitals that acquire highly specialized digital technologies must at least internalize a basic level of digital capability (e.g., a small but skilled IT team) to effectively orchestrate the external ecosystem, as seen in our case. Conversely, when digital transformation is driven by internal resources, as in the public hospital

studied, the internal team must establish partnerships for technologies they cannot develop at market speed (e.g., cybersecurity solutions). Thus, our findings highlight the priorities of each model, rather than suggesting that resources are purely internal or external - we focus on the predominant approach in each case.

We also contribute to the literature by identifying a third model of digital resource management in HCDT: hybrid governance, which our findings associate with research hospitals and their focus on R&D and innovation. We are not the first to explore this direction, as Franco et al. (2024) recently demonstrated that hybrid governance is a key strategy for blending digital resources in supply chain management. Their study highlights how hybrid governance helps balance internal and external strengths and weaknesses to achieve a more robust digital resource structure. However, our findings suggest that, in the case of research hospitals, hybrid governance is driven by a different need. We show that these hospitals base much of their development on R&D grants and combine internal development with external partnerships through an open innovation approach, which helps justify funding sources. This governance structure aligns well with the specific business model of research hospitals. While Franco et al. (2024) emphasize hybrid governance as a means of balance, our study highlights its role as a funding-driven strategy, revealing a distinct rationale behind its adoption.

The TOE perspective also helped us identify new elements not explicitly addressed in prior literature. First, we show that the organizational and external environment dimensions differ significantly among the three hospital types, particularly in their ecosystem approaches and organizational structures. While the healthcare literature has discussed the value of ecosystems and network activity in fostering innovation (Secundo et al., 2019; Schiavone et al., 2021; Dal Mas et al., 2023; Aloni et al., 2023), we demonstrate that HCDT ecosystem engagement differs between private and research hospitals. Private hospitals orchestrate technology providers to ensure interoperability and seamless integration into healthcare processes. Here, orchestration involves consolidating and aligning external actors, with the hospital as the central coordinator and focal point of the ecosystem (Schiavone et al., 2021). In contrast, research hospitals, which are innovation-driven, engage in horizontal collaborations with external institutions and stakeholders, treating them as partners rather than suppliers. These two ecosystem models are fundamentally different: the former aims to extract maximum value from external partners, while the latter focuses on co-developing value with external stakeholders. Finally, public hospitals exhibit a more centralized approach, focusing on internal coordination across functional departments. Given their scale and complexity, they resemble

internal ecosystems, managing a diverse array of employees and processes within the organization.

Although the digital technologies adopted by these hospitals are fundamentally similar – consistent with prior studies on HCDD technologies (Tortorella et al., 2020; Tortorella et al., 2022; Raimo et al., 2023; Mauro et al., 2024) – we observed differences in their technological approaches. The public hospital's HCDD is driven by internal organizational needs, with digital transformation being primarily influenced by healthcare professionals' demands. This explains why an internal digital resource structure is often the most appropriate for this model. In contrast, the private hospital adopts a more aggressive approach, with HCDD being propelled by performance and competitiveness objectives. External digital resources are more suitable for this model, as they enable rapid adaptation to the best solutions available in the market. Lastly, the research hospital pursues an innovative technological approach, taking greater risks in experimenting with new technologies. These distinct profiles define different innovation pathways within HCDD.

2.6 CONCLUSION

This study investigated how three distinct hospital models—public, private, and research-oriented—manage digital resources in their HCDD efforts. Rather than treating HCDD as a uniform process, we analyzed how each hospital's business model, financial structure, and institutional logic shape their approach, using the TOE framework and insights from RBV, RV, and hybrid governance theory. Our findings reveal distinct configurations of digital resource governance—internal, external, and hybrid—each aligned with specific organizational contexts. We highlight the importance of ecosystem orchestration, the complementarity between internal and external resources, and the unique role of hybrid governance in research hospitals. By offering a typology of HCDD strategies and explaining their underlying mechanisms, this study refines the conceptual understanding of digital transformation in healthcare and provides practical guidance for tailoring digital strategies to different hospital models.

2.6.1 Theoretical implications

Our study contributes to the healthcare management literature by focusing on an unexplored perspective of digital transformation. We demonstrate that HCDD strategies are highly sensitive to the hospital's business model, challenging the prevailing notion of a uniform

digital transformation path in healthcare. We also identified an interrelated mechanism that generates greater advantage when combining the hospital's operation model with the form of managing the digital resources. Therefore, our study challenges the usual assumption of HCDT literature by demonstrating that managerial decisions regarding the internalization or outsourcing of digital resources are not as freely open to make-or-buy choices as healthcare scholars might assume. Instead, these decisions are highly dependent on the hospital's organizational structure and the nature of its financial resources. This dependency also influences the TOE dimensions, which vary significantly across different business model types: private healthcare organizations, public healthcare organizations, and research-oriented healthcare organizations.

From a theoretical perspective, we show that the resource-based view (RBV), relational view (RV), and hybrid governance serve as valuable lenses for describing each organizational structure we identified. However, none of them fully capture the dynamics of all hospital business models. For instance, applying RBV to analyze a research hospital may lead to a limited or even biased perspective, as our findings suggest that hybrid governance better explains the underlying structure of HCDT in such organizations. Similarly, adopting a relational view to examine public hospitals with bidding processes may fall short, as our results indicate that focusing on internal resources provides a more accurate depiction of this profile. Thus, our study contributes by characterizing hospital business models in relation to their HCDT strategies and corresponding management structures. Furthermore, we highlight how the institutional profile also shapes the hospital's connection to the broader business environment. Ecosystem dynamics vary depending on the HCDT approach, and orchestration capabilities differ significantly, particularly between private and research hospitals.

2.6.2 Managerial implications

From a managerial perspective, hospitals can use these structural insights as a foundation for digitizing specific smart healthcare dimensions, such as operations, workforce management, consumption, supply chain, and services. As observed in our investigation, digital resource management configurations also take into account the need to establish a healthcare digital ecosystem. This approach helps reduce sociocultural barriers and enables resource sharing, facilitating both the internal and external development and adoption of HCDT (e.g., as seen in the case of Hospital C). The digital transformation of healthcare organizations presents

an innovative scenario where the actual contributions of external actors to organizational structures for HCDD management can be verified. Recognizing these contributions can help convince key stakeholders—both internal and external—of the importance of implementing HCDD. Furthermore, policymakers can leverage our findings to design policies that foster the evolution of HCDD across different healthcare environments, whether public, private, or research-oriented. This, in turn, can support the digital transformation of hospitals that are still in the pre-digital stage, ultimately improving the performance of medical service delivery at a national level.

2.6.3 Limitations and future research

This study presents several limitations that also suggest avenues for future research. First, our analysis focused on a single hospital for each type of digital resource management strategy. Although we conducted an external validation to demonstrate that our findings are not necessarily limited to the original context, the limited availability of literature—both academic and business—on the role of internal versus external digital resources in healthcare indicates that this remains an underexplored area. Consequently, our results are still subject to the inherent limitations of a qualitative study based on a small number of cases, which restricts broader generalization. While the selected cases offer useful models for understanding how the HCDD process unfolds, it is essential to examine additional cases that reflect incremental developments in HCDD. Such efforts would provide further evidence on resource management strategies and highlight potential challenges that hospitals may encounter throughout the transformation journey. Moreover, future research should explore these dynamics in countries with different healthcare systems, as national institutional structures and policies also influence how individual hospitals pursue digital transformation.

Moreover, not all hospitals have a digital strategy as clearly defined as those analyzed in this study. Future research could replicate our data collection and analysis in hospitals that do not follow these digital resource management strategies for HCDD, enabling meaningful comparisons on how digital resource management influences hospital operational performance. Second, another limitation of our study is its macro perspective on the organizational management structure of HCDD in hospitals. While this broad view is valuable, hospitals consist of multiple departments with distinct functions and characteristics, such as emergency, diagnostic, and outpatient services. Although our study provides an overarching perspective,

future research should explore the specific nuances within different hospital departments. Finally, future studies could adopt quantitative approaches with larger samples to further expand the insights generated here and identify potential patterns. For instance, quantitative analyses could assess the relationship between organizational structure, digital strategy, and digital resources in the context of HCDDT, offering a more systematic understanding of these dynamics.

2.7 REFERENCES

- Abdel-Basset, M., Chang, V., & Nabeeh, N. A. (2021). An intelligent framework using disruptive technologies for COVID-19 analysis. *Technological Forecasting and Social Change*, 163(120431), 120431. <https://doi.org/10.1016/j.techfore.2020.120431>
- Adner, R. (2017). Ecosystem as structure: an actionable construct for strategy. *Journal of Management*, 43, 39–58. <https://doi.org/10.1177/0149206316678451>
- Aloini, D., Benevento, E., Stefanini, A., & Zerbino, P. (2023). Transforming healthcare ecosystems through blockchain: Opportunities and capabilities for business process innovation. *Technovation*, 119, 102557. <https://doi.org/10.1016/j.technovation.2022.102557>
- Aranyossy, M., & Halmosi, P. (2024). Healthcare 4.0 value creation – The interconnectedness of hybrid value propositions. *Technological Forecasting and Social Change*, 208(123718), 123718. <https://doi.org/10.1016/j.techfore.2024.123718>
- Ayala, N. F., Paslauskis, C. A., Ghezzi, A., & Frank, A. G. (2017). Knowledge sharing dynamics in service suppliers' involvement for servitization of manufacturing companies. *International Journal of Production Economics*, 193, 538–553. <https://doi.org/10.1016/j.ijpe.2017.08.019>
- Barney, J. (1991). Firm Resources and Sustained Competitive Advantage. *Journal of Management*, 17(1), 99–120. <https://doi.org/10.1177/014920639101700108>
- Barratt, M., Choi, T. Y., & Li, M. (2011). Qualitative case studies in operations management: Trends, research outcomes, and future research implications. *Journal of Operations Management*, 29(4), 329–342. <https://doi.org/10.1016/j.jom.2010.06.002>
- Baudier, P., Kondrateva, G., Ammi, C., Chang, V., & Schiavone, F. (2023). Digital transformation of healthcare during the COVID-19 pandemic: Patients' teleconsultation acceptance and trusting beliefs. *Technovation*, 120, 102547. <https://doi.org/10.1016/j.technovation.2022.102547>
- Benitez, G. B., Ayala, N. F., & Frank, A. G. (2020). Industry 4.0 innovation ecosystems: An evolutionary perspective on value cocreation. *International journal of production economics*, 228, 107735. <https://doi.org/10.1016/j.ijpe.2020.107735>
- Beaulieu, M., & Bentahar, O. (2021). Digitalization of the healthcare supply chain: A roadmap to generate benefits and effectively support healthcare delivery. *Technological Forecasting and Social Change*, 167(120717), 120717. <https://doi.org/10.1016/j.techfore.2021.120717>
- Beane, M. (2019). Shadow Learning: Building Robotic Surgical Skill When Approved Means Fail. *Administrative Science Quarterly*, 64(1), 87–123. <https://doi.org/10.1177/0001839217751692>

- Blomqvist, K., Kyläheiko, K., & Virolainen, V.-M. (2002). Filling a gap in traditional transaction cost economics: *International Journal of Production Economics*, 79(1), 1–14.
[https://doi.org/10.1016/S0925-5273\(00\)00095-5](https://doi.org/10.1016/S0925-5273(00)00095-5)
- Cannavacciuolo, L., Capaldo, G., & Ponsiglione, C. (2023). Digital innovation and organizational changes in the healthcare sector: Multiple case studies of telemedicine project implementation. *Technovation*, 120(102550), 102550. <https://doi.org/10.1016/j.technovation.2022.102550>
- Capron, L., & Mitchell, W. (2004). Where firms change: internal development versus external capability sourcing in the global telecommunications industry. *European Management Review*, 1(2), 157–174. <https://doi.org/10.1057/palgrave.emr.1500017>
- Carter, N., Bryant-Lukosius, D., DiCenso, A., Blythe, J., & Neville, A. J. (2014). The Use of Triangulation in Qualitative Research. *Oncology Nursing Forum*, 41(5), 545–547.
<https://doi.org/10.1188/14.ONF.545-547>
- Cerchione, R., Centobelli, P., Riccio, E., Abbate, S., & Oropallo, E. (2023). Blockchain's coming to hospital to digitalize healthcare services: Designing a distributed electronic health record ecosystem. *Technovation*, 120, 102480. <https://doi.org/10.1016/j.technovation.2022.102480>
- Chaudhuri, A., Naseraldin, H., & Narayanamurthy, G. (2023). Healthcare 3D printing service innovation: Resources and capabilities for value Co-creation. *Technovation*, 121, 102596.
<https://doi.org/10.1016/j.technovation.2022.102596>
- Chaudhuri, R., Chatterjee, S., Mariani, M. M., & Wamba, S. F. (2024). Assessing the influence of emerging technologies on organizational data driven culture and innovation capabilities: A sustainability performance perspective. *Technological Forecasting and Social Change*, 200, 123165. <https://doi.org/10.1016/j.techfore.2023.123165>
- Chwiłkowska-Kubala, A., Cyfert, S., Malewska, K., Mierzejewska, K., & Szumowski, W. (2023). The impact of resources on digital transformation in energy sector companies. The role of readiness for digital transformation. *Technology in Society*, 74, 102315.
<https://doi.org/10.1016/j.techsoc.2023.102315>
- Corbin, J., & Strauss, A. (2014). Basics of qualitative research: Techniques and procedures for developing grounded theory. Sage Publications.
- Dal Mas, F., Massaro, M., Rippa, P., & Secundo, G. (2023). The challenges of digital transformation in healthcare: An interdisciplinary literature review, framework, and future research agenda. *Technovation*, 123, 102716. <https://doi.org/10.1016/j.technovation.2023.102716>
- Dalenogare, L. S., Le Dain, M.-A., Ayala, N. F., Pezzotta, G., & Frank, A. G. (2023). Building digital servitization ecosystems: An analysis of inter-firm collaboration types and social exchange mechanisms among actors. *Technovation*, 124, 102756.
<https://doi.org/10.1016/j.technovation.2023.102756>
- Dalenogare, L. S., Le Dain, M.-A., Benitez, G. B., Ayala, N. F., & Frank, A. G. (2022). Multichannel digital service delivery and service ecosystems: The role of data integration within Smart Product-Service Systems. *Technological Forecasting and Social Change*, 183, 121894.
<https://doi.org/10.1016/j.techfore.2022.121894>
- Dicuonzo, G., Donofrio, F., Fusco, A., & Shini, M. (2023). Healthcare system: Moving forward with artificial intelligence. *Technovation*, 120, 102510.
<https://doi.org/10.1016/j.technovation.2022.102510>

- Dubé, & Paré. (2003). Rigor in Information Systems Positivist Case Research: Current Practices, Trends, and Recommendations. *MIS Quarterly*, 27(4), 597. <https://doi.org/10.2307/30036550>
- Dyer, J. H., & Singh, H. (1998). The Relational View: Cooperative Strategy and Sources of Interorganizational Competitive Advantage. *The Academy of Management Review*, 23(4), 660. <https://doi.org/10.2307/259056>
- Elia, S., Giuffrida, M., Mariani, M. M., & Bresciani, S. (2021). Resources and digital export: An RBV perspective on the role of digital technologies and capabilities in cross-border e-commerce. *Journal of Business Research*, 132, 158–169. <https://doi.org/10.1016/j.jbusres.2021.04.010>
- Fernandez-Vidal, J., Antonio Perotti, F., Gonzalez, R., & Gasco, J. (2022). Managing digital transformation: The view from the top. *Journal of Business Research*, 152, 29–41. <https://doi.org/10.1016/j.jbusres.2022.07.020>
- Furstenau, L. B., Zani, C., Terra, S. X., Sott, M. K., Choo, K.-K. R., & Saurin, T. A. (2022). Resilience capabilities of healthcare supply chain and supportive digital technologies. *Technology in Society*, 71(102095), 102095. <https://doi.org/10.1016/j.techsoc.2022.102095>
- Franco, C. W., Benitez, G. B., Sousa, P. R. de, Kliemann Neto, F. J., & Frank, A. G. (2024). Managing resources for digital transformation in supply chain integration: The role of hybrid governance structures. *International Journal of Production Economics*, 278(109428), 109428. <https://doi.org/10.1016/j.ijpe.2024.109428>
- Galletta, A. (2013). *Mastering the semi-structured interview and beyond*. New York University Press.
- Gambal, M.-J., Asatiani, A., & Kotlarsky, J. (2022). Strategic innovation through outsourcing – A theoretical review. *The Journal of Strategic Information Systems*, 31(2), 101718. <https://doi.org/10.1016/j.jsis.2022.101718>
- Gawer, A., Cusumano, M.A., (2014). Industry platforms and ecosystem innovation. *Journal of Product Innovation Management*. 31, 417–433. <https://doi.org/10.1111/jpim.12105>
- Gioia, D. A., Corley, K. G., & Hamilton, A. L. (2013). Seeking qualitative rigor in inductive research: Notes on the Gioia methodology. *Organizational Research Methods*, 16(1), 15–31. <https://doi.org/10.1177/1094428112452151>
- Goffin, K., Åhlström, P., Bianchi, M., & Richtnér, A. (2019). Perspective: State-of-the-Art: The Quality of Case Study Research in Innovation Management. *Journal of Product Innovation Management*, 36(5), 586–615. <https://doi.org/10.1111/jpim.12492>
- Halmosi, P., & Aranyossy, M. (2024). Exploring macro-environmental catalysts and barriers of healthcare 4.0 transformation in Central-Eastern European countries: A comprehensive study in Hungary. *Technology in Society*, 78(102620), 102620. <https://doi.org/10.1016/j.techsoc.2024.102620>
- Huy, Q. N. (2001). In praise of middle managers. *Harvard Business Review*, 79(8), 72–79.
- Iyanna, S., Kaur, P., Ractham, P., Talwar, S., & Najmul Islam, A. K. M. (2022). Digital transformation of healthcare sector. What is impeding adoption and continued usage of technology-driven innovations by end-users? *Journal of Business Research*, 153, 150–161. <https://doi.org/10.1016/j.jbusres.2022.08.007>
- Kamalaldin, A., Linde, L., Sjödin, D., & Parida, V. (2020). Transforming provider-customer relationships in digital servitization: A relational view on digitalization. *Industrial Marketing Management*, 89, 306–325. <https://doi.org/10.1016/j.indmarman.2020.02.004>

- Kaplan, H. C., Brady, P. W., Dritz, M. C., Hooper, D. K., Linam, W. M., Froehle, C. M., & Margolis, P. (2010). The Influence of Context on Quality Improvement Success in Health Care: A Systematic Review of the Literature. *Milbank Quarterly*, 88(4), 500–559. <https://doi.org/10.1111/j.1468-0009.2010.00611.x>
- Ketokivi, M., & Choi, T. (2014). Renaissance of case research as a scientific method. *Journal of Operations Management*, 32(5), 232–240. <https://doi.org/10.1016/j.jom.2014.03.004>
- Ketokivi, M., & Mahoney, J. T. (2020). Transaction Cost Economics As a Theory of Supply Chain Efficiency. *Production and Operations Management*, 29(4), 1011–1031. <https://doi.org/10.1111/poms.13148>
- Kinkel, S., Baumgartner, M., & Cherubini, E. (2022). Prerequisites for the adoption of AI technologies in manufacturing – Evidence from a worldwide sample of manufacturing companies. *Technovation*, 110, 102375. <https://doi.org/10.1016/j.technovation.2021.102375>
- KPMG (2024). Healthcare horizons revisited: Trailblazers, transformers and the ‘how to’ of healthcare transformation. Report. <https://kpmg.com/healthcarehorizonsrevisited>
- Kraus, S., Schiavone, F., Pluzhnikova, A., & Invernizzi, A. C. (2021). Digital transformation in healthcare: Analyzing the current state-of-research. *Journal of Business Research*, 123, 557–567. <https://doi.org/10.1016/j.jbusres.2020.10.030>
- Liu, D. Y., Chen, S. W., & Chou, T. C. (2011). Resource fit in digital transformation: Lessons learned from the CBC Bank global e-banking project. *Management Decision*, 49(10), 1728–1742.
- Malik, M., Andargoli, A., Chavez Clavijo, R., & Mikalef, P. (2024). A relational view of how social capital contributes to effective digital transformation outcomes. *Journal of Strategic Information Systems*, 33(2), 101837. <https://doi.org/10.1016/j.jsis.2024.101837>
- March, J. G. (1991). Exploration and Exploitation in Organizational Learning. *Organization Science*, 2(1), 71–87.
- Marcon, É., Marodin, G. A., & Frank, A. G. (2025). Combining organizational and social factors to support industry 4.0 implementation: A sociotechnical and configurational analysis of technology adopters. *IEEE Transactions on Engineering Management*, 72, 146–160. <https://doi.org/10.1109/tem.2024.3506991>
- Massaro, M. (2023). Digital transformation in the healthcare sector through blockchain technology. Insights from academic research and business developments. *Technovation*, 120, 102386. <https://doi.org/10.1016/j.technovation.2021.102386>
- Mauro, M., Noto, G., Prenestini, A., & Sarto, F. (2024). Digital transformation in healthcare: Assessing the role of digital technologies for managerial support processes. *Technological Forecasting and Social Change*, 209(123781), 123781. <https://doi.org/10.1016/j.techfore.2024.123781>
- Merriam, S. B., Tisdell E. J. (2015). Qualitative research: A guide to Design and Implementation. John Wiley & Sons.
- Mura, M., Micheli, P., & Longo, M. (2021). The effects of performance measurement system uses on organizational ambidexterity and firm performance. *International Journal of Operations & Production Management*, 41(13), 127–151. <https://doi.org/10.1108/IJOPM-02-2021-0101>

- Oliveira, M., Zancul, E., & Salerno, M. S. (2024). Capability building for digital transformation through design thinking. *Technological Forecasting and Social Change*, 198, 122947. <https://doi.org/10.1016/j.techfore.2023.122947>
- Pham, P., Zhang, H., Gao, W., & Zhu, X. (2024). Determinants and performance outcomes of artificial intelligence adoption: Evidence from U.S. Hospitals. *Journal of Business Research*, 172(114402), 114402. <https://doi.org/10.1016/j.jbusres.2023.114402>
- Parmigiani, A. (2007). Why do firms both make and buy? An investigation of concurrent sourcing. *Strategic Management Journal*, 28(3), 285–311. <https://doi.org/10.1002/smj.580>
- Piccoli, G., Grover, V., & Rodriguez, J. (2024). Digital transformation requires digital resource primacy: Clarification and future research directions. *The Journal of Strategic Information Systems*, 33(2), 101835. <https://doi.org/10.1016/j.jsis.2024.101835>
- Raimo, N., De Turi, I., Albergo, F., & Vitolla, F. (2023). The drivers of the digital transformation in the healthcare industry: An empirical analysis in Italian hospitals. *Technovation*, 121, 102558. <https://doi.org/10.1016/j.technovation.2022.102558>
- Rocco DePietro, Edith Wiarda, & Mitchell Fleischer. (1990). The context for change: Organization, technology and environment. In L. G. Tornatzky & M. Fleischer (Eds.), *The processes of technological innovation* (pp. 151–175). Lexington Books: Lexington.
- Rosa, V. M., Saurin, T. A., Tortorella, G. L., Fogliatto, F. S., Tonetto, L. M., & Samson, D. (2021). Digital technologies: An exploratory study of their role in the resilience of healthcare services. *Applied Ergonomics*, 97(103517), 103517. <https://doi.org/10.1016/j.apergo.2021.103517>
- Rubbio, I., Bruccoleri, M., Pietrosi, A., & Ragonese, B. (2019). Digital health technology enhances resilient behaviour: evidence from the ward. *International Journal of Operations & Production Management*, 40(1), 34–67. <https://doi.org/10.1108/ijopm-02-2018-0057>
- Saha, E., & Rathore, P. (2024). The impact of healthcare 4.0 technologies on healthcare supply chain performance: Extending the organizational information processing theory. *Technological Forecasting and Social Change*, 201(123256), 123256. <https://doi.org/10.1016/j.techfore.2024.123256>
- Schiavone, F., Mancini, D., Leone, D., & Lavorato, D. (2021). Digital business models and ridesharing for value co-creation in healthcare: A multi-stakeholder ecosystem analysis. *Technological Forecasting and Social Change*, 166, 120647. <https://doi.org/10.1016/j.techfore.2021.120647>
- Secundo, G., Toma, A., Schiuma, G., & Passiante, G. (2019). Knowledge transfer in open innovation: A classification framework for healthcare ecosystems. *Business Process Management Journal*, 25(1), 144–163. <https://doi.org/10.1108/BPMJ-06-2017-0173>
- Steinhauser, S., Dobliger, C., & Hüsiger, S. (2020). The Relative Role of Digital Complementary Assets and Regulation in Discontinuous Telemedicine Innovation in European Hospitals. *Journal of Management Information Systems*, 37(4), 1155–1183. <https://doi.org/10.1080/07421222.2020.1831778>
- S&S Insider. (2022). Healthcare IT outsourcing market size, share & segmentation by application (Provider IT outsourcing, Payer IT outsourcing, IT infrastructure outsourcing, Life Science IT outsourcing, Operation IT outsourcing), by end user, by region: Global forecast 2024–2032 (Report No. SNS/HC/2428, p. 210). July, 2022. <https://www.snsinsider.com/reports/healthcare-it-outsourcing-market-2428>

- S&S Insider. (2025). Hospital outsourcing market size, share & segmentation by service (Healthcare IT, transportation services, clinical services, business services, others), by type (public, private), by region: Global forecast 2024–2032 (Report No. SNS/HC/5834, p. 210). February, 2025. <https://www.snsinsider.com/reports/hospital-outsourcing-market-5834>
- Tabim, V. M., Ayala, N. F., & Frank, A. G. (2021). Implementing Vertical Integration in the Industry 4.0 Journey: Which Factors Influence the Process of Information Systems Adoption? *Information Systems Frontiers*. <https://doi.org/10.1007/s10796-021-10220-x>
- Thuemmler, C., & Bai, C. (2017). Health 4.0: Application of Industry 4.0 Design Principles in Future Asthma Management. In *Health 4.0: How Virtualization and Big Data are Revolutionizing Healthcare* (pp. 23–37). Springer International Publishing. https://doi.org/10.1007/978-3-319-47617-9_2
- Tonetto, L. M., Saurin, T. A., Fogliatto, F. S., Tortorella, G. L., Narayanamurthy, G., da Rosa, V. M., & Tengskawan, J. (2023). Information and communication technologies in emergency care services for patients with COVID-19: a multi-national study. *International Journal of Production Research*, 61(24), 8384–8400. <https://doi.org/10.1080/00207543.2021.1967501>
- Tortorella, G. L., Fogliatto, F. S., Mac Cawley Vergara, A., Vassolo, R., & Sawhney, R. (2019). Healthcare 4.0: trends, challenges and research directions. *Production Planning & Control*, 31(15), 1245–1260. <https://doi.org/10.1080/09537287.2019.1702226>
- Tortorella, G. L., Fogliatto, F. S., Espôsto, K. F., Vergara, A. M. C., Vassolo, R., Mendoza, D. T., & Narayanamurthy, G. (2020). Effects of contingencies on healthcare 4.0 technologies adoption and barriers in emerging economies. *Technological Forecasting and Social Change*, 156, 120048. <https://doi.org/10.1016/j.techfore.2020.120048>
- Tortorella, G. L., Fogliatto, F. S., Espôsto, K. F., Mac Cawley Vergara, A., Vassolo, R., Tlapa Mendoza, D., & Narayanamurthy, G. (2020). Measuring the effect of Healthcare 4.0 implementation on hospitals' performance. *Production Planning & Control*, 33(4), 386–401. <https://doi.org/10.1080/09537287.2020.1824283>
- Tortorella, G. L., Saurin, T. A., Fogliatto, F. S., Rosa, V. M., Tonetto, L. M., & Magrabi, F. (2021). Impacts of Healthcare 4.0 digital technologies on the resilience of hospitals. *Technological Forecasting and Social Change*, 166. <https://doi.org/10.1016/j.techfore.2021.120666>
- Tortorella, G. L., Fogliatto, F. S., Sunder M, V., Cawley Vergara, A. M., & Vassolo, R. (2021). Assessment and prioritisation of Healthcare 4.0 implementation in hospitals using Quality Function Deployment. *International Journal of Production Research*, 60(10), 3147–3169. <https://doi.org/10.1080/00207543.2021.1912429>
- Tortorella, G. L., Fogliatto, F. S., Kurnia, S., Thürer, M., & Capurro, D. (2022). Healthcare 4.0 digital applications: An empirical study on measures, bundles and patient-centered performance. *Technological Forecasting and Social Change*, 181, 121780. <https://doi.org/10.1016/j.techfore.2022.121780>
- Tortorella, G. L., Fogliatto, F. S., Saurin, T. A., Tonetto, L. M., & McFarlane, D. (2022). Contributions of Healthcare 4.0 digital applications to the resilience of healthcare organizations during the COVID-19 outbreak. *Technovation*, 111, 102379. <https://doi.org/10.1016/j.technovation.2021.102379>
- Tortorella, G. L., Fogliatto, F. S., Tlapa Mendoza, D., Pepper, M., & Capurro, D. (2022c). Digital transformation of health services: a value stream-oriented approach. *International Journal of Production Research*, 61(6), 1814–1828. <https://doi.org/10.1080/00207543.2022.2048115>

- Verhoef, P. C., Broekhuizen, T., Bart, Y., Bhattacharya, A., Qi Dong, J., Fabian, N., & Haenlein, M. (2021). Digital transformation: A multidisciplinary reflection and research agenda. *Journal of Business Research*, 122, 889–901. <https://doi.org/10.1016/j.jbusres.2019.09.022>
- Voss, C., Tsikriktsis, N., & Frohlich, M. (2002). Case research in operations management. *International Journal of Operations & Production Management*, 22(2), 195–219. <https://doi.org/10.1108/01443570210414329>
- Wang, Y., & Hajli, N. (2017). Exploring the path to big data analytics success in healthcare. *Journal of Business Research*, 70, 287–299. <https://doi.org/10.1016/j.jbusres.2016.08.002>
- Wang, Y., Kung, L., & Byrd, T. A. (2018). Big data analytics: Understanding its capabilities and potential benefits for healthcare organizations. *Technological Forecasting and Social Change*, 126, 3–13. <https://doi.org/10.1016/j.techfore.2015.12.019>
- Yin, R. K. (2012). *Case Study Research: Design and Methods*. Sage.
- Yin, R. K. (2013). Validity and generalization in future case study evaluations. *Evaluation*, 19(3), 321–332. <https://doi.org/10.1177/1356389013497081>
- Yu, W., Zhao, G., Liu, Q., & Song, Y. (2021). Role of big data analytics capability in developing integrated hospital supply chains and operational flexibility: An organizational information processing theory perspective. *Technological Forecasting and Social Change*, 163(120417), 120417. <https://doi.org/10.1016/j.techfore.2020.120417>
- Zhang, X., Denicol, J., Chan, P. W., & Le, Y. (2024). Designing the transition to operations in large inter-organizational projects: Strategy, structure, process, and people. *Journal of Operations Management*, 70(1), 107–136. <https://doi.org/10.1002/joom.1275>

Appendix A - Representative studies in top-tier journals

Authors	Journal	The focus of the study	Methodology	Main findings
Abdel-Basset et al. (2021)	TFSC	Framework using disruptive technologies for COVID-19 analysis in healthcare	Qualitative - Case study	The study provides oversight on how to adopt technologies to reduce the impact of unprecedented outbreaks for COVID-19.
Aloini et al. (2023)	TECH	Opportunities for Blockchain-driven healthcare innovation and BPM capabilities	Qualitative - Case studies	Blockchain fosters healthcare innovation and enhances collaboration.
Aranyossy & Halmosi (2024)	TFSC	Healthcare 4.0 value creation	Qualitative - Case studies	Healthcare 4.0 reduces burdens, boosts revenue, and fosters trust through interconnected innovation.
Baudier et al. (2023)	TECH	Factors predicting the intention to use medical teleconsultation	Survey (n =1233) + PLS	Effort and performance drive teleconsultation; social and facilitating factors challenge theory.
Beaulieu & Bentahar (2021)	TFSC	Roadmap for healthcare supply chain digitalization	Observations + Literature review	A roadmap for internal and external digitalization enhances efficiency.
Cannavacciuolo et al. (2023)	TECH	Implementation of telemedicine projects	Qualitative - Case studies	Technology, acceptance, and organization drive telemedicine adoption.
Cerchione et al. (2023)	TECH	Platform integrating electronic medical records into a private blockchain	Design Science Research	Blockchain ensures trust, traceability, and transparency in healthcare services.
Chaudhuri et al. (2023)	TECH	Value dynamics and resources for co-creation in healthcare 3DP services	Qualitative - Case studies	Healthcare 3DP providers and surgeons co-create value through exploitative and explorative capabilities, enhancing innovation and hospital adoption.
Dal Mas et al. (2023)	TECH	Challenges of digital transformation in healthcare	Structured Literature Review	Digital health, stakeholder engagement, and value impact are key research areas.
Dicuonzo et al. (2023)	TECH	AI for effective and efficient healthcare management	Qualitative - Case study	AI optimizes healthcare resource management and improves health outcomes.
Furstenau et al. (2022)	TIS	Digital technologies supporting healthcare supply chain resilience	Qualitative - Case studies	A framework explains how technology adoption strengthens healthcare supply chain resilience.
Halmosi & Aranyossy (2024)	TIS	Macro-environmental factors influencing Healthcare 4.0 adoption in Hungary	Qualitative - Case study	Barriers and enablers in healthtech adoption are crucial for policymaking.
Iyanna et al. (2022)	JBR	Barriers to e-health adoption and continued usage	Qualitative - Case study	Healthcare providers, organizations, and patients face specific adoption barriers.
Kraus et al. (2021)	JBR	Stakeholder implementation of digital technologies for healthcare management	Systematic literature review	Prior research on digital healthcare transformation falls into five key clusters.
Malik et al. (2024)	JSIS	Relational view of organizations for network-based external complementarities	Survey (n = 183) + Mediation analyses	Digital transformation requires leveraging external knowledge for innovation.
Massaro (2023)	TECH	Comparison of blockchain-related papers and patents to assess implementation expectations in healthcare	Qualitative - Content analysis	Insights include dominance, value creation, and human standards.
Mauro et al. (2024)	TFSC	Impact of digital technologies on managerial and administrative processes	Qualitative - Case study	IoT, AI, and ML have the most significant impact on healthcare administration.
Oliveira et al. (2024)	TFSC	Design thinking for digital transformation and innovation in hospitals	Design thinking	Design thinking enhances digital transformation and innovation capabilities.
Pham et al. (2024)	JBR	Determinants and performance outcomes of AI adoption	Secondary dataset + OLS regression	AI adoption decisions are shaped by various determinants and yield multiple hospital benefits.
Raimo et al. (2023)	TECH	Digital transformation levels in Italian hospitals and influencing factors	Survey (n = 103) + OLS regression	Hospital characteristics directly affect digital transformation levels.
Rosa et al. (2021)	AE	Role of Healthcare 4.0 technologies in resilient performance	Survey (n = 109) + Interviews	Four H4.0 technologies significantly enhance service resilience.

Rubbio et al. (2019)	IJOPM	Capabilities enabling healthcare resilience and digital technology support	Qualitative - Case studies	Five dynamic capabilities are crucial for managing operational failures.
Saha & Rathore (2024)	TFSC	Impact of Healthcare 4.0 technologies on the supply chain	Survey (n = 119) + PLS-SEM method	Healthcare 4.0 adoption improves supply chain activities and performance.
Schiavone et al. (2021)	TFSC	Value creation, delivery, and capture in healthcare platform ecosystems	Qualitative - Case study	A multilevel framework explains the healthcare multi-stakeholder ecosystem.
Secundo et al. (2021)	JBR	Framework on digital technologies addressing COVID-19 challenges in healthcare	Quantitative - Optimization model	Policy implications emerge from decision support system (DSS) analysis.
Tonetto et al. (2023)	IJPR	ICT contributions to emergency care for COVID-19 patients	Survey (n = 109) + interviews	The most useful digital technologies for COVID-19 treatment have been identified.
Tortorella et al. (2019)	PPC	Trends, challenges, and gaps in Healthcare 4.0 implementation	Scoping review of the literature	H4.0 literature lacks structured academic alignment and practical orientation.
Tortorella et al. (2020)	PPC	Technology bundles, implementation barriers, and performance impact in H4.0	Survey (n = 181) + PCA + ANOVA	H4.0 technology bundles positively influence hospital performance.
Tortorella et al. (2020)	TFSC	Effect of contingency factors and barriers on H4.0 adoption in emerging economies	Survey (n = 156) + ANOVA + MANOVA	Contingencies significantly impact H4.0 adoption and associated barriers.
Tortorella et al. (2021)	TFSC	Impact of ten H4.0 technologies on four resilience abilities in hospitals	Survey (n = 109) + Cluster analysis	Four H4.0 technologies play a major role in resilience capabilities.
Tortorella et al. (2021)	IJPR	Quality Function Deployment for prioritizing technology integration	Qualitative - Case studies	A systematic method supports I4.0 integration in healthcare organizations.
Tortorella et al. (2022)	TFSC	H4.0 technology bundles, adoption measures, and performance impact	Survey (n = 119) + OLS regression	Four validated H4.0 applications enhance patient-centered performance.
Tortorella et al. (2022)	TECH	Digital applications contributing to healthcare resilience during COVID-19	Qualitative - Case studies	Supply chain and diagnosis applications contribute to healthcare resilience.
Tortorella et al. (2022)	IJPR	Value stream mapping for selecting H4.0 digital applications in lean healthcare	Qualitative - Case study	Three prioritized H4.0 applications improve value stream processes.
Wang & Hajli (2017)	JBR	Big data analytics-enabled business value model development	Qualitative - Case studies	Developing big data analytics capabilities drives healthcare transformation.
Wang et al. (2018)	TFSC	Historical development, architecture, and functionalities of big data analytics	Qualitative - Case studies	A big data analytics architecture has been established for healthcare.
Yu et al. (2021)	TFSC	Big data analytics in hospital supply chain integration and operational flexibility	Survey (n = 105) + SEM	Big data enhances hospital supply chain integration, mediating flexibility through patient and supplier collaboration.

* Journals considered in this review are those top ranked in the Academic Journal Guide (AJG) 2024 with 3 and 4 stars in the Operations Management field: IJOPM – International Journal of Operations and Production Management; JSIS – Journal of Strategic Information Systems; IJPR: International Journal of Production Research; PPC: Production Planning and Control; JBR: Journal of Business Research; AE: Applied Ergonomics, TFSC: Technological Forecasting and Social Change; TECH: Technovation; TIS: Technology in Society.

Appendix B - Interview guideline

1 – Information of the organizational context and professional background.

2 – Questions about the management of healthcare digital transformation, with focus on the management of digital resources.

a) What digital resource management strategy does the organization adopt? What motivated your organization to follow this resource management format? Please justify your answer.

b) What digital resources does the hospital use to support adopting digital technologies? Comment on their role and origin.

c) Please indicate which digital technologies your organization has already started to implement or use. If there are any technological projects in the early stages, please comment.

d) How are the demands for the adoption of digital technologies structured internally?

e) Comment on the basic infrastructure that the hospital has established for healthcare digital transformation.

f) How does the hospital organize itself to maintain the quality of digital resources and the use of digital technologies in clinical and operational procedures? Please justify your answer and provide examples.

g) Does your organization have external partners that support healthcare digital transformation? If yes, describe the role of each one and why they were chosen.

3. ALIGNING DIGITAL STRATEGY AND RESOURCES TO CREATE HEALTHCARE 4.0 ORGANIZATIONS: IMPLICATIONS FOR SMART OPERATIONS AND SMART WORKING

Lucas Miguel Alencar de Morais Correia, Paulo Renato de Sousa, Alejandro Germán Frank

This article has been published in Technological Forecasting & Social Change (JCR: 13,3)

<https://doi.org/10.1016/j.techfore.2025.124339>

Abstract

This study investigates the organizational foundations required to implement Healthcare 4.0 (H4.0) by examining how digital strategy, internal and external digital resources, and network embeddedness support the development of smart operations and smart working in healthcare organizations. While prior research has predominantly focused on technology-driven improvements in healthcare processes, less attention has been given to the sociotechnical integration of people, processes, and technologies, particularly from a strategic and managerial perspective. Drawing on a survey of 303 hospitals, we assessed the role of digital strategy and digital resources in advancing H4.0, conceptualized through the dual lenses of smart operations (technology for process automation) and smart working (technology to support workers). Using hierarchical regression and bootstrapping techniques, we find that digital strategy positively influences both smart operations and smart working. However, internal and external digital resources mediate this relationship only in the context of smart working. This indicates that while process automation relies primarily on strategic clarity, supporting workers requires stronger resource structures. Additionally, we find that healthcare network embeddedness positively moderates the relationship between digital strategy and external digital resources, but not internal resources. This suggests that external collaborations enhance a hospital's ability to translate strategic intent into actionable use of external resources, particularly in support of smart working. Our findings contribute to a deeper understanding of how strategic alignment and network embeddedness shape sociotechnical systems in healthcare. They offer guidance for managers seeking to design holistic digital transformation strategies to build H4.0 organizations.

Keywords: Digital Strategy; Digital Resources; Healthcare 4.0; Sociotechnical Systems.

3.1 INTRODUCTION

The digital transformation of healthcare organizations has been characterized by the application of advanced digital technologies such as the Internet of Things, Cloud Computing, Big Data, and Artificial Intelligence in medical devices and healthcare information systems (Kraus et al., 2021), resulting in what is called as 'Healthcare 4.0' (H4.0) (Thuemmler & Bai, 2017). The introduction of these technologies enables real-time personalization of patient and workers care services, facilitating the transition to a patient-centered environment (Wang et al., 2018). This transformation involves the interaction of people, processes, and digital technologies, forming micro-foundations of operational capability (Roscoe et al., 2019). In this

sense, the concept of H4.0 considers the interaction of two main dimensions that form a sociotechnical healthcare system: from the technical side, the use of digital technologies to automate operational processes is commonly referred to as *healthcare smart operations* (Tortorella et al., 2020), while the social side considers the use of digital technologies to support healthcare worker activities, usually known as *healthcare smart working* (Klein & Watson-Manheim, 2021). When H4.0 is conceived to create a balanced sociotechnical system between workers, organizational processes, and technologies, both smart working and smart operations need to be developed as part of the healthcare digital transformation strategy (Grover et al., 2018).

The literature has focused on understanding which digital technologies are used and how digital transformation can contribute to achieving better performance outcomes in healthcare systems (Tortorella et al., 2022a; Tortorella et al., 2022b). Overall, this literature has been predominantly oriented toward the techno-centric view of smart operations. For example, Dicuonzo et al. (2023) explore how artificial intelligence can support the process of the healthcare system; Cannavacciuolo et al. (2023) address the organizational changes through the implementation of telemedicine in healthcare; and Basile et al. (2024) consider the role of big data analytics to improve the quality of healthcare services. These studies are concerned with how digital technologies can boost different operational processes. On the other hand, an emerging stream has highlighted the need for integrating people with processes and technologies, considering that there is a sociotechnical interplay between them (Grover et al., 2018). For instance, Klinker et al. (2020) investigate the use of augmented reality technologies to increase healthcare workers' capabilities and Naslund et al. (2019) consider the use of digital technologies to scale up healthcare through nonspecialist workers. These are a few of the cases when digital technologies are considered for smart working in healthcare, bringing a social perspective to the technological view. However, the literature has missed the consideration of what is necessary in the context of strategy and organizational design when smart operations and smart working are in the spotlight of Healthcare 4.0.

When discussing the adoption and use of digital technologies, the influence of strategic and managerial aspects is often overlooked (Chanias et al., 2019; Verhoef et al., 2021) particularly in terms of their roles to balance the organizational system from a sociotechnical perspective. In this context, achieving organizational clarity is essential to support the effective adoption of digital technologies (Marcon et al., 2022; Marcon et al., 2025). Otherwise, the adoption of these technologies may be unsuccessful or limited to isolated implementations (Zimmer et al., 2023). This organizational support has often been regarded as a matter of

establishing a digital strategy and managing digital resources (e.g., technological infrastructure, people skills, and financial investments for digitalization) within organizations (Verhoef et al., 2021; Franco et al., 2024). In this context, the vision of strategic alignment between strategy and resources is essential (Edelman et al., 2005). This is because H4.0 can be supported through an appropriate digital strategy, backed by suitable resources, and, as a result, can facilitate the adoption of healthcare smart operations and healthcare smart working concepts. The alignment between strategy and resources can be further strengthened when an organization is embedded within a network of external healthcare actors. Such embeddedness facilitates access to critical knowledge and resources through ongoing interactions (Granovetter, 1985), including those that support digitalization efforts (Xing et al., 2023). In this context, network embeddedness represents a key component of the environmental dimension of sociotechnical systems that underpin digital transformation processes (Marcon et al., 2024).

However, little is known about the way digital strategy, resources and the connection with the external environment are associated in the implementation of healthcare sociotechnical systems in the context of digital transformation. It is well known that organizations need to define strategies for digital transformation (Chanias et al., 2019) and that managing resources is necessary for the effective implementation of digital solutions (Dalenogare et al., 2023), but it is not clear the role these organizational elements play in the development of smart operations and smart working concepts for H4.0. Since smart operations and smart working focus on different digitalization objects – the first focuses on the process and the latter on people – do organizational aspects like strategy and resources play the same role for both? And are the outcomes for both different when healthcare organizations are closer to external healthcare actors that can help with H4.0 development? The literature has not yet considered these organizational elements connected with the implementation of digital technologies for H4.0 regarding smart operations and smart working. Also, it is critical to understand the role and need of network embeddedness in H4.0, as hospitals are naturally geographically dispersed and not always close to external related actors. In such cases, the digital-resources alignment may need a different approach to deal with. Therefore, we propose the following research questions: *What is the role of digital strategy and digital resources – as organizational strategic alignment – in the development of Healthcare 4.0 sociotechnical dimensions, namely smart healthcare operations and smart healthcare working? And what is the effect of healthcare network embeddedness in this context?*

To answer these questions, we conducted a quantitative survey involving 303 hospitals. We consider the main elements of digital strategies in organizations and the development of

internal and external digital resources as antecedents of both smart operations (digital technologies for healthcare processes) and smart working (digital technologies to support workers). Furthermore, we analyze the moderating role of healthcare network embeddedness in the strategy-resources relationship. To investigate these relationships, we use hierarchical regression and bootstrapping techniques. Our results show that a stronger healthcare digital strategy positively affects the adoption of digital technologies for healthcare smart operations and healthcare smart working. On the other hand, the development of internal and external digital resources for H4.0 positively mediates only between digital strategy and smart working but there is no evidence for the same mediation in smart operations. We argue that this differentiates the focus between process automation (smart operations) and people support (smart working) in healthcare organizations. Smart working demands a stronger resource structure to embrace workers and reduce tensions related to the use of new technologies, but smart operations is more technical and less dependent on resource management, needing only a clear strategic focus for its implementation. Additionally, we found that healthcare network embeddedness positively moderates only the relationship between healthcare digital strategy and external digital resources. This indicates that a hospital's digital strategy is more effectively oriented toward leveraging external resources when it is more embedded in a network of external actors, who help clarify how such resources can be accessed and utilized. In contrast, we found no evidence that network embeddedness enhances the alignment between strategy and internal digital resources. These findings suggest that supporting smart working through external digital resources is particularly complex, as it depends not only on the availability of these resources but also on the hospital's contextual embeddedness – that is, having the necessary external connections to shape and operationalize a focused digital strategy for external resource use. Therefore, as main contribution, our study provides the theoretical basis for structuring the organizational aspects of digital transformation in healthcare by defining the main elements that consolidate a digital strategy and digital resources in H4.0. Our findings also contribute to the understanding of how to align the strategy with resources to support the implementation of digital technologies in sociotechnical healthcare systems. Healthcare managers can use these findings to understand how to create a holistic system of digital transformation aiming to build a Healthcare 4.0 organization.

3.2 THEORETICAL BACKGROUND

3.2.1 Healthcare 4.0 and the strategic alignment of sociotechnical systems

With the advent of the Fourth Industrial Revolution, also known as Industry 4.0, new information and communication technologies (ICTs) have been incorporated into organizations to digitally transform environments, enabling greater efficiency and flexibility in processes, services, and products (Liao et al., 2017). Due to its potential, I4.0 has expanded into other sectors, including healthcare, denoted as Healthcare 4.0 (H4.0) (Tortorella et al., 2022a). H4.0 adapts the principles and applications of Industry 4.0 to the healthcare sector, allowing real-time personalization of patient care services, professionals, and caregivers, facilitating the transition to a patient-centered environment (Thuemmler and Bai, 2017).

H4.0 has been predominantly considered from the technical point of view with the analysis of digital technologies to increase healthcare performance (Tortorella et al., 2022a; 2022b). However, a growing emphasis has been given to the development of sociotechnical systems in healthcare organizations (Grover et al., 2018). The sociotechnical view is based on the premise that for an organization or a workplace to achieve its goals, the social and technical systems of the organization need to be interconnected and treated together (Clegg, 2000). In this sense, the literature has highlighted that digital transformation involves the integration of processes, people, and technologies in an interplay of sociotechnical elements (Roscoe et al., 2019). In such interplay, digital technologies not only modify processes but also the way human work is performed and how people interact with them to carry out their tasks (Zimmer et al. 2023). Therefore, when H4.0 is considered from the perspective, it can be represented from two main dimensions of technology application: smart operations and smart working.

Healthcare *smart operations* considers the use of digital technologies to automate operational processes of the healthcare system (Tortorella et al., 2020a; 2020b). The focus of technology in this dimension is on task execution. It aims to increase the level of interoperability and efficiency, with more integrated and flexible processes to provide greater value to healthcare services and improve patient flow, reducing the possibility of errors arising from manual operations (Wang et al., 2018; Basile et al., 2024). For example, healthcare smart operations consider the use of cloud computing to store data generated by the organization, mobile applications for automatic communication, as well as the use of enterprise ICTs, such as ERP, to integrate different departments and internal and external areas (Raimo et al., 2023; Tortorella et al., 2022b). Additionally, it includes the use of telemedicine to increase the number

of remote clinical consultations (Cannavacciuolo et al., 2023), the use of electronic record systems, such as electronic medical records for patient management, and digital traceability for patient and medication tracking through RFID, QR codes, or the use of process simulation to predict demands and optimize operational processes (Cerchione et al., 2023).

On the other hand, Healthcare *smart working* considers the use of digital technologies to support healthcare worker activities. In this sense, it comprises a sociotechnical evolution of digital technologies and how they are used by healthcare workers. It is based on the concept of industrial smart working proposed by Frank et al. (2019) and Cagliano et al. (2019). Healthcare smart working aims to increase ergonomics and productivity through the interplay between the worker and technology (Dornelles et al., 2022). The concept goes beyond the operational efficiency of the organization, as it also seeks to improve the well-being, health, and safety of workers, enabling the enhancement or creation of new capabilities and skills (Klein & Watson-Manheim, 2021; Oliveira et al., 2024). From a technology perspective, it considers the use of wireless or biomedical/digital sensors, wearable devices, big data/analytics, artificial intelligence, augmented or virtual reality, or collaborative robots to assist workers in their tasks (Wang et al., 2018; Tortorella et al. 2022b; Raimo et al., 2023). These workers include nurses, doctors, healthcare technicians, and pharmacists, who work in a diversity of areas such as emergency care, outpatient clinics, hospital wards, pharmacies, or operating rooms, dealing directly with patient care throughout the entire process of diagnosis, treatment, and follow-up (Tortorella et al. 2022b).

Strategic alignment plays an important role in the design of sociotechnical systems, as this alignment is related to the organizational structure that supports sociotechnical dimensions (Marcon et al., 2022). Such alignment has been considered a matter of establishing strategies, managing resources, and defining the form of how both are combined (Chanias et al., 2019; Verhoef et al., 2021). In the context of digital transformation, the strategy considers a combination of ICT and business model goals (Chanias et al., 2019). The healthcare digital strategy can rely on digital resources to drive the digitization of organizations (Constantiou & Kallinikos, 2015; Vial, 2019). Digital resources include hardware, software, network, and data analytics resources (e.g., code programming or data science), which deal with the collection, storage, retrieval, sharing, and use of data, as well as financial resources for investments in basic infrastructure. Furthermore, they include related managerial domains, such as new capabilities, which typically reside in human, informational, or organizational capital, and new digital skills, both supported by a digital culture (Fernandez-Vidal et al., 2022). The

organizational structure can adopt different models to manage these digital resources, implemented according to the organization's capacity and ownership. For example, resources can be developed *internally* or *outsourced* to external partners (Parmigiani, 2007; Chaudhuri et al., 2023). All these elements when combined will consolidate a H4.0 organizational system. Thus, the strategic alignment between digital strategy and resources, and the development of healthcare smart operations and smart working will define a form of how digital transformation can be faced in healthcare organizations to create a balanced system in the digital context of organizations.

3.2.2 Network embeddedness theory

The Network Embeddedness Theory (NET) explains that the economic and organizational activities and decisions of firms are deeply rooted in the social networks in which they are embedded (Granovetter, 1985). The term embeddedness refers to the extent to which actors' actions are embedded in networks of relationships (Granovetter, 1985). These networks—or influential intermediaries—facilitate faster and more qualified access to information, resources, and control mechanisms, which become available through the relational and structural embedding of organizations (Granovetter, 1985; Dahlke et al., 2024; Chen et al., 2025). Relational embeddedness involves direct ties between actors, considering the frequency of interaction and reciprocity, which can enhance loyalty and reduce transaction costs. Structural embeddedness, in turn, goes beyond relational ties by emphasizing that the transfer of information and/or resources in social networks does not depend solely on direct contacts, but also on the overall network structure, including indirect connections with actors close to the organization (Chen et al., 2025). Furthermore, structural embeddedness refers to an actor's position and proximity within the network, and how this influences their behavior, access to information, resources, and power (Granovetter, 1985). In addition to these elements, NET considers trust to be an essential operational component. Since institutional arrangements alone are not sufficient to fully prevent opportunistic behavior or fraud, trust among network actors often allows formal contracts to be replaced, thereby reducing the risks associated with partnerships (Granovetter, 1985). Therefore, network embeddedness can be a mechanism through which organizations seek to adapt to or align with the external environment and acquire essential resources for specific purposes (Gnyawali & Madhavan, 2001).

Recent studies in the field of digital transformation, such as Chen et al. (2024), highlight the role of network embeddedness in enabling digital capabilities through technologies aimed at production customization. Similarly, Chen et al. (2025) demonstrated the contribution of network embeddedness in supply chains to strengthening operational resilience through digitalization. In addition, other studies, such as Lyu et al. (2020), identified the positive effects of network embeddedness on the relationship between open innovation and innovation radicalness, while Xing et al. (2023) proposed an approach using the TOE framework, positioning network embeddedness as an environmental-level factor that supports digital transformation and enhances innovation performance in manufacturing. Dahlke et al. (2024) also found evidence that the adoption of artificial intelligence is related to three epidemic effect mechanisms of network embeddedness. Collectively, these studies demonstrate how network embeddedness can support both digital transformation and organizational performance and innovation. Therefore, network embeddedness can be seen as an additional external dimension of sociotechnical systems applied to DT (Xing et al., 2023). This is because, beyond emphasizing the importance of organizations being embedded in social networks, this perspective expands the understanding of DT by incorporating the external environment as a sociotechnical dimension (Marcon et al., 2022; Xing et al., 2023; Marcon et al., 2025). However, the literature has not yet explored how the degree of network embeddedness can support digital transformation in the healthcare context, especially when viewed as a socio-technical system.

In this context, the structuring of H4.0 in health systems requires an appropriate environment to support its implementation, as highlighted by the sociotechnical literature (Marcon et al., 2022; Xing et al., 2023; Marcon et al., 2024), which also emphasizes the importance of external context and participation in the digitalization process. As noted in the previous section, from a technical standpoint, this includes the dimension of healthcare smart operations; from the social perspective, healthcare smart working; and from the organizational side, the alignment between digital strategy and digital resources. However, it is evident that the external environment of healthcare systems still needs to be incorporated to fully realize the concept of a sociotechnical system—integrating social, technical, organizational, and environmental aspects (Marcon et al., 2025; 2024). In this regard, network embeddedness can serve as a reliable approach to acting as the environmental dimension of H4.0, as it can facilitate the alignment between strategy and digital resources by enabling healthcare systems to access

essential digital information and resources—through direct or indirect connections with various network actors—to support the implementation of the smart dimensions of H4.0.

3.3 HYPOTHESES DEVELOPMENT

3.3.1. Healthcare digital strategies and the development of Healthcare 4.0 dimensions

The implementation of digital technologies demands a strategic definition to set goals and priorities of which technologies should be applied and how they must support processes and people (Enrique et al., 2023). From an organizational perspective, a digital strategy needs to establish digital goals for the organizational processes that will be changed with the support of digital technologies (Vial, 2019). When a sociotechnical system is aimed to be supported by digital technologies, the consolidation of this digital environment with clear rules will become the organizational driver of the digitalization process (Marcon et al., 2022). Without a clear digitalization strategy, technologies tend to be implemented to solve local problems, but at the risk of missing the big organizational picture that addresses the systemic problems through digital solutions (Baiyere et al., 2020).

When it comes to H4.0, the development of a digital strategy should be a priority for the implementation of both healthcare smart operations and smart working. First, smart operations consider digital technologies applied to healthcare processes, but which processes are there? Of course, we can refer to the whole healthcare system and its processes, but these are too many and too complex to consider ideally a complete automated system. Healthcare is a sociotechnical organization especially because it is embedded in human-related activities that are performed based on intensive use of high-skilled knowledge (Beane et al., 2019). Therefore, it cannot be simply automated – the digital strategy would be the organizational roadmap to define priorities and needs regarding the technologies that should be implemented to address processes that are the aim of digitalization (Baptista et al., 2020; Fischer et al., 2020). Another aspect is the competition for resources: a digital strategy creates financial priorities to invest in digital technologies of the healthcare organization; otherwise, a corporate strategy without digitalization as a priority will dilute financial resources in many other demands (Piccoli et al., 2024). Furthermore, it is important to consider that the digital strategy can also have as an aim the benefit of healthcare workers, especially in healthcare systems where workers' resilience is necessary and technologies can play an important role in it (Tortorella et al., 2022c). A clear digital strategy can set workers as a keystone of the sociotechnical healthcare system and define

the development of smart working as a priority, as it also happens in industrial contexts (Marcon et al., 2022). Therefore, when the healthcare digital strategy is positively associated with both smart operations and smart working, it can be considered a strategic system of the organization that has a clear vision of creating a sociotechnical balance. We hypothesize this as follows:

H1: The implementation of an organizational digital strategy is positively associated with higher levels of adoption of healthcare 4.0 technologies

H1a: The implementation of a digital strategy is positively associated with higher levels of healthcare smart operations (use of technologies to automate healthcare processes)

H1b: The implementation of a digital strategy is positively associated with higher levels of healthcare smart working (use of technologies to support healthcare workers)

3.3.2. The alignment of healthcare digital strategy and resources and the role of network embeddedness

Verhoef et al. (2021) describe strategic imperatives of digital transformation, considering digital resources as an element to be combined with the strategy and the goal of the digitalization process. This line of research establishes that for a digital transformation strategy to be developed, it needs to consider digital resources as a central element of its digital technology architecture (Piccoli et al., 2024). It means that digital transformation is not just about defining what to digitalize (goals) and adopting the technologies (as considered in H1). Instead, a set of organizational digital resources needs to be established to provide support to the whole sociotechnical system. These digital resources are necessary to act as support for the digitalization process which tends to be challenging due to several organizational barriers that need to be overcome during the digitalization (Verhoef et al., 2021). In this sense, a strategic alignment is necessary between the digital strategy and the resources to manage the digitalization processes (Fischer et al., 2020). Some authors have considered the digital strategy as a bundle of goals, organizational structure, and resources (Verhoef et al., 2021; Fischer et al., 2020), but this may be a consequence of a qualitative analysis of how companies establish the strategy and organizational requirements. We follow a classic stream of information systems that differentiates between strategy and resources and consider strategy as the antecedent for the development of the resources (Kearns and Leder, 2000; Yeow et al., 2018). According to this view, both are independent but interrelated elements of organizational development (Yeow et al., 2018). Furthermore, these resources, which can comprise a vast type of elements including financial, human, and infrastructure factors (Verhoef et al., 2021), can be part of the

established internal digital transformation or can be acquired from external partners, as a make-or-buy condition (Parmigiani, 2007). This is more evident in the digital era, as organizations do not master the whole set of digital resources necessary to develop the digitalization process (Dalenogare et al., 2023). This is the case of healthcare systems, which are not essentially born-digital. Instead, healthcare organizations need to set new strategies for the digital transformation process (Denicolai and Previtali, 2023). Consequently, healthcare organizations will also need to build and internalize digital resources that are supported by the strategy and that can help to implement the digitalization goals (Denicolai and Previtali, 2023). Therefore, we establish the following hypothesis:

H2: The implementation of an organizational digital strategy is positively associated with higher levels of development of digital resources

H2a: The implementation of a digital strategy is positively associated with higher levels of internal digital resources

H2b: The implementation of a digital strategy is positively associated with higher levels of external digital resources

Network embeddedness suggests that the position an organization occupies within a network influences its access to information and resources (Lin, 2001). This occurs because such information and resources are transmitted or reinforced through a chain of interconnected nodes—where each node represents an external actor—and the proximity and degree of connection among these actors facilitate the receiving organization's ability to understand its strategy and focus on resource development (Granovetter, 1985; Mazzola et al., 2015). Network embeddedness requires organizations to actively locate, process, and interact with different actors in the network that are geographically close and can provide the required information and resources (Dahlke et al., 2024). Geographic proximity refers to the physical distance between organizations, which can encourage more frequent interactions and, consequently, foster the formation of stronger healthcare environments. Being physically and cognitively close to these sources can facilitate the understanding of the best pathways for digital transformation, helping to the formulation of more effective strategies aligned with the necessary resources for the digitalization of healthcare (Chen et al., 2025).

On the one hand, geographic proximity to external network actors can help an organization develop an outward-looking perspective, fostering learning from other actors and, based on that, formulating a digital strategy aimed at strengthening its internal digital resources.

This occurs because, by being embedded in a network, the organization has the opportunity to observe which digital resources are being used by other actors—or which are necessary to support digitalization—and based on this learning, develop and manage such resources internally (Miroshnychenko et al., 2021; Xing et al., 2023; Chen et al., 2025). For example, a healthcare organization may capture knowledge about the development of technological solutions from another organization, whether public or private, within the same network and apply that knowledge internally by training its professionals to create an innovative and customized solution (Miroshnychenko et al., 2021). Alternatively, it may establish partnerships with universities and research institutions to access information and experiences related to digital transformation.

On the other hand, proximity to external network actors can assist an organization (e.g., a hospital) in identifying the most suitable partners for full access to external resources, enabling effective strategies to obtain those resources (Granovetter, 1985). This is because network embeddedness demonstrates that an organization can observe the trust levels and potential shortcomings among its potential suppliers—allowing for better comparisons—evaluating competitive advantages and exercising greater flexibility in accessing information and resources (Granovetter, 1985). For example, by occupying a central position in a network, an organization may gain access to suppliers offering more innovative solutions within its geographic context (Chen et al., 2025); it may also engage in collaborative environments that enable mechanisms for knowledge transfer and digital technologies (Lyu et al., 2020; Chen et al., 2024); or even establish contact with healthcare operators (such as insurers, health plans, and provider networks), whose regional commercial demands can drive the development of new digital healthcare solutions.

Based on these arguments, network embeddedness plays a crucial role in the digital strategy-resources alignment, as it provides a source of information and resources which is important to develop the digital transformation of healthcare activities better. Thus, we propose the following hypothesis:

H3: The level of network embeddedness hospital can have in a healthcare ecosystem will positively moderate and strength the alignment between digital strategy and resources

H3a: The level of network embeddedness positively moderates the relationship between digital strategy and internal digital resources

H3b: The level of network embeddedness positively moderates the relationship between digital strategy and external digital resources

3.3.3. Contribution of digital resources for the development of Healthcare 4.0 dimensions

A good strategy-resource alignment should help to develop the organizational information system for digital transformation (Yeow et al., 2018). Prior research on digital transformation has defended the need for structuring also digital resources to enhance technological implementation in the digital transformation process (Liu et al., 2011; Yeow et al., 2018). In the H4.0 perspective, this means that digital resources, when aligned to the digital strategy should support the development of the sociotechnical H4.0 system. It is similar to what was previously argued by Marcon et al. (2022) in industrial systems, where the combination of organizational strategies and resources (sociotechnical environment) provided the basis for Industry 4.0 technologies implementation. In this same vein, digital resources should provide the basis for the development of smart operations and working technologies. For instance, digital resources include the organizational culture and informational structure necessary to implement digital technologies in processes and change the form such processes are executed (Fischer et al., 2020; Marcon et al., 2022; Gillani et al., 2024), as proposed for healthcare smart operations. Also, digital resources include skills and capabilities necessary to support people, when they need to learn how to use new digital technologies for their task execution, as considered in healthcare smart working (Verhoef et al., 2021; Fernanez-Vidal et al., 2022). Additionally, digital resources can be managed internally and externally, as these are totally different sources and can require different strategic management approaches to deal with (Dalenogare et al., 2023). Consequently, they can have different contributions to smart operations and smart working. External digital resources can consider the orchestration of knowledge and new skills through collaboration that can change the way business processes are executed, affecting smart operations. It also can create new mindsets in workers helping to shape the way technologies are used in healthcare activities (Fischer et al., 2020). On the other hand, internal digital resources can help consolidate systems legacy and information systems teams and knowledge that can provide a basis for automating processes through digital technologies (Liu et al., 2011) and for expanding workers skills to the use of new digital technologies (Verhoef et al., 2021). Consequently, these different arguments point to a mediating role of healthcare digital internal and external resources in the strategic alignment for the development of an H4.0 sociotechnical system. We summarize this in the following hypotheses H4 (for internal resources) and H5 (for external resources):

H4: The development of internal healthcare digital resources is positively associated with higher levels of adoption of healthcare 4.0 technologies

H4a: The development of internal healthcare digital resources is positively associated with higher levels of healthcare smart operations (use of technologies to automate healthcare processes)

H4b: The development of internal healthcare digital resources is positively associated with higher levels of healthcare smart working (use of technologies to support healthcare workers)

H5: The development of external healthcare digital resources is positively associated with higher levels of adoption of healthcare 4.0 technologies

H5a: The development of external healthcare digital resources is positively associated with higher levels of healthcare smart operations (use of technologies to automate healthcare processes)

H5b: The development of external healthcare digital resources is positively associated with higher levels of healthcare smart working (use of technologies to support healthcare workers)

All the hypotheses presented in this section are summarized in the conceptual model of Figure 8. This figure represents, in its first part, the strategic alignment between healthcare digital strategy and the management of healthcare digital resources (internal and external) and the moderating role of healthcare network embeddedness. Then, in its second part, Figure 8 describes the mediating role of healthcare digital resources between healthcare digital strategy and the dimensions of Healthcare 4.0, namely smart operations and smart working.

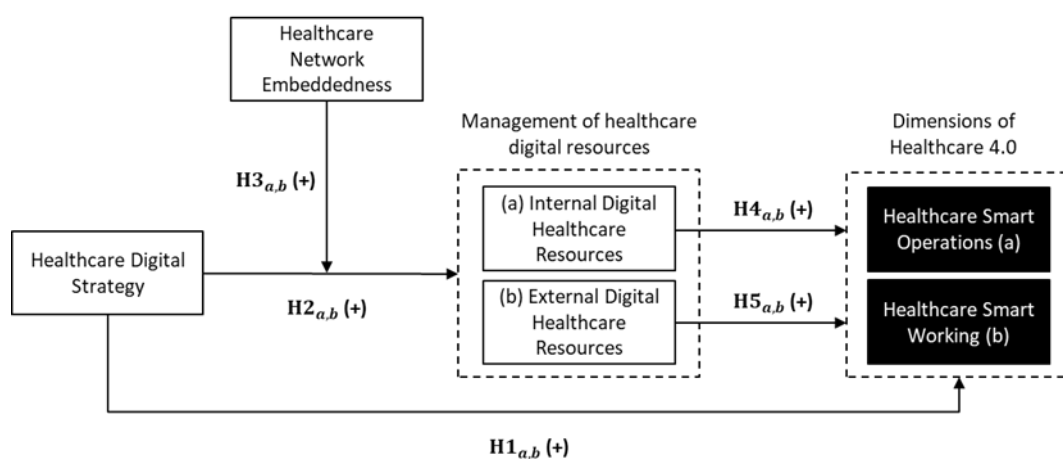


Figure 8. Conceptual research model

3.4 RESEARCH METHOD

3.4.1 Sampling

We focused our research unit on hospital management. Therefore, our target respondents are middle and senior-level healthcare leaders (directors, managers, or supervisors) who deal with healthcare technologies for hospital management. Also, we focused on internal elements of the hospitals, like the level of development of digital strategies and resources and the level of adoption of digital technologies. Therefore, our data source was the single respondent, as secondary data is not available for this research aim and multiple respondents would not be able to respond to issues related to digital transformation, which are usually under the coordination of technology/operations managers (Frank et al., 2019). This should be less biased than selecting experts from specific departments to analyze the system (Ketokivi, 2019). We defined the population for our sample based on the Annual Report on the Hospital Scenario in Brazil, developed by the Brazilian Hospital Federation (FBH). To mitigate the socioeconomic effect, we restricted the sample of our study to Brazilian hospitals. Although this criterion may limit the generalization of our findings, it strengthens the internal validity of our results. The population consists of 7,242 hospitals registered in the database of the National Registry of Health Establishments (CNES), DataSUS, and the Ministry of Health (MS). Considering a 5% error and a 90% confidence interval, it was determined that the minimum sample size needed to research hospitals and achieve national representativeness is 263. From this population, the initial sample available with the target contacts consisted of 3,823 hospitals from all regions of Brazil.

Data collection was carried out through three channels. Firstly, the survey was emailed to as many hospitals as possible identified in the sample. Secondly, a search for the sampled hospitals was conducted on the LinkedIn platform. Thirdly, the survey was manually delivered in printed forms. Three waves of the questionnaire were sent over three months. As a result, 316 initial responses were obtained; after excluding incomplete and duplicate questionnaires, we obtained 303 considered valid (response rate of 8%) for this research. The final sample size can be considered representative of Brazilian hospitals, as it exceeds the minimum requirement of 263 (90% confidence and 5% error), representing the proportions of the population. Table 5 presents the study sample profile.

Table 5. Sample composition (n=303 hospitals)

Number of beds	(%)	Respondent's role	(%)
≤150	100 33%	Middle level (e.g., supervisors)	249 82.18%
>150	203 67%	Senior level (e.g., manager or director)	54 17.82%
Number of employees		Age Hospital	
1 - 100	12 3.96%	<20	12 4.0%
101 - 200	14 4.62%	20-50	150 49.50%
201 - 500	110 36.30%	>50	141 46.50%
501 - 1000	115 37.95%	Hospital's profile	
1001 - 3000	36 11.88%	Teaching	33 10.89%
> 3000	16 5.28%	Public	110 36.30%
		Private	160 52.81%

3.4.2 Survey instrumentation

Our research instrument was developed based on constructs from the literature and adapted to the specific objectives of our study. The proposed instrument consists of six blocks of questions (see appendix A): (i) the level of implementation of healthcare digital strategies (STRATEGY), adapted from Nasiri et al. (2020) and Verhoef et al. (2021); (ii) the level of ownership of Internal Digital Healthcare Resources (INTERNAL), adapted from Barney (1986, 1991) and Franco et al. (2024); (iii) the level of outsourcing of External Digital Healthcare Resources (EXTERNAL), adapted from Dyer and Singh (1998, 2018), Franco et al. (2024); (iv) the level of healthcare network embeddedness (EMBEDDEDNESS), adapted from Xing et al. (2023), Chen et al. (2024), Chen et al. (2025); (v) the level of implementation of H4.0 digital technologies in healthcare management activities, proposed by Frank et al. (2019), Dornelles et al. (2022), Tortorella et al. (2022a; 2022b) and Deloitte (2020); (vi) information about the interviewees and their hospitals was collected to determine the demographic profile of the sample. Respondent information (e.g., role) was used to verify the legitimacy of their opinions and ensure a broader understanding of the investigated phenomenon. Meanwhile, hospital information allowed for controlling some contextual characteristics (e.g., number of beds, number of employees, hospital age in operation years, and ownership) that may influence the healthcare digital transformation process (Raimo et al., 2023).

To ensure the feasibility of survey administration, the instrument items were assessed and pre-tested by fifteen randomly selected experts (three academics and twelve healthcare practitioners) to ensure the face and content validity of the scale items (Taherdoost, 2016). The experts recommended minor terminological changes to enhance the clarity and comprehensibility of the survey items. The questions considered in this study were measured

on a five-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree) for evaluating blocks (i), (ii), and (iii), and from 1 (never) to 5 (very frequently) to assess the usage of digital technologies in sub-block (iv).

3.4.3 Constructs definition and validation

H4.0 digital technologies are typically organized into bundles of applications representing formative constructs (e.g., Tortorella et al., 2022a). We collected the information regarding the technologies usage through the questionnaire without using predefined bundles, as they were determined after through the use of Exploratory Factor Analysis (EFA). We conducted the EFA using orthogonal Varimax rotation to minimize potential ambiguities in the distribution of variables (Hair et al., 2018). We selected those factors with eigenvalues >1 (Hair et al., 2018), representing 67.953% of the total variance in responses, as shown in Table 6. We also assessed parameters validating the use of EFA for our selected dataset. These parameters include the Kaiser-Meyer-Olkin (KMO) test (which should be $> .500$), Bartlett's sphericity test (which should be $< .05$), the anti-image matrix for Measures of Sampling Adequacy (MSA) (which should be $> .500$), and the percentage of explained variance (which should be $> 60\%$). All our obtained parameters were in line with the suggested thresholds in the literature (KMO = 0.930; Bartlett < 0.000 ; MSA ≥ 0.879 ; % of variance = 67.953%) (Hair et al., 2018).

Analyzing the results of the factor loadings in each component, we labeled the two digital application bundles based on the composition of technologies and their predominant focus, following a procedure similar to previous literature (Tortorella et al., 2022b). As a result: Healthcare Smart Working (WORK) and Healthcare Smart Operations (OPERATIONS). Table 6 also displays Cronbach's alpha coefficients for these constructs obtained from the EFA, confirming reliability with values exceeding 0.9. Subsequently, we used the factorial scores of the two main factors obtained through the Thurstone method. More specifically, standardized scores were used in the subsequent linear regression analysis.

Table 6. Exploratory Factor Analysis for digital technologies

Variables	WORK	OPERATIONS	Comm.
Big Data/analytics	<u>0.742</u>	0.448	0.752
Artificial intelligence/machine learning	<u>0.763</u>	0.420	0.759
Blockchain	<u>0.828</u>	0.169	0.714
Augmented and virtual reality	<u>0.717</u>	0.206	0.557
Collaborative robotics (e.g., surgical robots, cobots, outpatient care, AGVs, AMRs, etc.)	<u>0.649</u>	0.448	0.622
Biomedical/digital wireless sensors	<u>0.746</u>	0.259	0.623
Digital patient relationship applications	<u>0.630</u>	0.433	0.584
Cloud Computing	0.539	<u>0.621</u>	0.677
Mobile apps	0.456	<u>0.619</u>	0.590
Corporate management systems (e.g., ERP - Enterprise Resource Planning)	0.250	<u>0.844</u>	0.774
Electronic health records (e.g., electronic prescription, electronic medical record)	0.146	<u>0.871</u>	0.780
Digital traceability (e.g., RFID, QR Code, etc.)	0.371	<u>0.790</u>	0.761
Simulation to support activities	0.502	<u>0.646</u>	0.669
Telemedicine	0.540	<u>0.599</u>	0.651
Extraction sums of squared loadings	8.355	1.158	
% of variance explained	59.682	8.271	
% of variance explained (cumulative)	59.682	67.953	
Rotation sums of squared loadings	4.971	4.543	
% of variance explained	35.504	32.449	
% of variance explained (cumulative)	35.504	67.953	
Cronbach's Alpha	0.906	0.918	

For the evaluation of healthcare digital strategy (STRATEGY) and healthcare digital resources (INTERNAL and EXTERNAL) and healthcare network embeddedness (EMBEDDEDNESS), we conducted a Confirmatory Factor Analysis (CFA) on constructs adapted from the literature. As indicated by the Comparative Fit Index (CFI), Tucker-Lewis Index (TLI), Root Mean Square Error of Approximation (RMSEA), Average Variance Extracted (AVE), Composite Reliability (CR), and Cronbach's Alpha, the analyzed constructs exhibited a good fit. All these values were considered acceptable and most exceeded predefined thresholds (Hair et al., 2018), as presented in Table 7. Additionally, the items demonstrated high factor loadings for each construct (Appendix A). We also ensured discriminant validity by applying the criteria of Fornell and Larcker (1981), which involve assessing correlation values with the square root of the AVE for each construct. Through this evaluation, we confirmed that the constructs built through the CFA procedure met the criteria proposed by Fornell and Larcker (1981). Additionally, we analyzed the means, standard deviations, and correlations between our constructs and the control variables incorporated into the model. We considered four control variables based on previous studies (Tortorella et al., 2022b; Raimo et al., 2023), namely: number of beds, number of employees, and two dummy variables, the first indicating whether the hospital is teaching, and the second indicating whether the hospital is public. Bed and

employee data were logarithmically transformed for inclusion in our regression model. Table 8 presents the descriptive statistics, Fornell-Larcker criteria, and bivariate correlations.

Table 7. Results of confirmatory factor analysis

Construct	CFI	TLI	RMSEA	AVE	CR	Cronbach's α
[STRATEGY]	0,990	0,983	0,070	0,685	0,928	0,928
[INTERNAL]	0,995	0,989	0,043	0,506	0,835	0,831
[EXTERNAL]	0,968	0,947	0,080	0,444	0,826	0,823
[EMBEDDEDNESS]	0,987	0,979	0,072	0,637	0,913	0,913

Table 8. Bivariate correlations

Variables	1	2	3	4	5	6	7	8	9	10
1 Ln_beds	-									
2 Ln_employees	0.849**	-								
3 Dummy_teaching	0.183**	0.262**	-							
4 Dummy_public	0.189**	0.156**	-0.082	-						
5 STRATEGY	0.007	0.077	0.193**	-0.579**	.830					
6 INTERNAL	0.013	0.024	0.189**	-0.461**	0.676**	.711				
7 EXTERNAL	0.054	0.074	0.100	-0.174**	0.358**	0.529**	.667			
8 EMBEDDEDNESS	0.147*	0.185**	0.140*	-0.260**	0.556**	0.508**	0.386**	.800		
9 WORK	-0.087	-0.138*	0.188**	-0.232**	0.295**	0.430**	0.360**	0.378**	-	
10 OPERATIONS	0.148*	0.261**	0.218**	-0.515**	0.628**	0.491**	0.269**	0.526**	0.000	-
Mean	5.347	6.423	0.116	0.396	3.448	2.807	2.937	3.049	0	0
S.D.	0.701	0.881	0.320	0.490	1.025	0.895	0.745	1.008	1	1

OBS1: ** $p < .01$; * $p < .05$. OBS2: Bold numbers in the diagonal report Fornell-Lacker criteria for discriminant validity.

3.4.4 Response bias and common method variance

To mitigate common method variance (CMV), we conducted a pre-test of the instrument's questions with a randomized subsample drawn from our target population. For this purpose, we pretested the questionnaire to ensure the clarity of our research instrument (as explained in Section 3.2). Additionally, we randomized the order of variables and questionnaire blocks to enhance their reliability and prevent order bias and primacy bias (Podsakoff et al., 2003). As pointed out by Ketokivi (2019), in surveys involving questionnaires, it is impossible to eliminate bias in the research. However, we implemented reasonable countermeasures to address this concern in our study.

3.4.5 Data analysis

To test our hypotheses, we conducted a series of hierarchical ordinary least squares (OLS) regressions. Before the regression analysis, we standardized the variables using a mean-centering Z-score. We employed a bootstrap approach based on the ‘Process’ macro proposed by Hayes (2017) to check for indirect mediation effects. To test the moderation effects (H3a and H3b), we standardized both the independent and moderating variables using mean-centering Z-scores and multiplied the moderator by the independent variable. This allowed us to create a multiplicative score to assess the interaction effect.

Before focusing on the interpretation of the results, we conducted a series of post hoc analyses to verify if all the conditions for OLS regression were met. Firstly, we assessed the normality condition required for OLS regression, employing three strategies based on the analysis of residual normality: we examined the density plot (histogram), the standardized normal probability (P-P) plot, and the quantile plot of the variable against the quantiles of a normal distribution. All these plots indicated a normal distribution of residuals for our OLS regression analysis. No model rejected the hypothesis of adherence to the normal distribution of residuals (Kolmogorov-Smirnov test p-value <0.01). Secondly, we checked linearity through partial regressions for the independent variables. Thirdly, we also assessed the homoscedasticity requirements for OLS using visual analysis with standardized residuals plotted against predicted values. Lastly, we examined the potential issue of multicollinearity in the OLS model through the Variance Inflation Factor (VIF). The results showed low multicollinearity, indicating a low VIF (< 3.847) for all independent variables, well below the threshold of 10 (Hair et al., 2018). Additionally, we examined the potential correlation of standardized residuals with independent variables and found no pattern suggesting multicollinearity.

3.5 RESULTS

The results of the hierarchical regression are presented in Table 9. In the first main stage, hierarchical regression models were executed in four steps. In the first step, we included only the control variables. In the second step, we included STRATEGY in both sources of digital healthcare resources (INTERNAL and EXTERNAL). In the third step, we add EMBEDDEDNESS, and in the fourth step, we test the moderation effect of EMBEDDEDNESS×STRATEGY on both sources of digital healthcare resources (INTERNAL

and EXTERNAL). In the second main stage, we tested the direct effect of STRATEGY on WORK and OPERATIONS (initially including only the control variables and then including the independent variable STRATEGY). Finally, in the third main stage, hierarchical regression models were executed, including the independent variable STRATEGY and the mediator's variables INTERNAL and EXTERNAL.

As shown in Table 9, the final model (3rd main stage) was statistically significant in both WORK ($F = 15.263$; $p < 0.01$) and OPERATIONS ($F = 43.986$; $p < 0.01$), explaining 24.8% and 49.9% of the variance (adjusted R^2), respectively. For the first main stage, regarding STRATEGY, our results showed a statistical association with INTERNAL ($B = 0.611$; $p < 0.000$) (H2a is supported) and EXTERNAL ($B = 0.375$; $p < 0.000$) (H2b is supported). The moderation model for EMBEDDEDNESS between STRATEGY and INTERNAL ($F = 42.537$; $p < 0.01$) explains 49.1% of the variance (adjusted $R^2 = 0.491$), but we found no statistical evidence supporting this moderation (H3a is not supported). On the other hand, the moderation model for EMBEDDEDNESS between STRATEGY and EXTERNAL ($F = 12.468$; $p < 0.01$) explains 21.0% of the variance (adjusted $R^2 = 0.210$). We observed a positive moderation effect between STRATEGY and EMBEDDEDNESS ($B = 0.231$, $p < 0.000$) (H3b is supported). To better understand the effect of this moderation, we plotted the slope in Figure 9, which shows the positive effect of STRATEGY on EXTERNAL when the level of EMBEDDEDNESS is high in hospitals. In other words, when a hospital has a high level of network embeddedness within a healthcare ecosystem, it strengthens the alignment between digital strategy and digital resources, and this effect is particularly positive in the relationship between digital strategy and external digital healthcare resources.

For the second main stage, the direct effect of STRATEGY was statistically significant for WORK ($B = 0.258$; $p < 0.01$) (H1a is supported) and OPERATIONS ($B = 0.416$; $p < 0.000$) (H1b is supported). Finally, for the third main stage, the results indicate a complete mediation effect only in the relationship between STRATEGY and WORK ($F = 15.263$; $p < 0.01$). Only the INTERNAL and EXTERNAL digital healthcare resources become the only statistically significant variables in the model.

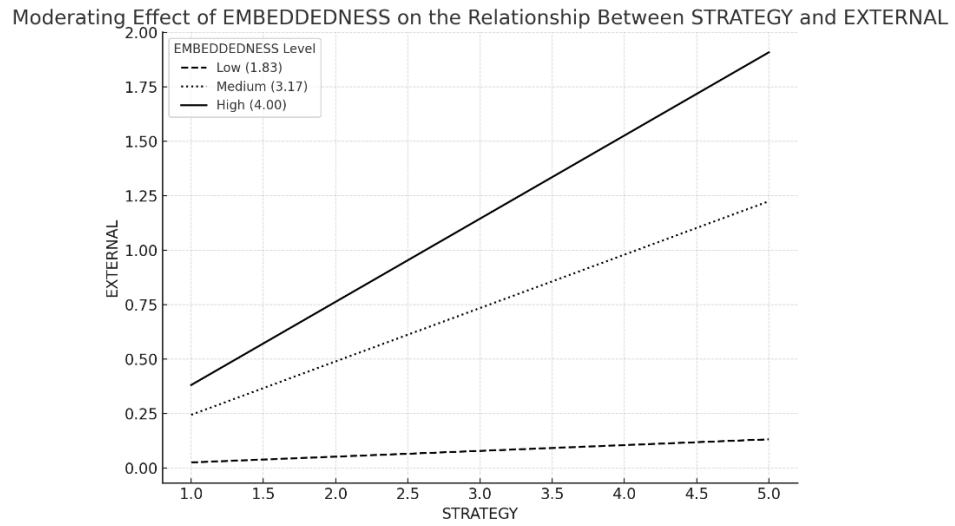


Figure 9. Slope for the moderating role of healthcare network embeddedness (EMBEDDEDNESS) on STRATEGY vs EXTERNAL.

Table 9. Regression results for OLS direct effects (a)

Variables	1st main stage								2nd main stage				3rd main stage	
	INTERNAL				EXTERNAL				WORK		OPERATIONS		WORK	OPERATIONS
	Model 1	Model 2	Model 3	Model 4	Model 1	Model 2	Model 3	Model 4	Model 1	Model 2	Model 1	Model 2	Model 1	Model 1
Dummy_hospital_research	0.434**	0.233	0.233	0.234	0.196	0.073	0.073	0.041	0.693**	0.608**	0.256	0.119	0.529**	0.096
Dummy_hospital_public	-0.948**	-0.214*	-0.235*	-0.235*	-0.373**	0.076	0.049	0.033	-0.404**	-0.094	-0.551**	-0.625**	-0.052	-0.606**
Ln_beds	0.089	0.137*	0.121	0.121	0.012	0.042	0.021	0.070	0.180	0.200*	-0.142	-0.110	0.154	-0.123
Ln_employees	-0.016	-0.143	-0.157*	-0.157*	0.076	-0.002	-0.020	-0.072	-0.318**	-0.371**	0.447**	0.360**	-0.332**	0.373**
STRATEGY		0.611**	0.493**	0.493**		0.375**	0.219**	0.326**		0.258**		0.416**	0.014	0.352**
INTERNAL													0.273**	0.094
EXTERNAL													0.206**	0.018
EMBEDDEDNESS			0.205**	0.205**			0.268**	0.258**						
EMBEDDEDNESS x STRATEGY				-0.001				0.231**						
F-test	23.685**	53.606**	49.795**	42.537**	3.462**	9.026**	10.818**	12.468**	9.757**	11.092**	49.014**	60.556**	15.263**	43.986**
R ²	0.241	0.474	0.502	0.502	0.044	0.132	0.180	0.228	0.116	0.157	0.397	0.505	0.266	0.511
Adjusted R ²	0.231	0.466	0.492	0.491	0.032	0.117	0.163	0.210	0.104	0.143	0.389	0.496	0.248	0.499
Change in R ²	0.241**	0.233**	0.028**	0.000	0.044**	0.088**	0.048**	0.048**	0.116**	0.042**	0.397**	0.108**	0.266**	0.511**

n = 303, **p < 0.01; *p < 0.05

(a) Unstandardized beta coefficients are reported, since the main variables were standardized before regression.

However, to assess mediation, indirect effects must be calculated (Preacher & Hayes, 2008). Therefore, we employed a procedure proposed by Hayes (2017), called PROCESS, to evaluate the mediation effects of our model. PROCESS involves a bootstrapping procedure to examine conditional indirect effects, making it more robust when compared to the Sobel z-test for testing mediation effects (Zhao et al., 2010). To operationalize PROCESS, as suggested by Preacher and Hayes (2008), we configured 5,000 bootstrap samples. Table 10 presents the direct effects, indirect effects (including standard deviation, significance level, and corresponding lower (LLCI) and upper (ULCI) confidence intervals), the total effect (which is given by the sum of direct and indirect effects), and the conclusion.

For OPERATIONS, the direct effect of STRATEGY was statistically significant (effect = 0.5446; $p < 0.001$). However, regarding the indirect effects, the bootstrap results indicated that for both INTERNAL and EXTERNAL, the values fall outside the 95% confidence interval. In this case, the hypothesis of mediation for each digital resource source is rejected, as the value "0" is within the interval (LLCI and ULCI), indicating that the indirect effect may not exist. However, this situation requires further investigation for more conclusive results. For WORK, the direct effect of STRATEGY was not statistically significant (effect = 0.0084; $p = 0.9041$). Regarding the indirect effects, the bootstrap results indicated significance for both INTERNAL and EXTERNAL, as both are within the 95% confidence interval, suggesting complete mediation for WORK.

Table 10. Bootstrapping results of the indirect effects

Mediation path	Direct effect		Indirect effect				Total Effect		Conclusion
	Effect	Sig.	Effect	SD	LLCI (95%)	ULCI (95%)	Effect	Sig.	
STRATEGY->INTERNAL->WORK	0.0084	0.9041	0.2205	0.0582	0.1113	0.34621	0.2951	0.0000	Complete
STRATEGY->EXTERNAL->WORK			0.0662	0.0240	0.0155	0.1108			Complete
STRATEGY->INTERNAL->OPERATIONS	0.5446	0.0000	0.0784	0.0464	-0.0107	0.1709	0.6278	0.0000	No mediation
STRATEGY->EXTERNAL->OPERATIONS			0.0048	0.0186	-0.0311	0.0422			No mediation

3.5.1 Power analysis

Based on the regression results reported in Table 9, we conducted a power analysis of the regression models using Cohen's f^2 estimation (Cohen et al., 2003). We also calculated the population effect size of R^2 using Cohen's f^2 estimation (Cohen et al., 2003) from the two final models in the last stage of our OLS hierarchical procedure. For the WORK, we obtained $f^2 =$

0.362, representing a statistical power of ≈ 1.00 at $\alpha = 0.01$. For the OPERATIONS, we obtained $f^2 = 1.045$, indicating a statistical power of ≈ 1.00 at $\alpha = 0.01$. Therefore, the result of the effect size for each model is considered an average effect for the explanatory variables (Cohen et al., 2003).

3.5.2 Robustness checks

We assessed the robustness of our model by applying various strategies to ensure the stability and consistency of the results. Following the approach of previous studies (e.g., Lerman et al. (2022) and Franco et al., 2023), we established an interactive protocol. First, we removed control variables and checked whether the results remained stable. The removal of control variables demonstrated that predictor variables were not artifacts of control variables. Second, we conducted an individual predictor analysis. Lastly, in the third step, we tested a competitive model, considering the healthcare digital strategy as a moderator between healthcare digital resources and the implementation of smart healthcare dimensions. However, we did not find statistical support for this competitive model. Overall, we found stable results, as there were no significant changes in the coefficients of all models altered in our hierarchical regression. All these strategies indicated a stable behavior of our proposed model.

3.6 DISCUSSIONS AND CONCLUSIONS

Our results demonstrate that H4.0 encompasses a sociotechnical view through two smart dimensions, which we refer to as healthcare smart operations and healthcare smart working. The former focuses on introducing technologies for task execution, aiming to make them more automated, especially those that are inherently mechanical. The second dimension focuses on workers, who are the actors responsible for task execution but are supported by the introduction of digital technologies, enabling an upskilling effect, which means enhancing worker competence (Dornelles et al., 2023). Both dimensions can be applied across hospital units. However, despite the digital technologies addressed in this study focusing on the first or second dimension, some of them can simultaneously encompass both. A potential explanation for this is the flexibility of application and the impact that technologies can have on different areas, such as telemedicine. Although our research suggests that telemedicine falls within the healthcare smart operations dimension, in practice, it also plays a significant role in the social context, as previously shown by Cannavacciuolo et al. (2023). Telemedicine is a technology that supports healthcare workers in providing real-time and virtual care, primarily in emergency

situations, allowing for the acceleration of clinical procedures. Furthermore, it benefits patients by reducing waiting times for care or the need to travel to a closer clinic. Therefore, our results confirm that smart healthcare dimensions have distinct focuses, one on processes/technical and the other on people/social, configuring a sociotechnical system. More importantly, our results add more details, showing that these dimensions should be present and can integrate into an H4.0 system. These findings elucidate that H4.0 is not only about the introduction of digital technologies applied to processes, as is typically presented in the literature (e.g., Tortorella et al., 2022b; Raimo et al., 2023) but also should be evaluated from the perspective of people (Grover et al., 2018; Roscoe et al., 2019). Otherwise, hospital environments may face challenges and struggle to succeed with the adoption of digital technologies, as highlighted by Marcon et al. (2022).

The results indicate that for both dimensions of H4.0 to work within a hospital organizational environment, the entire organizational structure is necessary to support them. The H4.0 literature has sometimes presented a list of digital technologies that impact operations (Tortorella et al., 2022a; Tortorella et al., 2022b). However, implementation requires organizational support for digital technologies to succeed, especially for the healthcare smart working dimension, which is directly related to people (Kleiner, 2008; Baptista et al., 2020; Klein et al., 2024). Our results show that both healthcare smart operations and healthcare smart working need alignment with the digital strategic structure for implementation (thus supporting H1). Hospitals that structure their digital strategy have higher levels of H4.0 development. The digital strategy role in H4.0 implementation highlights that IT strategy and business strategy for digital transformation are essential to conduct the process of consolidating a sociotechnical system (Chanas et al., 2019; Fischer et al., 2020). Furthermore, our findings show that digital strategy is an antecedent of the definition of digital resources, as hypothesized in H2. Prior research considered digital strategy and resources as a bundle of the strategic initiatives of digital transformation (Fischer et al., 2020; Verhoef et al., 2021), but our findings point to a sequential relationship of strategic alignment in which digital strategy is defined and will support the definition of digital resources. Also, we show that both internal and external digital resources matter for the development of this strategic alignment. This reinforces the qualitative prior findings from Fernandez-Vidal et al. (2022) that highlighted the need for governance structures able to work with internal and external sources that can support the digital transformation due to its complexity and variety of requirements that need to be attended.

We examined whether hospitals' level of network embeddedness within the healthcare ecosystem moderates the alignment between digital strategy and digital resources. The results reveal a significant positive moderating effect for external digital resources (supporting Hypothesis H3b), but not for internal resources (contrary to Hypothesis H3a). This distinction suggests that healthcare digital strategies aimed at leveraging external resources—particularly to support people-oriented digital transformation (smart working)—require strong ties to external actors. Although the digital transformation literature has emphasized the importance of external resources for digitalization (Franco et al., 2024), our findings indicate that, in the hospital context, this alignment is only effective when network embeddedness is high. That is, the ability to access and mobilize external resources depends not only on strategic intent but also on the hospital's level of connectivity within the broader healthcare network—especially in a sector where digital solutions are highly customized (Bavafa et al., 2021). In contrast, our results show no moderating effect of network embeddedness on the alignment between strategy and internal digital resources. While we initially expected that proximity to external actors would support internal digitalization through knowledge diffusion and learning, the findings suggest otherwise. Internal resources appear to be more directly shaped by the hospital's internal strategic direction, without strong dependence on external connections. Given the complexity of healthcare systems (Stevens and Schaik, 2020) and the context-specific nature of internal capabilities (Mukherjee and Sinha, 2020), these resources may be inherently less susceptible to external influence. As a practical implication, hospitals aiming to develop smart working infrastructures should consider their position within healthcare networks. Those with high embeddedness can pursue strategies that leverage external digital resources. However, hospitals with low embeddedness may benefit more from focusing on internally driven digital resource development, thereby reducing uncertainties linked to limited external support. It should be emphasized that, in our analysis, we chose not to examine the moderating effect of network embeddedness on the relationship between digital strategy and the H4.0 smart dimensions because we understand that network embeddedness acts as a factor that enhances the relationship between strategy and digital resources and that, when integrated, can facilitate the implementation of the H4.0 dimensions. Our focus was to analyze its conditional effect, considering that this moderating effect does not act solely on digital strategy in isolation, but rather relationally, shaping how strategy and digital resources combine to generate digital capabilities that enable the hospital to develop H4.0.

We also tested whether the alignment between healthcare digital strategy and healthcare digital resources indeed allows for the development of higher levels of healthcare smart operations and healthcare smart working, as hypothesized in H3. We identified that healthcare digital resources (internal or external) mediate the relationship between healthcare digital strategy and the healthcare smart working dimension (thus supporting H3a), but we did not find statistical evidence for this mediation between digital strategy and smart operations. One possible explanation for this is grounded in the sociotechnical view (Clegg, 2000). Mostly, resources are more focused on supporting workers than on processes, regardless of how these resources are managed, internally or externally. Digital resources should be orchestrated to provide the best possible conditions so that technologies can be well-employed by workers (Piccoli et al., 2024). These resources create the condition for workers to be capable of dealing with new technologies, developing new skills, and integrating better their empirical knowledge of healthcare operations with the technology usage (Fischer et al., 2020; Gillani et al., 2024). On the other hand, healthcare smart operations involve only the introduction of technologies to perform tasks, most of the time without the presence of workers, making the need for resources to support them less critical. Of course, digital resources would be relevant for technology acquisition and implementation (Liu et al., 2011; Yeow et al., 2018; Verhoef et al., 2021), but not for its adoption which is dependent on the daily use in healthcare activities. Moreover, another possible explanation for the absence of evidence regarding this mediation may be that internal or external resources are less impactful due to regulatory, organizational, or technical constraints related to DT focused on process automation, which tends to be more flexible. In this case, certain types of digital resources to support technologies are not necessarily decisive, since process automation does not always involve human aspects, which might primarily require compliance with specific regulatory constraints, such as user data protection, or organizational and technical requirements, such as training and capacity-building for technology handling. Therefore, our findings suggest that the adoption of technologies aimed at healthcare smart operations may not necessarily demand a large concentration of institutional resources and efforts, as regulatory restrictions or internal organizational changes may not be strong enough to prevent, hinder, or condition the implementation of technological projects.

Our results also highlight a critical aspect, often overlooked but essential in digital strategies for the implementation of technologies such as IoT, electronic health records, and artificial intelligence (healthcare smart operations): security risks and cyber threats (Schmitt, 2023; Marcon et al., 2025). In a digital, intelligent, and interconnected environment, the act of

automating processes requires the protection of sensitive data in the face of increasing threats (Schmitt, 2023). This concern is even more relevant in the healthcare sector, which is characterized by complex systems and non-standardized processes (Tortorella et al., 2020b; Stevens & Schaik, 2020; Uysal, 2022). Although our study does not delve deeply into this topic, the results reinforce the need to integrate cybersecurity structures — enabled by AI, for example, as proposed by Schmitt (2023) into digital strategies, to mitigate risks associated with the large volume of sensitive data (Aceto et al., 2020; Uysal, 2022) generated by both healthcare smart operations and healthcare smart working. Small vulnerabilities can lead to significant losses and compromise the confidentiality, integrity, and availability of organizational information, which requires robust strategic cybersecurity measures to protect both the organization, and the actors involved (Marcon et al., 2025). Therefore, there is a need to explore and develop cybersecurity resilience in the digital strategy dimension, a vital element for the security of healthcare organizations.

Finally, our findings align with the study by Edelman et al. (2005), which identified a positive association between the alignment of strategy and resources for performance improvement in organizations. They also resonate with the studies by Verhoef et al. (2021), presenting digital resources as a crucial strategic imperative for digital transformation. Furthermore, they are also related to the findings of a special edition presented by Baptista et al. (2020) on the strategic implications across various layers of technologies in the workplace and organizational structure. Gillani et al. (2024) also highlight the need for organizations to strategically navigate, including the establishment of an organizational structure as a facilitator for digital transformation. However, our results add a counterpoint to these studies by differentiating the role of resources in the strategic alignment, being different when it is associated with smart working or with smart operations.

3.6.1 Theoretical implications

Our study offers several contributions to existing literature and has various theoretical implications, as detailed below.

First, our results provide insights into organizational structures that may be associated with the implementation of digital transformation. In this regard, we extend previous literature that suggested considering strategic and management aspects in the adoption of digital technologies, both for processes and people. We empirically demonstrate these relationships in the context of healthcare. Additionally, existing literature has primarily focused on the

implementation of digital technologies or how digital transformation is associated with improvements in operational performance. We broaden this perspective by highlighting the implementation of healthcare smart working and healthcare smart operations as outcomes of sociotechnical initiatives. While some researchers have found positive relationships between organizational factors and performance, we can further expand the sociotechnical view to achieve higher levels of digital technology implementation by considering social elements.

Second, we offer a theoretical contribution by combining the alignment perspective of digital strategy with digital resource management. Digital strategy adopts an “integrated view” approach, where both information technology strategy and business strategy are considered. Therefore, digital strategies in healthcare settings must be identified, understood, and translated into social initiatives that can support the implementation of H4.0 dimensions. We contribute to the theory by demonstrating that digital resources can serve as an operational means between digital strategy and H4.0 implementation. In sum, an organizational structure that aligns between digital strategy and digital resources facilitates the implementation of digital technologies to support healthcare workers, enabling hospitals to adequately consider the sociotechnical aspects of digital transformation in healthcare. Thus, we expand existing research by highlighting that alignment between strategy and digital resources leads to higher levels of technological support for healthcare workers, adding the lens of the sociotechnical view.

Third, our study clarifies the role of organizational support in the digitization of healthcare environments. In this case, we demonstrate that this organizational support can emerge from the management of digital resources, enabling the provision of support for digital transformation. We show that even though digital resources may be managed in different ways, they can still be capable of providing this support for adoption and development, especially in the dimension of smart healthcare workers within H4.0. In other words, internally, hospitals can develop the necessary digital resources to support this process, just as hospitals can also outsource these resources since both the internal and external dimensions were positively associated with smart working. This represents possible pathways that reduce the pursuit of initiatives to support workers as an additional effort for their performance in carrying out activities. In this regard, an innovative finding is that organizational support through digital resources aligned with digital strategy can follow different management perspectives, considering not only the processes but also the people’s perspective for changes in healthcare professionals’ job roles. In sum, we encourage existing literature to place organizational support

in the context of digital transformation, as well as its implications and changes in work in the digital era.

3.6.2 Managerial implications

Our study has several implications for managerial practice related to sociotechnical, organizational, and digitization aspects in healthcare, which are presented as follows.

Our findings show hospitals can consider not only the technocentric perspective to integrate and automate processes but also the sociotechnical view that involves people in implementing H4.0 in their operations. So far, the focus has been predominantly on processes. Expanding the potential of the H4.0 concept, sociotechnical initiatives should also be considered by hospitals to assist healthcare workers, both in the requirements for implementing technologies and changes in job roles and in maintaining the activities they perform over time, supported by digital technologies. To achieve this, hospitals must establish a well-defined digital strategy and have a set of digital resources to support the operationalization of these technologies by workers. When hospitals seek to increase interoperability and improve tasks performed by professionals in a high-complexity context, implementing H4.0 dimensions can help capture real-time information from both processes and patients to support decision-making, as well as worker support. Similarly, patients and society, among others, can be included in this context, allowing the integration of all stakeholders.

Secondly, organizational structure can play a significant role in mitigating potential obstacles to implementing H4.0. From a general perspective, our study recommends that healthcare managers structure the digital strategy to align with digital resources. This combination acts as an organizational pathway to implement digital technologies to support healthcare workers. Managers should also consider digital strategies that define operational targets for technology implementation, as digital strategy is also strongly related to digital technologies for healthcare smart operations. By considering these elements, managers can follow the path toward a sociotechnical organizational development of Healthcare 4.0.

Finally, our findings indicate that hospitals planning to implement smart working infrastructures should carefully assess their position within healthcare networks. Hospitals with strong external connections—such as partnerships with other institutions, technology providers, or public agencies—are better positioned to adopt strategies that rely on external digital resources. These networks can offer critical support and expertise during digital transformation efforts. In contrast, hospitals with limited embeddedness in such networks may benefit more

from focusing on internally driven resource development. This approach can reduce uncertainty and increase control over implementation, ensuring that smart working initiatives are grounded in the hospital's own capabilities.

3.6.3 Limitations and future research

This study presents some methodological limitations that point to opportunities for future research. Regarding the sample data of the study, all interviewees were from hospitals located in Brazil, a country classified as an emerging economy. This may limit the generalizability of our findings to different socioeconomic contexts. While we have identified significant and consistent relationships, we suggest that this study could be conducted in other socioeconomic contexts, involving hospitals from various locations. This would allow for an international comparative analysis that could contribute to a global understanding, aiding in identifying international differences and strategies for generating social value. Also, our constructs of digital resources evaluate the resources used by hospitals to support the digitization of their operational processes and workers' activities. However, we are limited to considering only which resources are appropriate for this support. As digital transformation progresses, other types of resources and/or workers' capabilities may become necessary to support the implementation and use of different new technologies. For example, the use of generative AI as a medical support tool is a promising field for future research. Therefore, identifying which new digital resources and capabilities are needed for this and other technologies can guide further future investigations.

Lastly, our constructs of digital strategy and digital resources allowed us to analyze the relationship between them in implementing the H4.0 smart dimensions. However, other sociotechnical aspects, such as worker learning analysis, technostress evaluations, and the development of new digital skills and capabilities, for example, could also be considered. This enables the development of new research. Additionally, future studies could also investigate the role of this alignment between digital strategy and digital resources in the performance of healthcare workers. Moreover, another limitation of our study concerns the scope of the questionnaire, which was restricted to organizational-level questions, without addressing individual-level aspects within the hospitals. For future research, it would be interesting to include additional questions specifically directed at healthcare professionals, exploring the complexity of their tasks, their digital experience, and information about their profiles. We also suggest grouping participants according to their respective hospital departments to deepen their understanding of the factors that influence the success of HCDDT.

Acknowledgements

The authors thank the National Association of Private Hospitals (ANAHp) for their assistance with data collection. The authors thank the Brazilian National Council for Scientific and Technological Development (CNPq – Productivity scholarship level PQ 1D) and the Research Coordination of the Brazilian Ministry of Education (CAPES – PhD scholarship) for the financial support received to conduct this research.

3.7 REFERENCES

- Ayala, N. F., Paslauskis, C. A., Ghezzi, A., & Frank, A. G. (2017). Knowledge sharing dynamics in service suppliers' involvement for servitization of manufacturing companies. *International Journal of Production Economics*, 193, 538–553. <https://doi.org/10.1016/j.ijpe.2017.08.019>
- Baiyere, A., Salmela, H., & Tapanainen, T. (2020). Digital transformation and the new logics of business process management. *European journal of information systems*, 29(3), 238–259. <https://doi.org/10.1080/0960085X.2020.1718007>
- Barney, J. (1991). Firm Resources and Sustained Competitive Advantage. *Journal of Management*, 17(1), 99–120. <https://doi.org/10.1177/014920639101700108>
- Barney, J. B. (1986). Organizational Culture: Can It Be a Source of Sustained Competitive Advantage? *The Academy of Management Review*, 11(3), 656. <https://doi.org/10.2307/258317>
- Baptista, J., Stein, M. K., Klein, S., Watson-Manheim, M. B., Lee, J. (2020). Digital work and organisational transformation: Emergent Digital/Human work configurations in modern organisations. *The Journal of Strategic Information Systems*, 29(2), 101618. <https://doi.org/10.1016/j.jsis.2020.101618>
- Basile, L. J., Carbonara, N., Panniello, U., Pellegrino, R. (2024). The role of big data analytics in improving the quality of healthcare services in the Italian context: The mediating role of risk management. *Technovation*, 133, 103010. <https://doi.org/10.1016/j.technovation.2024.103010>
- Bavafa, H., Savin, S., & Terwiesch, C. (2021). Customizing primary care delivery using E-visits. *Production and Operations Management*, 30(11), 4306–4327. <https://doi.org/10.1111/poms.13528>
- Beane, M. (2019). Shadow learning: Building robotic surgical skill when approved means fail. *Administrative Science Quarterly*, 64(1), 87–123. <https://doi.org/10.1177/0001839217751692>
- Cagliano, R., Canterino, F., Longoni, A., Bartezzaghi, E. (2019). The interplay between smart manufacturing technologies and work organization: The role of technological complexity. *International Journal of Operations & Production Management*, 39(6/7/8), 913–934. <https://doi.org/10.1108/IJOPM-01-2019-0093>
- Cannavacciuolo, L., Capaldo, G., Ponsiglione, C. (2023). Digital innovation and organizational changes in the healthcare sector: Multiple case studies of telemedicine project implementation. *Technovation*, 120, 102550. <https://doi.org/10.1016/j.technovation.2022.102550>
- Cerchione, R., Centobelli, P., Riccio, E., Abbate, S., Oropallo, E. (2023). Blockchain's coming to hospital to digitalize healthcare services: Designing a distributed electronic health record ecosystem. *Technovation*, 120, 102480, 2023.
- Chanas, S., Myers, M. D., & Hess, T. (2019). Digital transformation strategy making in pre-digital organizations: The case of a financial services provider. *Journal of Strategic Information Systems*, 28(1), 17–33. <https://doi.org/10.1016/j.jsis.2018.11.003>
- Chaudhuri, A., Naseraldin, H., & Narayanamurthy, G. (2023). Healthcare 3D printing service innovation: Resources and capabilities for value Co-creation. *Technovation*, 121, 102596. <https://doi.org/10.1016/j.technovation.2022.102596>
- Chen, Yihua, Yang, S., Zhang, Z., & Shen, H. (2024). How does digital enablement affect product customization? The roles of innovation capability and network embeddedness. *Technological Forecasting*

- and *Social Change*, 201(123272), 123272. <https://doi.org/10.1016/j.techfore.2024.123272>
- Chen, Y., Li, B., & Huo, B. (2025). Building operational resilience through digitalization: The roles of supply chain network position. *Technological Forecasting and Social Change*, 211(123918), 123918. <https://doi.org/10.1016/j.techfore.2024.123918>
- Clegg, C. W. (2000). Sociotechnical principles for system design. *Applied Ergonomics*, 31, 463-477. [https://doi.org/10.1016/S0003-6870\(00\)00009-0](https://doi.org/10.1016/S0003-6870(00)00009-0)
- Cohen, J., Cohen, P., West, S. G., Aiken, L.S. (2003). *Applied Multiple Regression/correlation Analysis for the Behavioral Sciences*, third ed. Routledge.
- Constantiou, I. D., & Kallinikos, J. (2015). New games, new rules: Big data and the changing context of strategy. *Journal of Information Technology*, 30(1), 44-57. <https://doi.org/10.1057/jit.2014.17>
- Dahlke, J., Beck, M., Kinne, J., Lenz, D., Dehghan, R., Wörter, M., & Ebersberger, B. (2024). Epidemic effects in the diffusion of emerging digital technologies: evidence from artificial intelligence adoption. *Research Policy*, 53(2), 104917. <https://doi.org/10.1016/j.respol.2023.104917>
- Dalenogare, L. S., Le Dain, M.-A., Ayala, N. F., Pezzotta, G., Frank, A. G. (2023). Building digital servitization ecosystems: An analysis of inter-firm collaboration types and social exchange mechanisms among actors. *Technovation*, 124, 102756. <https://doi.org/10.1016/j.technovation.2023.102756>
- Deloitte (2020). Digital Transformation: Shaping the Future of European Healthcare. Retrieved from <https://www2.deloitte.com/content/dam/Deloitte/uk/Documents/life-sciences-health-care/deloitte-uk-shaping-the-future-of-european-healthcare.pdf>
- Denicolai, S., Previtali, P. (2023). Innovation strategy and digital transformation execution in healthcare: The role of the general manager. *Technovation*, 121, 102555. <https://doi.org/10.1016/j.technovation.2022.102555>
- Gnyawali, D. R., Madhavan, R. (2001). Cooperative Networks and Competitive Dynamics: A Structural Embeddedness Perspective. *The Academy of Management Review*, 26(3), 431-445.
- Dornelles, J. de Assis, Ayala, N. F., Frank, A. G. (2023). Collaborative or substitutive robots? Effects on workers' skills in manufacturing activities. *International Journal of Production Research*, 61(22), 7922-7955. <https://doi.org/10.1080/00207543.2023.2240912>
- Dornelles, J. de Assis, Ayala, N. F., Frank, A. G. (2022). Smart Working in Industry 4.0: How digital technologies enhance manufacturing workers' activities. *Computers and Industrial Engineering*, 163, 107804. <https://doi.org/10.1016/j.cie.2021.107804>
- Dicuonzo, G.; Donofrio, F.; Fusco, A.; Shini, M. (2023). Healthcare system: Moving forward with artificial intelligence. *Technovation*, 120, 102510. <https://doi.org/10.1016/j.technovation.2022.102510>
- Dyer, J. H., & Singh, H. (1998). The Relational View: Cooperative Strategy and Sources of Interorganizational Competitive Advantage. *Academy of Management Review*, 23(4), 660-679. <https://doi.org/10.2307/259056>
- Dyer, J. H., Singh, H., & Hesterly, W. S. (2018). The relational view revisited: A dynamic perspective on value creation and value capture. *Strategic Management Journal*, 39(12), 3140-3162. <https://doi.org/10.1002/smj.2785>
- Edelman, L. F., Brush, C. G., Manolova, T. (2005). Co-alignment in the resource-performance relationship: strategy as mediator. *Journal of Business Venturing*, 20(3), 359-383. <https://doi.org/10.1016/j.jbusvent.2004.01.004>
- Elia, S., Giuffrida, M., Mariani, M. M., & Bresciani, S. (2021). Resources and digital export: An RBV perspective on the role of digital technologies and capabilities in cross-border e-commerce. *Journal of Business Research*, 132, 158-169. <https://doi.org/10.1016/j.jbusres.2021.04.010>
- Enrique, D. V., Marodin, G. A., Santos, F. B. C., Frank, A. G. (2023). Implementing industry 4.0 for flexibility, quality, and productivity improvement: technology arrangements for different purposes. *International Journal of Production Research*, 61(20), 7001-7026. <https://doi.org/10.1080/00207543.2022.2142689>
- FBH – BRAZILIAN HOSPITAL FEDERATION. (2023). Annual Report on the Hospital Scenario in Brazil 2023. Brasília: DF, [s.d.]. Available at: <https://indd.adobe.com/view/da2a8973-0675-4109-9aba-dc367df1f720>. Accessed 1st Jun 2023.
- Fernandez-Vidal, J., Perotti, A. F., Gonzalez, R., Gasco, J. (2022). Managing digital transformation: The view

- from the top. *Journal of Business Research*, 152, 29–41. <https://doi.org/10.1016/j.jbusres.2022.07.020>
- Fornell, C., Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal Marketing Research*, 18(1), 39. <https://doi.org/10.1177/002224378101800104>
- Franco, C. W., Benitez, G. B., Sousa, P. R. de, Kliemann Neto, F. J., & Frank, A. G. (2024). Managing resources for digital transformation in supply chain integration: The role of hybrid governance structures. *International Journal of Production Economics*, 278(109428), 109428. <https://doi.org/10.1016/j.ijpe.2024.109428>
- Franco, W. C., Benitez, G. B., Sousa, P. R., Neto, F. J. K., Frank, A. G. (2023). A contingency-configurational view of purchasing operations: The mediating role between supplier relationship and firm performance. *Journal of Purchasing and Supply Management*, 29, 100815. <https://doi.org/10.1016/j.pursup.2022.100815>
- Frank, A. G., Dalenogare, L. S., and Ayala, N. F. (2019). Industry 4.0 technologies: Implementation patterns in manufacturing companies. *International Journal of Production Economics*, 210, 15-26. <https://doi.org/10.1016/j.ijpe.2019.01.004>
- Fischer, M., Imgrund, F., Janiesch, C., Winkelmann, A. (2020). Strategy archetypes for digital transformation: Defining meta objectives using business process management. *Information & Management*, 57(5), 103262. <https://doi.org/10.1016/j.im.2019.103262>
- Gillani, F., Chatha, K. A., Jajja, S. S., Cao, D., Ma, X. (2024). Unpacking Digital Transformation: Identifying key enablers, transition stages and digital archetypes. *Technological Forecasting and Social Change*, 203, 123335. <https://doi.org/10.1016/j.techfore.2024.123335>
- Granovetter, M. (1985). Economic Action and Social Structure: The Problem of Embeddedness. *American Journal of Sociology*, 91(3), 481–510. <http://www.jstor.org/stable/2780199>
- Grover, P., Kar, A. K., Davies, G. (2018). “Technology enabled Health” – Insights from twitter analytics with a socio-technical perspective. *International Journal of Information Management*. 43, 85-97. <https://doi.org/10.1016/j.ijinfomgt.2018.07.003>
- Hair, J. F., Bakc, W. C., Babin, B. J., & Anderson, R. E. (2018). *Multivariate Data Analysis* (8th ed.). Cengage.
- Hayes, A. F. (2017). *Introduction to Mediation, Moderation, and Conditional Process Analysis. A regression-based approach*. Guilford Press.
- Kearns, G., Lederer, A. (2000). The effect of strategic alignment on the use of IS-based resources for competitive advantage. *Journal of Strategic Information Systems*, 9(4), 265-293. [https://doi.org/10.1016/S0963-8687\(00\)00049-4](https://doi.org/10.1016/S0963-8687(00)00049-4)
- Ketokivi, M. (2019). Avoiding bias and fallacy in survey research: a behavioral multilevel approach. *Journal Operations Management*, 65(4), 380–402. <https://doi.org/10.1002/joom.1011>
- Klein, S. P., Spieth, P., Söllner, M. (2024). Employee acceptance of digital transformation strategies: A paradox perspective. *Journal of Product Innovation Management*, 2024, 1-23. <https://doi.org/10.1111/jpim.12722>
- Klein, S., Watson-Manheim, M.B. (2021). The (re-)configuration of digital work in the wake of profound technological innovation: Constellations and hidden work. *Information and Organization*, 31(4), 100377. <https://doi.org/10.1016/j.infoandorg.2021.100377>
- Kleiner, B. M. (2008). Macroergonomics: work system analysis and design. *Human Factors*, 50(3), 461-467. <https://doi.org/10.1518/001872008X288501>
- Klinker, K., Wiesche, M., Krcmar, H. (2020). Digital Transformation in Health Care: Augmented Reality for Hands-Free Service Innovation. *Information Systems Frontiers*, 22, 1419–1431. <https://doi.org/10.1007/s10796-019-09937-7>
- Kraus, S.; Schiavone, F.; Pluzhnikova, A.; Invernizzi, A. C. (2021). Digital transformation in healthcare: analyzing the current state-of-research. *Journal Business Research*, 123, 557-567. <https://doi.org/10.1016/j.jbusres.2020.10.030>
- Lerman, L. V., Benitez, G. B., Müller, J. M., de Sousa, P. R., Frank, A.G. (2022). Smart green supply chain management: a configurational approach to enhance green performance through digital transformation. *Supply Chain Management: International Journal*, 27(7), 147–176. <https://doi.org/10.1108/SCM-02-2022-0059>
- Liao, Y., Deschamps, F., Loures, E. F. R., Ramos, L. F. P. (2017). Past, Present and Future of Industry 4.0: A

- Systematic Literature Review and Research Agenda Proposal. *International Journal of Production Research*, 55(12), 3609–3629. <https://doi.org/10.1080/00207543.2017.1308576>
- Lindell, M. K., Whitney, D.J. (2001). Accounting for common method variance in cross-sectional research designs. *Journal of Applied Psychology*, 86(1), 114–121. <https://psycnet.apa.org/doi/10.1037/0021-9010.86.1.114>
- Liu, D., Chen, S. and Chou, T. (2011). Resource fit in digital transformation: Lessons learned from the CBC Bank global e-banking project, *Management Decision*, 49(10), p. 1728-1742. <https://doi.org/10.1108/00251741111183852>
- Lin, N. (2001). *Social Capital: A Theory of Social Structure and Action*. Cambridge University Press, New York, NY.
- Lyu, Y., Zhu, Y., Han, S., He, B., & Bao, L. (2020). Open innovation and innovation “Radicalness”—the moderating effect of network embeddedness. *Technology in Society*, 62(101292), 101292. <https://doi.org/10.1016/j.techsoc.2020.101292>
- Marcon, E., Soliman, M., Ayala, N. F., Sturgeon, T., & Frank, A. G. (2025). A system view of how balancing sociotechnical factors in Industry 4.0 implementation: Evidence from interviews and ethnographic research. *Computers & Industrial Engineering*, 203(111047), 111047. <https://doi.org/10.1016/j.cie.2025.111047>
- Marcon, E., Marodin, G. A., & Frank, A. G. (2024). Combining organizational and social factors to support industry 4.0 implementation: A sociotechnical and configurational analysis of technology adopters. *IEEE Transactions on Engineering Management*, 72, 146–160. <https://doi.org/10.1109/tem.2024.3506991>
- Marcon, E., Soliman, M., Gerstlberger, W., Frank, A. G. (2022). Sociotechnical factors and Industry 4.0: an integrative perspective for the adoption of smart manufacturing technologies. *Journal of Manufacturing Technology Management*, 33(2), 259-286. <https://doi.org/10.1108/JMTM-01-2021-0017>
- Mazzola, E., Perrone, G., & Kamuriwo, D. S. (2015). Network embeddedness and new product development in the biopharmaceutical industry: The moderating role of open innovation flow. *International Journal of Production Economics*, 160, 106–119. <https://doi.org/10.1016/j.ijpe.2014.10.002>
- Miroshnychenko, I., Strobl, A., Matzler, K., & De Massis, A. (2021). Absorptive capacity, strategic flexibility, and business model innovation: Empirical evidence from Italian SMEs. *Journal of Business Research*, 130, 670–682. <https://doi.org/10.1016/j.jbusres.2020.02.015>
- Mukherjee, U. K., & Sinha, K. K. (2020). Robot-assisted surgical care delivery at a hospital: Policies for maximizing clinical outcome benefits and minimizing costs. *Journal of Operations Management*, 66(12), 227-256. <https://doi.org/10.1002/joom.1058>
- Nasiri, M., Ukko, J., Saunila, M. and Rantala, T. (2020). Managing the digital supply chain: the role of smart technologies, *Technovation*, 96-97, 102121. <https://doi.org/10.1016/j.technovation.2020.102121>
- Naslund, J. A., Shidhaye, R., Patel, V. (2019). Digital Technology for Building Capacity of Nonspecialist Health Workers for Task Sharing and Scaling Up Mental Health Care Globally. *Harvard Review of Psychiatry*, 27(3), 181-192. <http://dx.doi.org/10.1097/HRP.0000000000000217>
- Oliveira, M., Zancul, E., Salerno, M. S. (2024). Capability building for digital transformation through design thinking. *Technological Forecasting and Social Change*, 198, 122947. <https://doi.org/10.1016/j.techfore.2023.122947>
- Parmigiani, A. (2007). Why do firms both make and buy? An investigation of concurrent sourcing. *Strategic Management Journal*, 28(3), 285–311. <https://doi.org/10.1002/smj.580>
- Piccoli, G., Grover, V., Rodriguez, J. (2024). Digital transformation requires digital resource primacy: Clarification and future research directions. *Journal of Strategic Information Systems*, 33(2), 101835. <https://doi.org/10.1016/j.jsis.2024.101835>
- Podsakoff, P. M., MacKenzie, S. B., Lee, J. Y., Podsakoff, N. P. (2003). Common method biases in behavioral research: a critical review of the literature and recommended remedies. *Journal of applied psychology*, 88(5), 879–903. <https://psycnet.apa.org/doi/10.1037/0021-9010.88.5.879>
- Preacher, K. J., Hayes, A. F. (2008). Asymptotic and resampling strategies for assessing and comparing indirect effects in multiple mediator models. *Behavior Research Methods*, 40(3), 879–891. <https://doi.org/10.3758/BRM.40.3.879>

- Raimo, N. Turi, I., Albergo, F. Vitolla, F. (2023). The drivers of the digital transformation in the healthcare industry: An empirical analysis in Italian hospitals. *Technovation*, 121, 102558. <https://doi.org/10.1016/j.technovation.2022.102558>
- Roscoe, S., Cousins, P. D., & Handfield, R. (2019). The microfoundations of an operational capability in digital manufacturing. *Journal of Operations Management*, 65(8), 774-793. <https://doi.org/10.1002/joom.1044>
- Stevens, M., & Schaik, J. (2020). Implementing new technologies for complex care: The role of embeddedness factors in team learning. *Journal of Operations Management*, 66(1-2), 112-134. <https://doi.org/10.1002/joom.1034>
- Taherdoost, H. (August 10, 2016). Validity and reliability of the research instrument; how to test the validation of a questionnaire/survey in a research. How to Test the Validation of a Questionnaire/survey in a Research. Available at SSRN: <https://ssrn.com/abstract=3205040> or <http://dx.doi.org/10.2139/ssrn.3205040>
- Thuemmler, C., Bai, C. (2017). Health 4.0: Application of industry 4.0 design principles in future asthma management. In: Healthcare 4.0: How Virtualization and Big Data are Revolutionizing Healthcare. Springer, Cham, London, 23-37. https://doi.org/10.1007/978-3-319-47617-9_2
- Tortorella, G. L., Fogliatto, F. S., Mac Cawley Vergara, A., Vassolo, R., & Sawhney, R. (2020a). Healthcare 4.0: trends, challenges and research directions. *Production Planning & Control*, 31(15), 1245-1260. <https://doi.org/10.1080/09537287.2019.1702226>
- Tortorella, G. L., Fogliatto, F. S., Espôsto, K. F., Vergara, A. M. C., Vassolo, R., Mendoza, D. T., Narayanamurthy, G. (2020b). Effects of contingencies on healthcare 4.0 technologies adoption and barriers in emerging economies. *Technological Forecasting and Social Change*, 156, 120048. <https://doi.org/10.1016/j.techfore.2020.120048>
- Tortorella, G. L., Fogliatto, F. S., Espôsto, K. F., Vergara, A. M. C., Vassolo, R., Mendoza, D. T., Narayanamurthy, G. (2022a). Measuring the effect of Healthcare 4.0 implementation on hospitals' performance. *Production Planning & Control*, 33(4), 386-401. <https://doi.org/10.1080/09537287.2020.1824283>
- Tortorella, G. L., Fogliatto, F. S., Kurnia, S., Thürer, M., Capurro, D. (2022b). Healthcare 4.0 digital applications: An empirical study on measures, bundles and patient-centered performance. *Technological Forecasting and Social Change*, 181, 121780. <https://doi.org/10.1016/j.techfore.2022.121780>
- Tortorella, G. L., Fogliatto, F. S., Saurin, T. A., Tonetto, L. M., McFarlane, D. (2022c). Contributions of Healthcare 4.0 digital applications to the resilience of healthcare organizations during the COVID-19 outbreak. *Technovation*, 111, 102379. <https://doi.org/10.1016/j.technovation.2021.102379>
- Verhoef, P. C., Broekhuizen, T., Bart, Y., Bhattacharya, A., Dong, J. Q., Fabian, N., Haenlein, M. (2021). Digital transformation: A multidisciplinary reflection and research agenda. *Journal of Business Research*, 122, 889-901. <https://doi.org/10.1016/j.jbusres.2019.09.022>
- Vial, G. (2019). Understanding digital transformation: A review and a research agenda. *The journal of strategic information systems*, 28(2), 118-144. <https://doi.org/10.1016/j.jsis.2019.01.003>
- Wang, Y., Kung, L., Wang, W., Cegielski, C. (2018). An integrated big data analytics-enabled transformation model: Application to health care. *Information & Management*, 55(1), 64-79. <https://doi.org/10.1016/j.im.2017.04.001>
- WHO (World Health Organization) (2007). Everybody's business—strengthening health systems to improve health outcomes: WHO's framework for action. Available at: https://apps.who.int/iris/bitstream/handle/10665/43918/9789241596077_eng.pdf, accessed 14th Mar 2024.
- Williams, L. J., Hartman, N., and Cavazotte, F. (2010). Method variance and marker variables: A review and comprehensive CFA marker technique. *Organizational research methods*, 13(3), 477-514. <https://doi.org/10.1177/1094428110366036>
- Xing, X., Chen, T., Yang, X., & Liu, T. (2023a). Digital transformation and innovation performance of China's manufacturers? A configurational approach. *Technology in Society*, 75(102356), 102356. <https://doi.org/10.1016/j.techsoc.2023.102356>
- Yeow, A., Soh, C., Hansen, R. (2018). Aligning with new digital strategy: A dynamic capabilities approach. *Journal of Strategic Information Systems*, 27(1), 43-58. <https://doi.org/10.1016/j.jsis.2017.09.001>

- Zimmer, M. P., Baiyere, A., Salmela, H. (2023). Digital workplace transformation: Subtraction logic as deinstitutionalising the taken-for-granted. *The Journal of Strategic Information Systems*, 32(1), 101757. <https://doi.org/10.1016/j.jsis.2023.101757>
- Zhao, X., Lynch, J. G., Chen, Q. (2010). Reconsidering baron and kenny: myths and truths about mediation analysis. *Journal Consumer Research*, 37(2), 197–206. <https://doi.org/10.1086/651257>

Appendix A: Questionnaire structure

Questionnaire items to assess Healthcare Digital Strategy (STRATEGY)

(Adapted from Nasiri et al. (2020) and Verhoef et al. (2021)). *Concordance Likert scale: 1 - strongly disagree to 5 - strongly agree. CFI = 0.990; TLI = 0.983; RMSEA = 0.070; AVE = 0.685; CR = 0.928; Cronbach = 0.928. Factor loadings are shown in parentheses.*

- a) Our hospital aims to excel in the efficiency of its processes through the use of digital technologies (0.752).
- b) Our hospital seeks to collect and integrate data from different internal areas through the use of digital technologies (0.816).
- c) Our hospital aims to create a strong communication network among different internal areas through the use of digital technologies (0.865).
- d) Our hospital aims to provide more efficient medical services through the use of digital technologies (0.884).
- e) Our hospital seeks to enhance contact with patients through the use of digital technologies (0.845).
- f) Our hospital aims to improve its relationship with suppliers through the use of digital technologies (0.795).

Questionnaire items to assess Internal Digital Healthcare Resources (INTERNAL)

(Adapted from Barney, 1986, 1991). *Concordance Likert scale: 1 - strongly disagree to 5 - strongly agree. CFI = 0.995; TLI = 0.989; RMSEA = 0.043; AVE = 0.506; CR = 0.835; Cronbach = 0.831. Factor loadings are shown in parentheses.*

- a) Our hospital has the necessary knowledge to develop healthcare digital transformation (0.730).
- b) Our hospital has the necessary information to develop healthcare digital transformation (0.759).
- c) Our hospital has the necessary human resources to develop healthcare digital transformation (0.796).
- d) Our hospital has the necessary financial resources to develop healthcare digital transformation (0.641).
- e) Our hospital has the necessary organizational culture to develop its own healthcare digital transformation (0.614).

Questionnaire items to assess External Digital Healthcare Resources (EXTERNAL)

(Adapted from Dyer and Singh, 1998, 2018). *Concordance Likert scale: 1 - strongly disagree to 5 - strongly agree. CFI = 0.968; TLI = 0.947; RMSEA = 0.080; AVE = 0.444; CR = 0.826; Cronbach = 0.823. Factor loadings are shown in parentheses.*

- a) Our hospital uses complementary knowledge from external partners to develop healthcare digital transformation (0.776).
- b) Our hospital has the necessary information from external partners to develop healthcare digital transformation (0.693).
- c) Our hospital utilizes human resources from external partners to develop healthcare digital transformation (0.641).
- d) Our hospital utilizes technological resources from external partners to develop digital transformation in healthcare (0.618).
- e) Our hospital utilizes financial resources from external partners to develop healthcare digital transformation (0.582).
- f) Our hospital has a culture of collaboration with external partners for the development of healthcare digital transformation (0.671).

Questionnaire items to assess Healthcare Network Embeddedness (EMBEDDEDNESS)

(Adapted from Xing et al., 2023; Chen et al., 2024; Chen et al., 2025). *Concordance Likert scale: 1 - strongly disagree to 5 - strongly agree. CFI = 0.987; TLI = 0.979; RMSEA = 0.072; AVE = 0.637; CR = 0.913; Cronbach = 0.913. Factor loadings are shown in parentheses.*

- a) Our hospital is located in an environment close to healthcare suppliers that develop innovative solutions for the healthcare sector (0.792).
- b) Our hospital is located in an environment close to other hospitals and clinics that develop innovative solutions for the healthcare sector (0.826).
- c) Our hospital is located in an environment close to healthcare operators (insurers, health plans, agreements, etc.) that develop innovative solutions for the healthcare sector (0.757).
- d) Our hospital is located in an environment close to public actors that develop innovative solutions for the healthcare sector (0.766).
- e) Our hospital is located in an environment close to universities and research centers that develop innovative solutions for the healthcare sector (0.862).

f) Our hospital is located in an environment where there are collaboration networks and mechanisms for the transfer of knowledge and technologies for innovation in the healthcare sector (0.782).

Questionnaire items to assess healthcare technologies for smart operations and smart working

(Adapted from Frank et al. (2019), Meindl et al. (2020), Dornelles et al. (2022), Tortorella et al. (2022a; 2022b) and Deloitte (2020)). *Concordance Likert scale: 1 - never to 5 – very frequently.*

- a) Our hospital uses big data/analytics to support activities.
- b) Our hospital uses artificial intelligence/machine learning to support activities.
- c) Our hospital uses blockchain to support activities.
- d) Our hospital uses augmented or virtual reality to support activities.
- e) Our hospital uses collaborative robotics (e.g., surgical robots, cobots, outpatient care, AGVs, AMRs, etc.) to support activities.
- f) Our hospital uses wireless biomedical/digital sensors to support activities.
- g) Our hospital uses digital patient relationship applications to support activities.
- h) Our hospital uses cloud computing in hospital processes.
- i) Our hospital uses mobile applications to support activities.
- j) Our hospital uses corporate management systems (e.g., ERP) for the integration of internal data and with suppliers.
- k) Our hospital uses electronic health records (e.g., electronic prescription, electronic medical records) to perform operational activities.
- l) Our hospital uses digital traceability (e.g., RFID, QR Code, etc.) to carry out activities.
- m) Our hospital uses simulation to support its activities.
- n) Our hospital uses telemedicine to provide services to patients.

Questionnaire items for control variables

- a) Number of beds in your hospital.
- b) What is the approximate number of employees working at your hospital.
- c) How many years ago was your hospital founded.
- d) Ownership of your hospital (two dummies):
 - Tier 1 – Teaching;
 - Tier 2 – Public;
 - Tier 3 – Private.
- e) Your role within your hospital:
 - Middle manager (e.g., supervisor or coordinator)
 - Senior manager (e.g., manager or director)

4. HEALTHCARE DIGITAL TRANSFORMATION AS A SOCIOTECHNICAL SYSTEM: SHAPING THE INTERACTIONS BETWEEN INDIVIDUALS, STRUCTURES, AND PROCESSES

Lucas Miguel Alencar de Moraes Correia, Matthias Thüerer, Alejandro Germán Frank

This paper will be submitted to International Journal of Operations & Production Management

Abstract

Emerging technologies are widely used to enhance administrative processes and foster a more positive experience for both patients and professionals in health care systems. Although this Healthcare Digital Transformation (HCDT) is a profoundly sociotechnical transformation, research on HCDT has primarily focused on a technocentric perspective. We redress the balance, examining how HCDT as a sociotechnical system shapes the microfoundations involving individuals, structures, and processes—viewed here as social aspects. We develop a theoretical model that explains how organizational, technical, and social aspects are integrated to implement HCDT and how new digital operating routines emerge. Data was collected through a qualitative case study conducted at a teaching and research hospital in southern Brazil. We found that changes in the interactions among individuals, structures, and processes caused by digital transformation enabled the emergence of new digital healthcare operating routines. This emergence is dependent on the number of professionals involved and the complexity of each area. This study contributes to sociotechnical theory by empirically demonstrating how technical, organizational, and social aspects can be integrated to digitalize operations in complex hospital settings. It also adds to the knowledge-based view, showing how healthcare professionals' knowledge can be leveraged to adapt their tasks across different configurations of flexible/rigid processes and authority/consensus-based structures.

Key Words: Healthcare; Digital transformation; Sociotechnical system; Operating routines.

4.1 INTRODUCTION

Technological innovations have played a beneficial and revolutionary role in operations (Mithas et al., 2022; Choi et al., 2022; Stark et al., 2023). This includes digital technologies, which advancement and growing implementation in organizational operations is referred to as Digital Transformation (DT) (Angelopoulos et al., 2023). DT redefines how interactions occur between workers, hierarchical structures, and processes (Roscoe et al., 2019; Mukherjee & Sinha, 2020; Choi et al., 2022; Angelopoulos et al., 2023; Bardhan et al., 2025). This may lead to conflicts between humans and machines (Kellogg et al., 2020; Abeliensky et al., 2024; Amaya & Holweg, 2024), making it necessary to establish implementation strategies that consider the human aspect in technology adoption to mitigate such challenges (Amaya & Holweg, 2024; Bendoly et al., 2025). DT in organizations should therefore be seen through the lens of sociotechnical dynamics (Heim et al., 2021), a perspective that enables an understanding

of the interaction and interdependence between social components (e.g., people, practices) and technical components (e.g., technology, tools) within an organizational system (Malhotra et al., 2021). The central idea is that social and technical aspects should not be treated as mutually exclusive but rather as integrated elements, as they are intrinsically interconnected (Bardhan et al., 2025). This study uses a healthcare setting to explore how routines can be used at the micro-level to achieve this integration to realize desired macro-level outcomes.

Different sectors adopt digital technologies to increase productivity and efficiency in their operations, and healthcare is no exception. Significant investments in digital technologies are made in healthcare system (Wani & Malhotra, 2018; Mukherjee & Sinha, 2020; Dai et al., 2022) to redefine both administrative and patient care processes (Kc et al., 2020; Bardhan et al., 2025) to improve patient care while increasing overall productivity (Devaraj et al., 2013; Gardner et al., 2015; Bradley et al., 2018; Queenan et al., 2019). This process is referred to as Healthcare Digital Transformation (HCDT) (Kraus et al., 2021) in the literature. Healthcare systems are characterized by high levels of technical knowledge—due to the specialized training of professionals—and intense interactions with patients (Sharma et al., 2016). Considering sociotechnical dynamics, the implementation of HCDT therefore requires a harmonious integration of technical and social aspects, as both are essential and inseparable components of healthcare systems (Tushe et al., 2024; Lowry et al., 2025).

The technical aspect includes defining what is the purpose and application of digital technologies, whether for clinical care or administrative operations within the organization (Bradley et al., 2018; Mukherjee et al., 2020; Stevens & Schaik, 2020). The social aspect includes understanding how professionals can contribute to the implementation of these technologies, which they will ultimately use, and how this is shaped by social factors (Sykes & Aljafari, 2022; Bardhan et al., 2025; Bendoly et al., 2025). For this, the so-called microfoundations of operating routines (Roscoe et al., 2019) need to be investigated. The microfoundations refer to the "interaction" effects that occur between individuals, hierarchical structures (which may be based on consensus or authority), and processes (which may be rigid or flexible), leading to macro-level outcomes (Felin et al., 2012). The notion of microfoundations originates from the Knowledge-Based View (KBV), which considers knowledge as the most valuable resource of a company, with individuals serving as the primary source of this knowledge (Grant, 1996). In this context, operating routines emerge when individuals collectively develop sequential interaction patterns that enable the integration of their specialized knowledge without the need for explicit communication (Grant, 1996).

Healthcare systems are widely recognized as complex systems (Stevens & Schaik, 2020). This complexity arises from the fact that they involve multiple interacting attributes, such as professionals, operating routines, departments, and areas, as well as non-standardized processes and high levels of uncertainty that hinder digitalization (Chandrasekaran et al., 2012; Stevens & Schaik, 2020). HCDDT as a sociotechnical system represents a potential pathway to achieving excellence and success in digitalization in this complex scenario, but could it also contribute to its failure? This question arises because HCDDT shapes the routines of healthcare professionals by incorporating additional digital tasks into their already demanding daily activities and may create difficulties related to digital illiteracy. Professionals should actively participate in the design, implementation, and continuous improvement of the use of digital technologies (Stevens & Schaik, 2020; Malik et al., 2024), which may require time, effort, knowledge, and availability, potentially leading to adverse effects (Stevens & Schaik, 2020).

How to develop effective strategies to structure and implement HCDDT therefore remains a relevant topic that needs to be further explored. There remains specifically a gap in understanding — at the micro level of operating routines — as to why some tasks in hospitals are easier to digitalize than others (Sharma et al., 2016; Tushe et al., 2024), why certain digitalization efforts tend to be more flexible (Mukherjee & Sinha, 2020; Bavafa et al., 2021), or why there is a higher adoption of digitalization in certain departments compared to others (Sharma et al., 2016). These questions may help explain variations — or even cases of excellence — in how HCDDT, understood as a socio-technical system, shapes the interactions among the microfoundations of operating routines, influencing their success or failure within organizations, and how digital changes in these micro-level interactions evolve into new digital healthcare operating routines.

To answer our research question – *How does HCDDT, as a sociotechnical system, shape microfoundations and generate new digital operating routines in healthcare?* – we conducted an in-depth atypical case study at a large Brazilian research hospital. The unit of analysis are healthcare professionals and the process of implementing an internally developed Hospital Management System (HMS) that is used by these professionals. We adopted a theoretical approach based on the sociotechnical perspective to explore how DT shapes the interactions of the microfoundations of operating routines in the hospital, and we used the KBV to explore how individual knowledge should be considered to create new digital healthcare operating routines. Data were collected through semi-structured interviews, participant observation, and documents provided by the hospital under investigation. These data were objectively verified

using Gioia's methodology to ensure rigor in qualitative research, following methods aligned with non-intrusive measures (Webb et al., 1999).

We found that changes in interactions between individuals, structures, and processes caused by HCDDT at the micro level vary according to the complexity of each area and the number of professionals involved. For example, professionals interact with processes as they carry out their roles, and under HCDDT, they are guided to map these processes to suggest improvements. When processes are flexible, such as in surgical departments, implementing HCDDT is more challenging due to the complexity of tasks, which demand a more conservative approach to change, longer learning periods, and more centralized decision-making. Conversely, in rigid processes (standardized processes), professionals have greater autonomy and freedom to adapt workflows, converting manual and analog tasks into digital ones using HMS modules and other associated technologies. Changes in the interaction between individuals and structures also vary. In departments with smaller teams, a consensus-based structure enables collaborative work and shared knowledge development to improve digital technologies. In areas with larger teams, a hierarchical authority-based structure is more appropriate. In such cases, the role of a project owner becomes essential to coordinate actions and represent the teams in decision-making forums. Regardless of the structure, the knowledge generated and operational feedback from professionals is fed back into the strategic levels of HCDDT, guiding new digital initiatives. This personalized model enabled the emergence of new digital healthcare operating routines, such as digitizing tasks previously performed on paper, ad hoc problem-solving, and active participation of professionals in digitalization projects. This study thus contributes to the sociotechnical theory literature by presenting how the aspects relate to each other and by offering new pathways for implementing HCDDT in complex systems, such as healthcare. It also advances KBV by demonstrating how the knowledge and experiences of healthcare professionals can be leveraged to develop personalized HCDDT aligned with their specific needs. Finally, from a practical perspective our findings provide valuable insights for hospital managers in designing digitalization strategies, as well as for policymakers in developing guidelines for healthcare digitalization in their respective regions.

The remainder of the article is organized as follows. We provide the theoretical background in Section 2, before Section 3 presents research design, data collection, and analysis methods. Findings are presented in Section 4 and discussed in Section 5. Finally, Section 6 concludes by outlining the study's contribution to theory and managerial practice and proposing potential areas for future research.

4.2 THEORETICAL BACKGROUND

4.2.1 Sociotechnical system

A sociotechnical system integrates social and technical aspects in the design and management of organizational systems (Marcon et al., 2025). It emphasizes the interdependence between people (social aspect) and technology (technical aspect) in work environments, arguing that both are equally important and should be managed in a balanced way to ensure long-term performance (Correia et al., 2025). The sociotechnical theory assumes that individuals can take on responsibilities and actively participate in the system, promoting flexibility, autonomy and participation (Malhotra et al., 2021), a systemic view of the environment (Easton & Rosenzweig, 2015), and an improvement in the quality of work life (Zhang et al., 2025). Following this theory, the interaction between people and technologies occurs through how work is designed and processes are executed in operating routines (Frank et al., 2024).

Social and technical aspects are immersed in a dynamic and inseparable organizational context, in which organizations are understood as organic systems that depend on the interaction between workers and technologies (Kellogg et al., 2020; Heiberg et al., 2022). In this sense, the operational performance of an organization is influenced by the harmonious interaction between various factors that compose these sociotechnical aspects (Kellogg et al., 2020; Zhang et al., 2025) and impact the functioning of the system (Heiberg et al., 2022). For example, social factors may include the skills, knowledge, experience, capabilities, and training of individuals, as well as their well-being, collaboration practices, and participation in group activities, among others (Rousseau, 1977; Malhotra et al., 2021). Technical factors focus on the typologies of technologies, purposes, and usability (Choi et al., 2022; Angelopoulos et al., 2023).

4.2.2 Organizational design for the HCDDT from a sociotechnical perspective

HCDDT introduces digital technologies for providing medical services to patients (Mukherjee & Sinha, 2020; Li et al., 2021; Delana et al., 2022; Wowak et al., 2023). During the transition process from analog to digital it is important to integrate technologies, organizational factors and healthcare professionals (e.g., Kc et al., 2020; Sykes & Aljafari, 2022; Tushe et al., 2024; Lowry et al., 2025). But the literature on the implementation of HCDDT has widely neglected the social aspect focusing mainly on the technical aspect (Bardhan et al., 2025). We therefore propose to consider HCDDT as a sociotechnical system based on the

principles of sociotechnical theory introduced above. This means that there are three main aspects that need to be considered.

The organizational aspect of HCDT considers digital strategy and digital resources. The digital strategy dimension refers to the formulation of paths and investment decisions (Dai et al., 2022; Bardhan et al., 2025), involving, for example, the development of IT master plans and/or digital transformation plans aimed at improving medical services, optimizing processes, and enhancing the experience of professionals, patients, and families. The formation of digital governance committees and IT management committees is also part of this dimension. To operationalize digital strategy, organizational resources, also known as Digital Resources, need to be leveraged. Technological infrastructure is the first key resource for enabling the use of digital technologies. It includes hardware, software, and the network structure necessary to implement digital solutions and data analytics tools, such as programming and data science (Gardner et al., 2015; Lowry et al., 2025). Human capital is a second key resource, encompassing the formation of qualified teams in digitalization, IT professionals, and programmers (Stevens & Schaik, 2020; Lowry et al., 2025). This is because DT involves managerial domains related to new capabilities, which often reside in human, informational, or organizational capital, as well as in the development of new digital skills (Mithas et al., 2022; Nwafor et al., 2023). In this context, a third intrinsic and essential resource for DT, although intangible, is the development of a digital culture, which fosters continuous improvement and adaptation to technological changes. Digital resources can be managed through different strategies (Correia & Frank, 2025). For example, the institution may choose internal development, promoting the complete internalization of its digital resources for the journey from analog to digital (Correia & Frank, 2025), or it can outsource these resources to external partners through direct contracts, establishing strategic alliances, and forming a collaborative ecosystem that provides the necessary digital solutions for the healthcare organization (Correia & Frank, 2025). Hybrid governance (Franco et al., 2024; Correia & Frank, 2025), which combines both internal development and external acquisition of digital resources, is also possible allowing for greater flexibility and adaptability to the demands of HCDT.

The technical aspect of HCDT considers which digital technologies should be adopted and their roles in healthcare operations. For example, the use of the Internet of Things (IoT) for data capture and real-time monitoring of patients and equipment, big data & analytics for predicting patient diseases and operational demands (Guha & Kumar, 2018; Nenova & Shang, 2022), artificial intelligence for automating medical tasks (Dai & Tayur, 2022), collaborative

robots in surgical procedures (Mukherjee & Sinha, 2020), telemedicine for online consultations (Bavafa et al., 2021; Delana et al., 2023), 3D printing for prototype development, or immersive treatments through the use of wearable devices, haptic devices, augmented reality, virtual reality, and extended reality (Lowry et al., 2025). These technologies aim to provide personalized medicine and value-based, connected healthcare to promote faster, safer, and patient-centered care (Kc et al., 2020). Technologies such as blockchain, electronic health records, and hospital management systems (Wani & Malhotra, 2018; Ransbotham et al., 2021; Nenova & Shang, 2022) have been adopted to enable vertical integration between different areas and departments within the hospital, as well as horizontal integration with external partners (Bradley et al., 2018; Kc et al., 2020; Nwafor et al., 2023).

The social aspect of HCDT considers the interactions between the microfoundations of operating routines, which encompass individuals, structures, and processes, operating at the micro level of the organization (Felin et al., 2012). These interactions involve changes in how professionals and their tasks are shaped by the introduction of digital technologies (Kellogg, 2022; Franke et al., 2024), often contingent on social factors such as the training of healthcare professionals, adaptation of hospital processes, level of educational background, profession, work experience, accumulated knowledge (Roscoe et al., 2019; Amaya & Holweg, 2024; Franke et al., 2024) and the role of professionals – whether individually or in groups – in the digital transformation process (Kellogg, 2022). A relevant example is the knowledge accumulated by individuals, which can dictate how technology can be developed and implemented in operating routines (Nwafor et al., 2023; Amaya & Holweg, 2024). This knowledge becomes a crucial element in defining the implementation paths of HCDT (Stevens & Schaik, 2020).

The resulting theoretical model of HCDT as a sociotechnical system is illustrated in Figure 10. In the first part of the model (organizational and technical aspects), HCDT is structured into three main dimensions: digital strategy, digital resources, and digital technologies. In the second part (social aspect), these dimensions are incorporated into the microfoundations of operational routines, which encompass the interaction between individuals, structures, and processes – it is at this level that HCDT is effectively operationalized.

The second part of the model presented in Figure 10 is based on the framework proposed by Roscoe et al. (2019), in which processes function as integration mechanisms, incorporating

individuals as sources of knowledge development and sharing. The interaction between individuals and processes enables the accumulation of experience and knowledge, which, when applied for a specific purpose, contributes to the continuous improvement of processes. Additionally, the model includes organizational structures that act as organizing mechanisms, determining how and how often individuals interact (Roscoe et al., 2019).

Finally, once embedded in the interactions of the microfoundations, new digital healthcare operating routines can be generated. Digital strategy, digital resources, and digital technologies precede the microfoundations, as these three dimensions establish the guidelines, prerequisites, and methodologies for the operationalization of HCDT, positioning them at a macro-management level. In contrast, the microfoundations represent the operational level of healthcare activities, being directly shaped by HCDT.

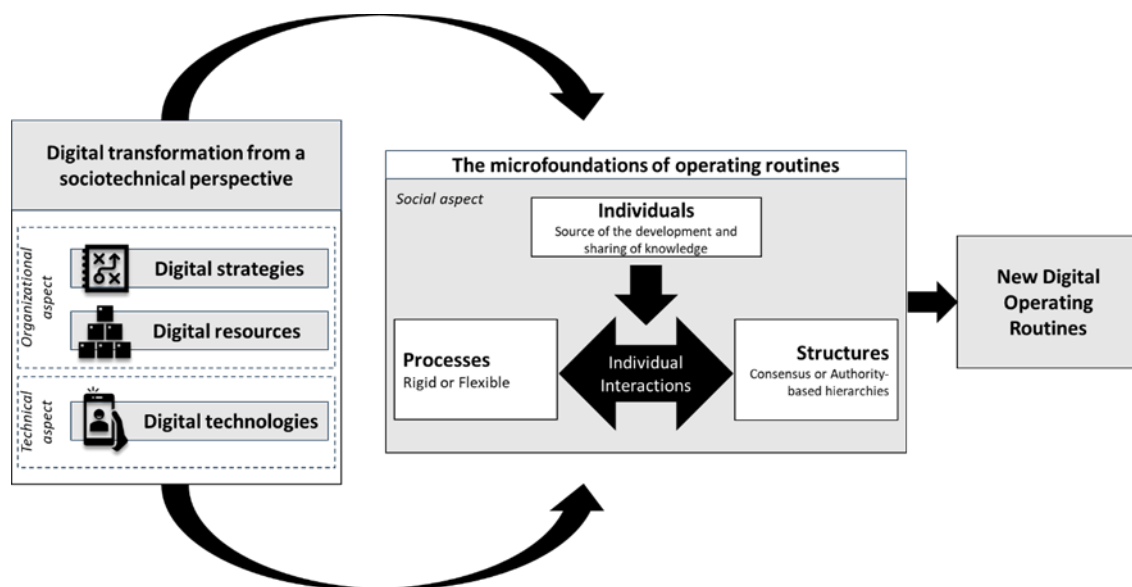


Figure 10. Theoretical model

4.2.3 The microfoundations of operating routines

Operating routines emerge when individuals, acting collectively, develop sequential patterns of interaction that facilitate the integration of their specialized knowledge without the need for explicit communication of such knowledge (Grant, 1996). When these routines stabilize over time, they enable the formation of operational capabilities (Barney & Felin, 2013, Roscoe et al., 2019). The microfoundations of operating routines refer to the interactions between individuals, processes, and structures, and their effects, that enable these macro-level

outcomes (Felin et al., 2012). They provide a detailed understanding how individuals act within organizations and their specific roles in creating and consolidating operating routines.

In this context, the KBV emerges as a relevant theory, as one of its fundamental premises is that knowledge is created by individuals (Grant, 1996; Nonaka, 1994). When individuals work in an organization, they bring with them knowledge acquired from previous experiences, e.g., through training, observation of colleagues, or experiential learning (Felin et al. 2012) in this or other organizations, and apply this knowledge to their new roles (Grant, 1996; Amaya & Holweg, 2024). They develop specialized knowledge in a productive task and use this knowledge to simplify the future execution of the same task, whether by themselves or by others (Grant, 1996). A practical example are surgical procedures (Mukherjee & Sinha, 2020; Stevens & Schaik, 2020), where professionals such as doctors and nurses accumulate knowledge from previous clinical cases, including information about the steps involved, operational methods, required resources, recovery time, and the likelihood of success or patient relapse. This specialized knowledge is then applied to streamline the surgical procedure, making the process more efficient (Stevens & Schaik, 2020). As a result, both the patient, who receives faster and more accurate care, and the professional, who improved its productivity, benefit.

But the microfoundations of operating routines are not limited to individuals in isolation: they also encompass the level and types of interactions that occur between individuals within an organization, which, over time, evolve to form a collective. In this context, interaction is defined as moments when two or more individuals mutually engage through verbal and non-verbal social mechanisms (Cousins et al., 2006; Felin & Hesterly, 2007). These mechanisms are understood as instruments or activities, whether natural or established, that facilitate the occurrence or execution of something (Franke et al., 2024). They can be classified as formal or informal socialization mechanisms (Cousins et al., 2006). In the case of formal mechanisms, they refer to structures specifically created to communicate expectations and share useful knowledge among individuals. Examples include regularly scheduled meetings, conferences, or team events (Cousins et al., 2006). Informal socialization mechanisms are less structured and involve spontaneous interactions, such as impromptu conversations in hallways or social gatherings after work hours. These activities help build trust among individuals and facilitate the sharing of ideas in a more organic way (Cousins et al., 2006).

Individuals and collectives also interact with the processes and structures of the organizational design. A process is defined as a sequence of interdependent events. It can function as integration mechanism, facilitating the combination of different organizational elements, such as individuals, teams, departments, or multifunctional knowledge resources (Chandrasekaran et al., 2012). Processes can be classified as rigid or flexible (Felin et al. 2012), and their design significantly impacts the pattern of interaction among individuals. Rigid processes refer to highly organized and controlled sequences of events, where individuals have well-defined roles and responsibilities. For example, in healthcare, this type of process occurs in surgical procedures, as there are strict protocols with well-defined rules and steps, rigorous control, and a focus on efficiency (Mukherjee et al., 2020). Another example is inventory control in hospital pharmacies, where procedures are carefully structured for the receipt, organization, and distribution of medications across different hospital departments. Flexible processes involve sequences of events that can be adapted by individuals to achieve a desired outcome. An example of this type of process in healthcare occurs in personalized patient care (Wang et al., 2023), where clinical treatments can follow different paths depending on individual needs — such as replacing in-person consultations with remote ones (Bavafa et al., 2021). Another example involves procedures aimed at developing new digital solutions for hospitals (Lowry et al., 2025). This approach provides individuals with the freedom to experiment and develop innovative solutions, adjusting the process as needed. The interaction between individuals and processes creates a self-reinforcing cycle: individuals learn from their mistakes and, by applying this newly acquired knowledge, continue to refine and improve the process, generating continuous improvement (Roscoe et al., 2019).

In this context, organizational structures are defined as conditions that enable or constrain processes and individual and collective interactions within an organization. A basic structure for organizing complex social activities involves the use of hierarchies (Grant, 1996) that can be classified as authority-based or consensus-based (Nickerson & Zenger, 2004). An authority-based hierarchy relies on centralized decision-making to optimize the transmission and processing of knowledge, for example, a senior manager that sets guidelines to steer the search for solutions without necessarily considering the knowledge of other individuals in detail. A consensus-based hierarchy, in contrast, involves the collective agreement of individuals on a search path and the creation of a common language that integrates different specialized knowledge (Roscoe et al., 2019). This promotes the building of a shared identity within the organization, which reduces communication-related costs and establishes

coordination rules that influence the direction of the search for and implementation of solutions. In consensus-based hierarchies, individuals may receive incentives—such as salary increases, bonuses, awards, or promotions—to encourage the search for solutions, the transfer of knowledge to others, and the generation of new knowledge (Grant, 1996; Stevens & Schaik, 2020; Nwafor et al., 2023). Such incentives can also motivate individuals to work collaboratively, whether in cross-functional teams or projects, to solve problems and propose improvements that enhance organizational performance (Rousseau, 1977; Kellogg, 2021; Nwafor et al., 2023).

Since processes necessarily unfold in a structure there is an overlap, and the main question is a question of fit or alignment. But this alignment alone does not result in the creation or accumulation of knowledge without the specific interaction of individuals who share knowledge. In other words, in processes, the presence of individuals is essential to carry out a series of productive tasks, while organizational structures lack a real purpose without people to occupy and perform roles. In this sense, the central argument of the microfoundations of operating routines is that individuals constitute the essential unit of analysis for understanding how organizations update or develop new operating routines, and this occurs when individuals operate at the intersection of processes and structures (Roscoe et al., 2019).

Based on this theoretical background, we developed the model presented in Figure 10, which proposes a new approach to addressing HCDDT. In the remainder of this study, we present a case where we aim to understand whether this approach—focused on explaining how microfoundations are shaped—contributes to the organization's success with HCDDT or, potentially, to its failures. Figure 10 suggests that HCDDT as a sociotechnical system shapes microfoundations and can explicitly encourage—or alternatively restrict—the development of new digital routines in hospital settings. By analyzing the phenomenon at the micro level, the model enables us to identify the key micro-level elements and to understand why some areas or departments are more susceptible to the adoption of digital technologies than others.

4.3 RESEARCH DESIGN

4.3.1. Case description

The Research Hospital studied is a public institution, part of the network of university hospitals under Brazil's Ministry of Education (MEC) and academically affiliated with a Federal University in the southern region of Brazil. It aims to provide high complexity,

humanized, and excellent care within the Unified Health System (SUS). The Hospital offers approximately 919 inpatient beds and more than 55 medical services. Over the last two years, it has conducted over 64,300 hospitalizations and 1,035,000 consultations and employs 6,800 healthcare professionals. Although the hospital primarily serves SUS patients, it also provides care for private insurance and self-paying individuals. The Hospital's strategic priorities are patient-centered care, resources and space optimization, and sustainability. To meet these strategic priorities, the hospital has established a management culture deeply rooted in innovation and digital transformation, aiming to enhance administrative and healthcare processes for patients and professionals through the implementation of new digital technologies. During the Covid-19 pandemic, for example, this Hospital innovated by offering new digital services, such as telemedicine and telehealth, as well as distance health education for remote communities.

The hospital's organizational structure consists of various directorates, committees, councils, coordinators, and units. Among the hierarchies related to innovation and DT, the Digital Governance Committee (DGC) stands out. This committee comprises representatives from the directorates and several areas of the institution. Its primary objective is to develop, propose, and monitor Digital Governance policies that address the institution's Information and Communication Technology (ICT) needs. Another relevant body is the Coordination of Information and Communication Technology Management (CICTM), whose purpose is to provide ICT services to the departments of the hospital. Both seek to enhance activities related to healthcare, management, education, and research, in alignment with the Hospital's master plan, ICT strategic planning, and digital transformation plan. Additionally, the Innovation and Technology Transfer Center (ITTC) was established to facilitate and enable initiatives for innovation and technology transfer. Figure 11 illustrates the bodies of ICT, innovation, and data established by the Hospital for HCDDT, which are divided into strategic management and tactical and operational management. The former encompasses digital governance and data governance, determining what needs to be done, while the latter involves the management of information and communication technologies and data management, translating into how the actions established in strategic management should be executed. Finally, at the intersection of Digital Governance, Data Governance, and Information and Communication Technology Management lies the Executive Board.

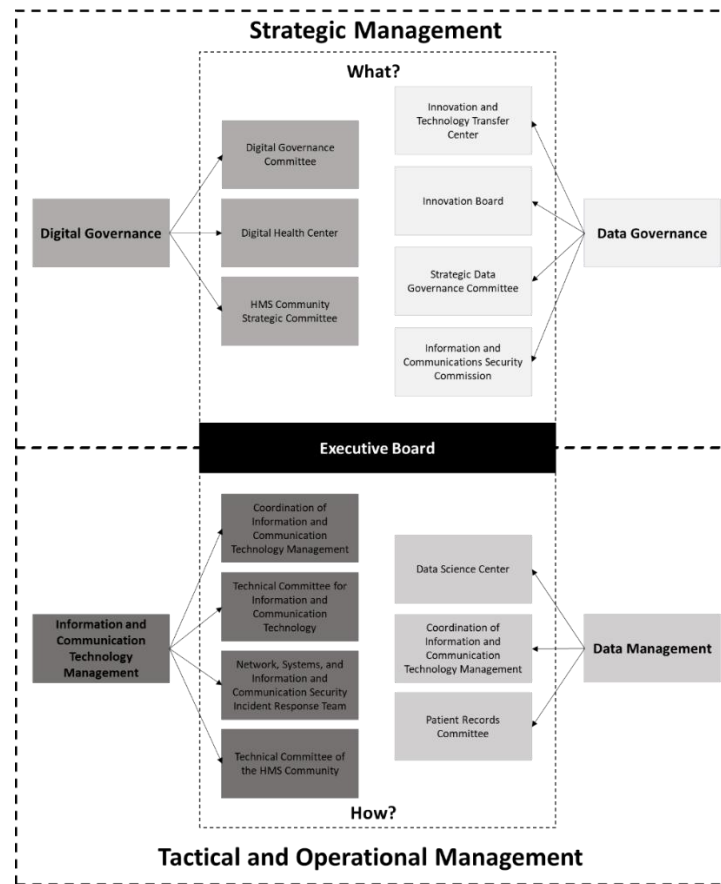


Figure 11. Bodies of information and communication technology, innovation, and data

The main driver for HCDT in our study is the HMS of the studied hospital. Operating since 1971, the studied hospital is well known for its efficient management. Its management model became a reference for creating a Brazilian hospital services company, which aims to enhance the structure and processes of all university hospitals within the MEC network. At the core of this management model is the HMS, a solution internally developed by the institution itself. The HMS has undergone an evolution spanning more than 40 years, marked by technological advancements and significant transformations. The journey began when the progressive growth of administrative demands at the Research Hospital highlighted the need to automate key processes. Efforts were initiated to internally develop a first system, with the first functionalities gradually implemented in a mainframe environment. In 1983, the first hospital applications emerged, such as the outpatient scheduling module, followed by the patient admission module. In the subsequent years, the focus shifted to patient-centered digitization efforts, leading to the creation of modules related to exams, medical and nursing prescriptions, supplies, and more. Starting in 2000, all previously developed systems underwent a migration from a character-based environment to a graphical environment, adopting a client-server

architecture. This transition gave rise to the HMS as it is known today. Among the most notable advancements were the implementation of the Online Medical Record, Surgical Portal, Obstetrics Center, Outpatient Module, Care Protocols, Progress Notes, Electronic Prescription Verification, and Therapeutic Sessions. In 2009, the HMS again underwent a significant transformation by becoming part of the National Program for the Restructuring of Federal University Hospitals, coordinated by MEC. This program aimed to restructure and revitalize federal university hospitals using the HMS from the Research Hospital as a model. By 2014, the HMS received a comprehensive update to its architecture, and its name was rebranded to reflect its new scope, which expanded beyond the internal confines of the hospital. In 2017, the HMS Community was established as a collaborative development initiative, making the system available for use across various healthcare institutions and levels of care.

Two committees were established to manage the HMS: the Strategic Committee, which is responsible for setting goals and priorities, and the Technical Committee, which is responsible for implementing updates. Currently, the HMS Community includes the Research Hospital featured in this study (serving as the coordinating entity), the country's military forces, universities, and state health departments. The HMS is structured into more than 16,000 macropoints and consists of 55 modules organized into three major categories: Assistance (24), administrative (13) and Interoperability (18). The clinical modules address the needs of patient diagnosis, treatment, and follow-up, while also assisting in the organization of healthcare professionals' tasks. The administrative modules focus on the organizational and operational management of the Research Hospital, ensuring efficiency and support for institutional activities, while interoperability is related to modules for integration with both internal and external systems of the institution.

4.3.2. Data collection procedure

To assess how HCDT driven by the HMS aligns with our theoretical model, a triangulated data collection strategy was employed (Yin, 2014) using three primary sources: (1) semi-structured interviews; (2) participant observation; and (3) analysis of primary and secondary hospital documents (Figure 12). Results of the semi-structured interviews were first corroborated through participant observation, before primary and secondary hospital documents were collected for further corroborating observations and contextualization. Combining different data sources enables more precise triangulation of information,

contributing to the development of more insightful theories and enhancing the robustness of resulting theoretical frameworks (Eisenhardt, 1989; Silva et al., 2025).

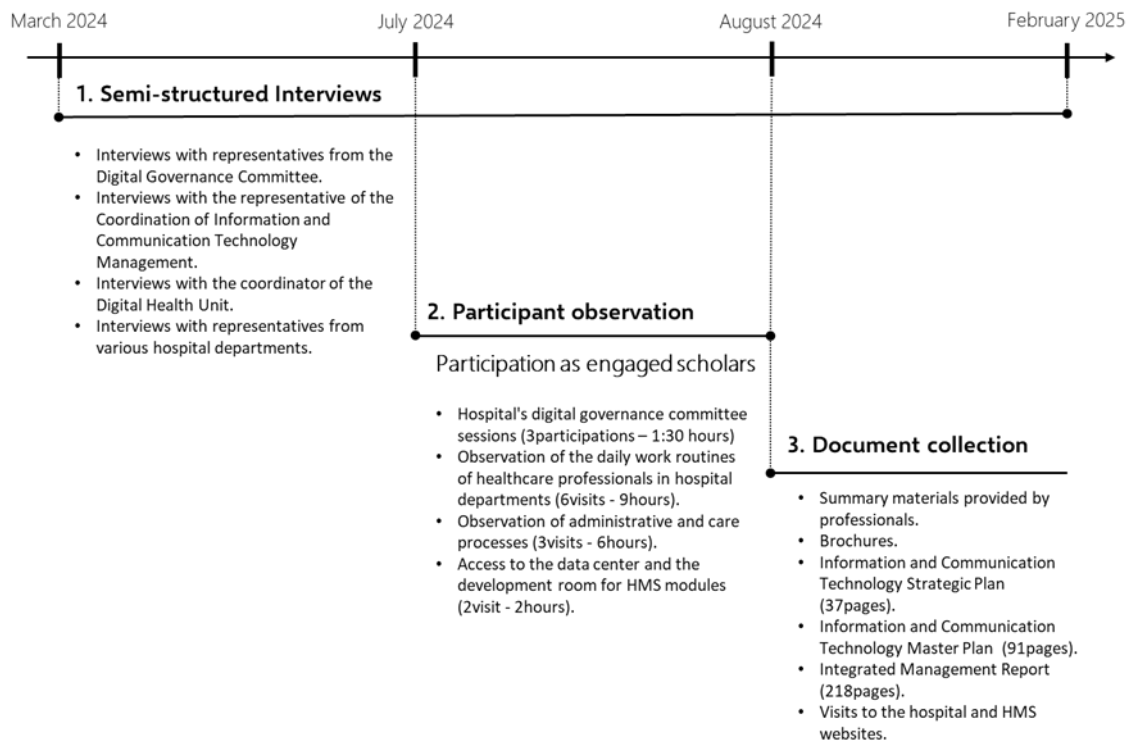


Figure 12. Data collection

4.3.2.1 Phase 1: Semi Structured Interviews

25 semi-structured interviews were conducted with 40 interviewees as detailed in the Table in Appendix A. The main objectives were: to understand the management of the hospital's digitalization with key leaders, including gaining insight into the digital strategies and resources utilized, to explore how microfoundations are influenced by the adoption of HCDDT as a sociotechnical system from a professionals' perspective, and to understand the role of these professionals in developing new digital healthcare operating routines. Our sampling logic was therefore guided by the collection of data from individuals working in different hospital departments who are directly involved in the use of HMS and other digital technologies in clinical and administrative processes. We aimed to cover various areas (outpatient clinic, surgical unit, pharmacy, emergency department) and roles within the institution, including directors, department heads, managers, and operational professionals (doctors, nurses, technicians, pharmacists and ICT technicians). The data collection from these multiple perspectives provided distinct insights into the same phenomenon, offering a detailed

understanding of how individuals, processes, and structures interact during HCDT and evolve into new digital healthcare operational routines. All interviewees have had at least 2 years of experience with the HMS. Some have used the system since their undergraduate studies over 20 years ago, which allows them to compare the latest HMS updates and the adoption of new technologies. Gathering the experiences of these key informants is important to ensure the success of case study-based research (Yin, 2014).

The interviews were conducted online, with durations ranging from 30 minutes to 2 hours and 30 minutes. All sessions were audio-recorded and later fully transcribed. Following the recommendations of Gioia et al. (2012), interviewees were informed at the beginning of each session about the anonymity of the information provided, including the confidentiality of their names and specific roles, to ensure research reliability and participant comfort. A semi-structured interview script was used in interviews with operational professionals. The script was divided into a few questions, which are detailed in Appendix B. Additionally, following the recommendations of Galletta (2013), open-ended and exploratory questions were posed during the interviews, going beyond the initial script. This approach provided greater flexibility to the research, enabling the discovery of unexpected information and enhancing the depth of the collected data (Gioia et al., 2012). The interviews were divided into three parts. First, the interviewees were invited to share information about their roles, length of service, and experiences with the evolution of the HMS in their positions. Second, the questions were directed toward the framework in Figure 10, asking the interviewees to describe in detail their interactions with processes, hierarchical structures, and their daily routines with the system. Finally, interviewees were asked to share their perspectives and explain important details, such as routines before the adoption of HCDT, the role of individuals in the design or update of new modules and digital technologies, and how the management of new digital technologies is conducted in the healthcare context. This interview structure allowed for the collection of open narrative data, as well as factual and specific information.

As engaged researchers, it was essential to maintain constant attention to potential biases, which could arise both from the researchers and the interviewees, to prevent participants from distorting or dramatizing their experiences and accounts (van Marrewijk & Dessing, 2019). Interviews were therefore scheduled at least one week in advance, ensuring greater comfort for the participants. The interview times were set according to the availability of the interviewees, ensuring they were well-rested and reducing the risk of responses being influenced by time constraints. Additionally, researchers avoided mentioning data or

information from previous interviews during subsequent conversations, eliminating potential biases in the responses. Moreover, the interviewers paid close attention to the research script, ensuring that it was focused on comprehensive questions (anticipating related topics that should be addressed) and that it did not contain leading questions (such as: "Don't you agree that...?") (Gioia et al., 2012). To further reinforce impartiality, at least two researchers that were directly involved in the study conducted the interviews, reducing the possibility of individual biases during the data collection and subsequent analysis.

In total, almost 20 hours of interview recordings were collected, resulting in over 690 pages of transcribed content. After conducting the 25 interviews, we observed that all participants shared common points in their responses, indicating that theoretical saturation had been reached, as highlighted by Lee (1999).

4.3.2.2 Phase 2: Participant Observation

To further corroborate the data obtained from the interviews participatory observation techniques were applied (Atkinson & Hammersley, 1994; Spradley, 2016; Silva et al., 2025). To understand, at a managerial level, how the process of prioritizing the adoption of digital technologies occurs, one researcher participated three times (1:30 hours) in the hospital's digital governance committee sessions. This allowed for evaluating, from a hierarchical perspective, how decisions are made regarding the assessment of digital projects, the definition of digital strategies, the allocation of technological resources, as well as the development or acquisition of new HMS modules and other digital technologies. The researcher also followed on six occasions the routines and tasks performed by healthcare professionals in different hospital departments, totaling nine hours. During this phase, it was possible to observe ongoing processes, the usability of the HMS modules and other digital technologies, as well as the professionals' interactions with these tools. Additionally, the researcher asked informal questions, took detailed notes, and collected, reviewed, and validated hospital documents. Additionally, three more visits were focused on administrative and care processes, totaling six hours, while two more visits were dedicated to the data center and the HMS module development room, lasting two hours. This approach allowed for objective verification of the data collected during the interviews and a deeper analysis of the points discussed in the digital governance committee sessions. In total, there were more than 18 hours of participant observation in this study.

4.3.2.3 Phase 3: Document Collection

We had access to a rich set of additional data to complement the interviews and participatory observations. This includes documents provided by hospital professionals, summarized materials, and a search on the hospital's and HMS's official websites for primary and secondary documents. Among these documents, the strategic planning and the IT and communication master plan stand out, since it was essential to understand the hospital's organizational structure and how it was reconfigured over the years in line with the development and evolution of the HMS.

4.3.3. Data analysis

Based on the records of the semi-structured interviews, participant observation notes, and collected documents, we analyzed the data following Gioia's methodology (Gioia et al., 2012). This methodology was chosen because it prioritizes the interviewees' interpretations, positioning researchers as responsible for providing an accurate account of their experiences. Data is categorized into first- and second-order categories to facilitate their subsequent structured presentation in an organized manner. We used the qualitative analysis software NVivo 15 to operate this methodological approach.

Data were first organized and categorized into first-order empirical concepts using the model presented in Figure 10 as a reference. Thematic analysis techniques (Braun & Clarke, 2006) combined with correspondence logic (Yin, 2014; Webb et al., 1999) was applied. Initially, there were many terms, concepts, and jargon provided by the informants that needed to be organized. During the analysis, we made an effort to stay faithful to the terms used by the informants and made little effort to distill the categories. As the analysis progressed, we began to identify similarities and differences between the categories or coding identifiers previously established (processes, structures, changes at work, expertise, experiential learning, and pre-existing knowledge). This process allowed for a reduction of the initial number of categories to a more manageable set, which was later labeled with short and precise descriptions (e.g., 'The same doctor who works in the ICU also holds management positions in the digital health unit, as well as on the digital governance committee').

We then identified second-order themes by consciously analyzing the similarities and differences between the first-order empirical concepts. Following the recommendations of Corbin and Strauss (2014), axial coding was applied. For example, from the first-order empirical concepts 'HMS modules aligned with professionals' workflows' and 'Learning from

process errors,' the second-order theme 'Process mapping' emerged. Finally, the second-order themes were grouped into overarching theoretical dimensions. For instance, the themes 'Digital records' and 'Ad hoc problem resolution' were grouped under the theoretical dimension 'New digital operating routines in healthcare.' When the first-order empirical concepts did not easily fit into the predefined coding scheme, new second-order themes were created.

The coding model of second-order themes was iteratively refined until the researchers achieved a final structure that provided a robust explanation of the case studied (Eisenhardt & Graebner, 2007). Figure 13 illustrates the coding process and data structure, providing examples of first-order empirical concepts, second-order themes, and theoretical dimensions. According to Gioia et al. (2012), this data structure not only allows for organizing the data in a coherent visual way but also provides a graphical representation of how we progressed from raw data to terms and themes during the analysis, ensuring rigor in qualitative research.

To ensure coding consistency and establish inter-rater reliability, a second member of the research team repeated the coding process for both the first-order empirical concepts and the second-order themes (Gioia et al., 2012). This approach reinforced the reliability of the analysis of the semi-structured interviews, as all participants answered the same structured questions in the same order. The independent analysis by the second researcher resulted in a high level of agreement regarding the key themes identified, reducing the risk of biases and further enhancing the rigor of the qualitative analysis process (Armstrong et al., 1997).

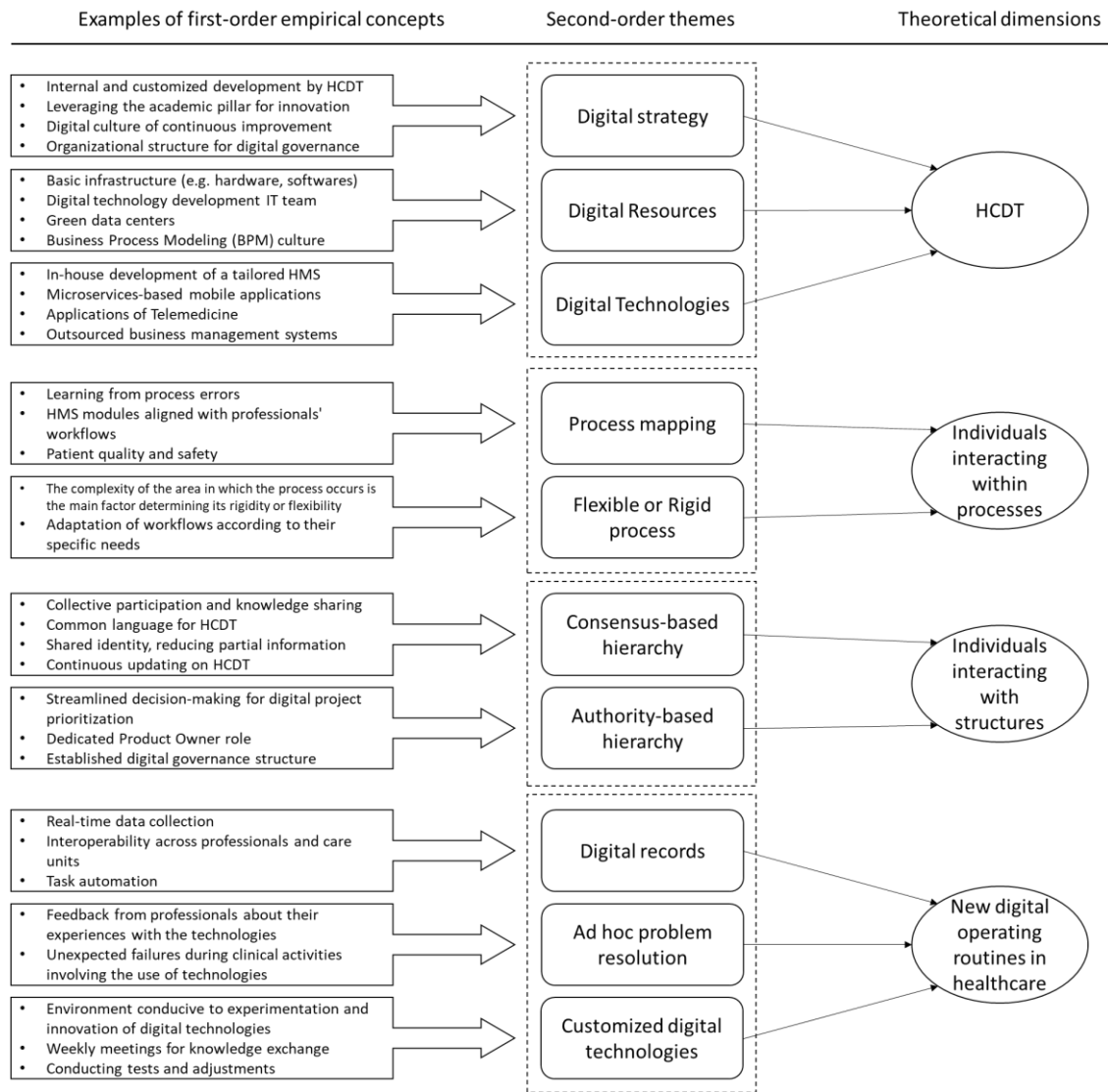


Figure 13. Coding process and data structure

4.4 RESULTS

Our results are presented based on the logic of the theoretical model put forward in Figure 10. From a sociotechnical perspective, HCDT initially requires the structuring of organizational aspects, including the definition of digital strategies and the identification of the necessary resources to operationalize parts of these strategies and support the digitalization process. Next, it is essential to determine which digital technologies will be implemented and how they will be used. In this study, we focus primarily on the HMS, the core technology developed internally by the hospital itself, which comprises different modules for specific departments but also allows integration with other digital technologies.

As argued previously, the organizational and technological aspect build the frame for the social aspect, being it people that manifest in a structure and that execute processes. The social aspect builds the link between structure and process, which main requirement is alignment. The presentation of results should be read with this contextualization in mind, building Section 4.4.1. and 4.4.2 the backdrop for Section 4.4.3, where we show how organizational and technical aspects shape the interactions of the microfoundations of operating routines, thus addressing the social aspect, and Section 4.4, where the role of these microfoundations in promoting continuous improvements is discussed.

4.4.1. HCDT as a sociotechnical system – organizational aspects

Digital strategy

Regarding the digital strategy dimension, the hospital established a set of key objectives and strategies to address HCDT. The main objectives include:

- Patient-centered care, aimed at maximizing the value delivered to patients by focusing on quality, safety, and efficiency. This also involves delivering care that is coordinated across the continuum of care pathways.
- Sustainability, which refers to expanding environmental and social sustainability initiatives, strengthening governance practices to achieve institutional goals, and fostering relationships to attract funding and resources.
- Optimization of resources and spaces, which involves implementing the hospital's master plan, including the mapping of productive capacities, adopting best practices for continuous improvement of internal processes, and promoting data science, its integration, and secure accessibility to support the hospital's operations.

Based on these objectives, the hospital established digital strategies to realize them. First, *internal and customized development of HCDT*, which means that digital technologies are developed in-house, leveraging available resources and encouraging the active participation of professionals. Second, *Leveraging the academic pillar for innovation*, which means that the hospital utilized its intrinsic characteristic as a teaching and research institution to develop its own digital technologies and/or other technological innovations, including its customized HMS, designed for both clinical care and professional workflows. Third, *Creation of an external collaborative community*, which means that the hospital established a collaborative network with external partners to expand the development of new HMS modules and

applications. Simultaneously, it also seeks to expand the HMS services beyond the hospital by making it replicable for other interested institutions, offering it as an HMS as a Service (SaaS). This creates a collaborative ecosystem of actors for both digital technology development and implementation. Fourth, *Digital culture of continuous improvement*, which includes ongoing interaction with engaged professionals through regular feedback rounds, ensuring the evolution of applications and optimization of user experience. Finally, *establishing an organizational structure for digital governance*, which refers to the creation of an organizational structure dedicated to digital governance that enables the ongoing management of initiatives focused on digitizing and innovating hospital operations, as well as the evaluation of results, goals, and indicators (e.g., patient-centered care, sustainability, and optimization of resources and space) related to HCDT projects. These digital strategies were explicitly mentioned by the coordinator of the Information and Communication Technology Management Coordination (R03) and reinforced by the coordinator of the Digital Health Unit (R01). The secondary data highlight these digital strategies (e.g., in the section “Definition of strategic pillars and indicators” [Information and Communication Technology Strategic Plan 2021]). It is worth noting that the hospital’s documents/reports related to digital transformation are operationalized and updated every four years.

According to interviewees R01, R02, and R03, which have leader roles, the main objective of these strategic actions is to increase process efficiency, collect and integrate data from different areas, strengthen communication between internal departments and external partners, improve the quality of medical services, optimize the tasks performed by professionals, and enhance patient interaction using digital technologies. Additionally, by adopting a customized HCDT model for the development of digital technologies and the HMS, the hospital aims to integrate all internal and external areas, centralizing patient clinical procedures, professional routines, and administrative activities. This approach also enables the incorporation of new digital technologies into the system, overcoming limitations often found in off-the-shelf HMS solutions. Another relevant factor is the independence from vendors and public procurement processes, which can require significant financial investment and time, making HCDT more expensive.

According to interviewee R03: “*The strategy of developing a customized HMS improves our internal operations by meeting the needs of both our professionals and patients, while also leveraging the pillars of education and research, as well as the resources available within the*

hospital.” He further adds: “Offering the HMS as a service to other external institutions opens several opportunities for the hospital. Becoming a software provider can serve as a source of revenue—not necessarily financial, but by reducing the IT team’s workload in developing new digital applications. Moreover, this strategy may attract more collaborators for the joint development of new solutions, driving the digitalization of other institutions.”

Interviewee R02 emphasized the importance of the strategy to establish a digital governance committee to manage HCDT within the hospital. According to him: *“This organizational structure includes professionals from different levels—operational (e.g., representatives from various clinical departments and the IT team), managerial, and strategic—who take part in decision-making regarding investments in technology projects, the definition of strategic directions, and the planning of digital transformation initiatives over time.”* He also highlighted: *“The organizational structure for HCDT enables the monitoring and growth of the HMS, the engagement of the HMS community, and the coordination of what each partner contributes related to the HMS.”* R02 further stressed the importance of a digitalization process that goes beyond just the technological aspect, also considering the hospital’s organizational structure, which is inherently geared toward education and research, as well as the human factor, including both patients and healthcare professionals.

Digital resources

Several digital resources were integrated preceding the digitalization process to ensure both the deployment and functioning of the HMS, as well as the incorporation of other digital technologies. It is important to note that to manage the digital resources of HCDT, the hospital adopted a hybrid governance model. According to interviewee R03, this model combines the internal development of resources with the acquisition of external resources through its collaborative partner network. The adoption of this model is facilitated by the hospital being a public institution focused on education and research, which allows for leveraging existing resources while also accessing external ones through R&D and innovation partnerships, or even through bidding processes with suppliers.

Among the resources, the hospital initially established the basic infrastructure. It developed a corporate Wi-Fi network and installed antennas to ensure device connectivity. More specifically, firewalls and routers, network points, Wi-Fi access points, telecom systems, and auditing tools were implemented across the hospital. Additionally, a technological environment was set up, including various hardware, such as workstations, computers and

microcomputers, printers, televisions, laptops, tablets, and QR code scanners and multiparameter monitors. These resources are essential for the operation of HMS modules, enabling real-time access and recording of patients' historical data and vital signs. They are used, for instance, during bedside checks by the nursing team and by physicians who need to record notes, patient progress, and observations within outpatient clinics and ICUs. The importance of these resources was emphasized by interviewees R11 and R12 and further supported by R09 and R10, who are involved in processes related to nursing, patient safety, and the quality of care provided. To validate and confirm these statements, the secondary data highlight the use of these digital resources (e.g., as presented in the "Hospital IT Ecosystem" section [Hospital Institutional Web Page 2024]).

The hospital also built green data centers (with high availability and energy efficiency), which mark a significant milestone because they enable secure data backups and the implementation of a robust information security infrastructure while contributing to its sustainability goals. However, it was necessary to adopt additional security measures, such as the development of "ethical hackers" to safeguard sensitive data. The hospital in this study also has 400 servers, various databases (Oracle, PostgreSQL, Microsoft SQL Server / MySQL), 0.5 PB of storage capacity, and a certified secure vault room. These resources allow for historical data queries, report generation, and indicator assessment, in addition to supporting the development of digital applications like dashboards and big data analytics for clinical and administrative decision-making.

Nurse R09 emphasizes the importance of ensuring the availability of digital resources before introducing new digital technologies: *“Digitalization is not just about software. A system cannot function without supporting hardware. To implement HCDDT in nursing processes, it was also necessary to invest in infrastructure, such as Wi-Fi networks, to enable system use at the bedside, which therefore required investments in hardware.”* She adds: *“There’s no point in having software without the necessary hardware, and this needs to be made clear when presenting a project. These items must be acquired in a planned and gradual way, as it’s not feasible to purchase them all at once.”*

In terms of human resources, the hospital has a large IT team composed of 130 professionals with diverse backgrounds, including artificial intelligence, computer science, information systems, and digital transformation according to interviewee R35. More specifically, the team is structured into coordination (9), technology (43), support and relationship (35), business (13), and third-party contractors (30). These specialists are allocated to different digital technology development and implementation projects, working on software

architecture, big data analysis, and programming, as well as the enhancement of HMS modules (presented in the "Hospital IT Ecosystem" section [HMS Institutional Web Page 2024]). Individual team constitutions consider the type of project, as well as the skills, competencies, experience, familiarity, and availability of each professional. Within digital technology projects, these teams play a crucial role in capturing insights from frontline professionals to ensure that system updates and operations meet the needs of both professionals and patients. Additionally, they are responsible for managing implementation, collecting feedback, and continuously maintaining the systems, ensuring that the HMS modules are consistently improved to improve hospital processes.

Finally, internally generated knowledge and information by the hospital's own IT team—as well as that acquired through external partners such as consulting firms, universities, and military entities, were identified as strategic assets for the development of HCDDT. These resources were considered by hospital management to be essential for advancing digital healthcare transformation. The hospital established, for example, a strong culture of Business Process Modeling (BPM), which became a key element in ensuring the success of HCDDT initiatives. Professionals are responsible for mapping processes and creating detailed process descriptions, known as "descriptive memorials". The importance of BPM in the context of HCDDT will be discussed in more detail in later sections.

4.4.2. HCDDT as a sociotechnical system – technical aspects

Digital technologies

According to interviewee R35, the HMS is the main technology adopted, functioning as a monolithic system. The HMS integrates the functionalities of an Electronic Medical Record (EMR), Electronic Health Record (EHR), Hospital Information System (HIS), and Enterprise Resource Planning (ERP), consolidating everything into a single platform. This integration is achieved through the various modules of the HMS, which include patient records, hospitalization, outpatient services, emergency, surgery, therapeutic sessions, exams, supplies, finance/accounting, hospital management, and clinical research. In this way, the hospital can promote vertical integration across different internal areas and horizontal integration with external partners. Additionally, because it is highly customized according to the hospital's needs, the HMS modules, as well as the implementation of digital technologies, consider the perceptions and experiences of professionals to enhance workflows through digitalization. This information was identified in the "HMS presentation brochures" available on the institutional

HMS website, where the integrated modules that make up the system are presented, along with their descriptions and objectives.

In addition to the development of the HMS, the interviews revealed that professionals use other digital technologies, some of which can be integrated into the HMS modules, while others operate independently. Although the hospital develops its own digital technologies, for some specific applications that cannot be developed internally, the hospital acquires them from external partners. Technologies mentioned in the interviews and their respective applications are summarized in Table 11.

Table 11. Digital technologies used by the hospital as mentioned in the interviews

Technology	Application
IoT and biomedical sensors and RFID, along with QR code and barcode identifiers	Used for tracking materials, medications, and patients in real time
Cloud computing with Oracle (private cloud) and Google (backoffice cloud)	Hospital data storage
Mobile microservices applications	Systems for scheduling, and accessing surgery and exam information
Big data-based applications	Solutions for patient risk classification and demand forecasting to support decision-making
Telemedicine	Used for remote consultations
Dashboards using tools like Microsoft Power BI	Real-time monitoring of process, service, and patient indicators
Artificial Intelligence and Generative AI (e.g., ChatGPT, OpenEvidence)	Support for professionals' routine activities (e.g., for disease prediction and other decision-making processes)
Augmented and Virtual Reality technologies	Used for clinical procedure simulations for healthcare professionals
Da Vinci robots	Used for surgical procedures
Other management systems for the hospital's business areas provided by third parties	
Technology	Application
Picture Archiving and Communication System (PACS)	Used for storing and managing medical images
Radiology Information System (RIS)	Used for workflow management in radiology departments
Laboratory Information System (LIS)	Used for controlling and managing laboratory processes, such as clinical tests and biomedical analyses
Platform for managing clinical and epidemiological data	Mainly used for analyzing and monitoring hospital and ICU indicators
Specialized management system for nephrology and dialysis services	Supports the management of patients with chronic kidney disease
Hospital transfer system	Used to organize hospital regulatory flow and patient transfers

Regarding the results achieved using HMS modules and integrated digital technologies, secondary data — specifically the HMS performance folders provided in 2023 and available on the hospital's institutional website — indicate significant progress. In the HMS supply module, for example, with the adoption of RFID technology, the time required to complete inventory was reduced from 720 minutes to just 10 minutes. Inventory accuracy increased from 84.5% to 98.45%. Restocking time dropped from 225 minutes to 21 minutes. Furthermore, the discrepancy rate about the total number of items withdrawn, previously at -30%, was reduced to 0%. The same secondary source also showed a substantial reduction in paper usage in

operations and storage: from 8,733 kg in 2018 to 2,675 kg in 2023 — demonstrating not only significant economic savings for the hospital but also sustainable gains.

4.4.3. HCDT as a socio-technical system – social aspects

Individuals interacting within processes

The starting point for changes in interactions between healthcare professionals and processes in the hospital was establishing a digital culture focused on the individual, in which the healthcare professional was seen as a key agent capable of driving change. This is because professionals possess knowledge of the processes, due to their daily interactions with patients and colleagues, and the ability to identify digital resources and propose technological solutions best suited to transform the processes they engage with. From this perspective, the hospital views healthcare professionals not merely as users of technology, but as active participants in the development and implementation of digital technologies, working alongside the IT team and hospital managers to shift from analog to digital processes. As a result, it becomes possible to modify interactions between professionals and processes, making them digitally tailored to the needs and expectations of each department, while also promoting patient well-being.

The interviewees strongly emphasized this aspect; for example, the IT coordinator (R03) commented: *“The relationship that the IT team has with professionals from the business areas in the implementation of HCDT is a key differentiator for the hospital. Often, we suggest solutions, but when we arrive in a specific department, the physician or nurse in charge has already developed the ideas and identified the necessary resources, using their knowledge and understanding of the workflow to propose improvements to the process and optimize daily routines.”* The nephrologist (R24) also reinforced this finding: *“Nephrology is a highly complex specialty with several branches. Therefore, when structuring a digital project for the area, it is essential to understand how it previously functioned. In general, only the professionals who work directly in the field can visualize the processes and their interactions, as well as anticipate challenges, thus avoiding future issues related to poor implementation of digital technologies or the lack of necessary support resources.”*

In this way, the introduction of HCDT generated a self-reinforcing cycle, in which professionals (e.g., IT professionals in the context of HMS’s module development, or doctors and nurses in daily tasks) learned from process-related mistakes and used this knowledge to continuously propose improvements. This includes developing new HMS modules and/or

integrating additional digital technologies, as well as building the foundational infrastructure for the processes carried out by professionals—generating knowledge through trial and error.

More specifically, the interviewees emphasized that changes in interactions between healthcare professionals and processes with the adoption of HCDDT begin with process mapping. They highlighted that accurately mapping healthcare professionals' processes and tasks is essential to identify gaps in interactions with patients and other professionals. Furthermore, it supports hospital managers in decision-making regarding investments and resource allocation related to HCDDT (e.g., IT teams, hardware acquisition, or outsourcing to an external partner) to meet these needs. Process mapping also helps determine whether processes are rigid or flexible, as they may follow different paths of change. According to interviewee R33, when properly conducted, process mapping facilitates the effective adoption of digital technologies, benefiting both the interactions between professionals and processes and the hospital's overall performance, making it more resilient and sustainable. Interviewees R33 and R20 emphasized that mapping processes alongside healthcare professionals strengthens their commitment to developing new digital solutions, such as HMS modules, dashboard applications for data visualization, electronic panels, exam scheduling, and telemedicine, and fosters the development of resilient skills among professionals, such as monitoring, anticipating, responding, and learning. For example, by participating in governance committees and working groups or by taking on Project Owner (PO) roles in digital projects, healthcare professionals actively contribute to process mapping and sharing of digital technology suggestions for discussion with other members, drawing on their own and their colleagues' prior experiences. This generates a sense of professional appreciation and helps expand DT across processes.

Operationally, professionals help reduce workload in processes, improving the use of time and space, and eliminate redundant procedures. Additionally, they enhance clinical decision-making and implement more structured information-recording processes, requiring less cognitive effort. On the IT side, this enables the development of HMS module architectures that better reflect actual workflows, with more intuitive interfaces, fewer clicks to perform clinical procedures, and easier access to historical data. Nurse and manager R09 provided an example of how healthcare professionals participate in changing process interactions: *"My team and I participated in the development of a new application within the risk assessment and prediction scales module in the HMS. We mapped all the processes and, in partnership with the IT team, developed this application within the HMS. Nursing diagnoses related to pressure injury and fall risks became more accurate with the updated HMS modules. In other words,*

patients at risk received the appropriate nursing diagnosis." She added: *"This customized digitalization supports healthcare professionals, especially nurses, in clinical reasoning. That's because when classifying the patient and applying the risk prediction scale, the system provides a risk score at the end, along with the level of risk—low, moderate, or high—which supports clinical decision-making."*

At the same time did interviewees point out that the absence or incomplete mapping of processes (especially based on experiences from previous technological projects), as well as not considering the participation of professionals and their interactions with the processes, can lead to failures in the development of new modules or updates to the HMS, or hinder the adoption of new digital technologies. This impacts the hospital, including the IT team, professionals, and patients. From the IT perspective, this includes rework for the team, errors, and pop-ups during the usability of the HMS, lack of information in the modules, unforeseen costs, waste of digital resources and labor, insecurity among IT professionals, processing time of the HMS, and delays in other projects. From the healthcare professional's perspective, some may bypass steps and clicks in the HMS modules, as they are not aligned with their workflow, or even avoid using the technologies due to issues such as the system's inadequate architecture, lack of process standardization, and unintuitive design. This can result in a return to manual paper-based processes, performing tasks outside the HMS, and difficulty in adapting, especially for older professionals, whether due to low digital literacy or age.

Additionally, such challenges can generate frustration, and technostress, and reinforce conservative barriers against the use of technologies, causing professionals to spend more time learning to operate the system than focusing on clinical care, which negatively impacts both their performance and that of the patient and the hospital. From the patient's perspective, the interviewees mentioned that the lack of proper process mapping and the absence of professionals' participation in the implementation of digital technologies can compromise the quality and safety of care, especially in high-complexity sectors such as nephrology, transplants, and chemotherapy units. For example, in interviews with nephrology professionals, it was reported that when a patient needs to move between departments — especially in urgent cases or to undergo dialysis — clinical procedures and decisions must be recorded in the HMS modules for documentation and future reference, as well as to ensure complete patient mapping. However, these processes are known only to the physicians and nurses in that area, who understand what needs to be done, what must be recorded, and how to integrate these steps into the system. Without the involvement of these professionals, as highlighted in the interview,

there is a risk of the patient moving between sectors without clarity on what has already been done, which can compromise future medical decisions, increase waiting times, reduce patient safety, and negatively impact the efficiency of care. In interview I07, interviewees R06, R07, and R08 highlighted that the positive or negative impacts of changes in interactions between professionals and processes, driven by elements of HCDDT, are directly influenced by proper process mapping and the incorporation of professionals' suggestions.

During the interviews, it was also observed that it is common for healthcare professionals to provide feedback on their interactions with processes after the implementation of digital technologies. This occurs through meetings with directors, team events, or even impromptu conversations in the hospital corridors, phone calls, and message exchanges. This practice reinforces the establishment of new digital strategies and identifies the necessary digital resources for their realization. With each update, professionals learn the new digital process and tend to suggest new improvement ideas, allowing for the generation of new digital operational routines thereby creating a cycle of continuous improvement for professionals, processes, and especially patients. These activities help build trust among healthcare professionals and facilitate the sharing of ideas for the continuous improvement of HCDDT in the hospital.

Finally, depending on the type of process, rigid or flexible, the interactions between professionals and processes are shaped by HCDDT in different ways. The interviews revealed that the complexity of the area where the process occurs is the main determining factor for its rigidity or flexibility. Areas such as administration, pharmacy, outpatient services, medical appointments and exams, for example, were classified as more rigid because they have clear standards, which makes it easier for professionals to implement and handle technologies with less effort, replacing analog and manual processes with digital solutions, and without requiring extensive training. It is in this type of process that improvements are more visibly achieved. A practical example occurred in the outpatient department, where the bedside check process, previously performed with clipboards and papers that required more time and cognitive effort from the professional to input into the database, was moved to tablets. This change allowed for the immediate recording of information in a personalized HMS module, developed with the contribution of professionals, enabling real-time access to patient data. These changes were particularly evident in interviews with the unit's nurses and were later validated during a technical visit to the hospital, where it was possible to observe, in practice, the interaction

between doctors, nurses, and the technologies implemented in the processes, as observed in the following observational data:

Observation data from an inpatient ward [08/06/24 – 11:10 AM]: While walking toward the ICU rooms, the researcher and the Digital Health Unit coordinator discussed changes in the routines and tasks of nurses and physicians. At that moment, the coordinator pointed to a section of an inpatient ward where four patients were receiving medication and highlighted the interactions of two nurses involved in the process. She explained: *“At this stage, At this stage, the nurses have already completed bedside electronic verification. They scanned the patient's ID wristband and the medication barcode using tablets, enabling electronic verification. They also reviewed the multiparameter monitors displaying the patients' vital information. The nurses check which medications have already been administered and update the clinical status of each patient in the system. Previously, they had to use paper clipboards, which were left hanging at the bedside for doctors to later check what had been done for each patient. This change has accelerated the patient assessment process for healthcare professionals and reduced the number of missing or incomplete records.”*

Another example of changes in interactions between professionals and rigid processes occurred in medical consultation services. Initially conducted exclusively in person, consultations were later offered virtually after the development of an in-house teleconsultation platform. This allowed doctors to remotely attend to patients for diagnoses and treatment follow-ups. In 2020, the number of teleconsultations surpassed 52,000 appointments (as presented in the "General Overview of the HMS" brochure [Hospital Institutional Web Page 2022]).

Processes considered complex within the hospital were classified as flexible, as they do not follow complete standardization but rather involve task customization that can vary according to the patient's stage, specific needs, and the professionals' individual experiences. This variability hinders both the adoption of digital technologies and the development of new HMS modules, as mentioned by interviewee R04 in reference to a module for surgery scheduling and management. Changes require greater commitment and resilience from professionals, especially during the development of HMS modules and training on the use of technologies, such as in the case of using Da Vinci robots in surgical procedures. Additionally, errors in this context can represent significant risks to patient safety. Examples of flexible and complex processes include organ transplants, heart surgeries, chemotherapy treatments, and nephrology procedures.

It was also observed that, in rigid processes, professionals tend to maintain the same process, using technologies mainly to enhance it. When flexible, the interaction between healthcare professionals and the introduction of HCDDT indicates that professionals often receive roles and responsibilities to decide what digital technologies to adopt, without collective participation, together with technological support to make decisions in their routines. Flexible

processes centralized the decision-making on technology adoption in a few individuals, while rigid processes encouraged collaborative adaptations to the HMS, allowing professionals to contribute their experience to drive transformations.

Individuals interacting with structures

At the strategic and managerial level, an organizational structure was established composed of members from all departments responsible for decision-making regarding HCDT (as described in section 3.1). However, at the operational level, interviews revealed that interactions between individuals and the type of hierarchical structure vary depending on the hospital department, especially based on the number of professionals involved in each department or process, as explained by the head of the hemodialysis unit, nurse R23, in interview I18: *"Depending on the department unit, when many people are involved, a PO is designated to lead the digital transformation project and represent them, concentrating greater decision-making authority on that individual. In such cases, involving everyone individually would make the process unproductive and disorganized. On the other hand, when fewer people are involved, actions are based on consensus, where everyone works together with the digital governance bodies as well as the IT team for the development and implementation of digital solutions."*

When the hierarchical structure is organized by consensus, technicians, nurses, nutritionists, physical therapists, pharmacists, and physicians, are encouraged to share ideas in operational teams to develop new HMS modules and suggest digital technologies that enhance their activities. In this type of hierarchy, professionals work in an engaged and collaborative manner, using a common language focused on improving daily tasks. This kind of interaction allowed for considering collective participation and knowledge when digitalizing routines, reducing potential tensions between professionals and technologies after implementation. More specifically, professionals actively contribute to mapping current processes, identifying problems and limitations, suggesting technological solutions, improving the architecture of HMS modules, and continuously providing feedback. A practical example was shared by pharmacist R22, medication and logistics manager: *"When a technological improvement proposal is defined by the pharmacy leadership, we first listen to the professionals' concerns and suggestions. Next, we conduct preliminary tests. However, before the technological implementation, we inform the team. After adoption, if any issues arise, we encourage users to*

report any unforeseen impacts, ensuring personalized adjustments and solutions developed by them." This makes the construction of HMS modules truly multidisciplinary, as further highlighted by the nursing manager R09. At the hospital, this hierarchical structure enabled the development of a shared identity, reducing partial information among professionals and establishing cooperation mechanisms that influence the pursuit of digital solutions.

According to the interviewees, professionals are constantly seeking to update themselves on HCDT within the consensus-based hierarchy, diving into various sources of knowledge to apply and share in collaborative DT projects. The main sources mentioned include events, congresses, lectures on digital transformation, reading articles in the medical field, LinkedIn, interactions with healthcare professional groups, previous experiences with digital technologies outside the hospital, benchmarking with other institutions, and short-term courses. Professionals who contribute to technological projects are encouraged and recognized through participation certificates, internal recognition within the hospital, and opportunities to take part in new projects due to the knowledge they have shared or innovations they have generated. However, there are no financial rewards, as this is a public teaching and research institution with a limited budget. This shows that professionals are motivated by a collaborative digital culture, driven by the desire to improve work routines. An interesting point highlighted was that when professionals actively participate in the development of DT, more improvement ideas tend to emerge. However, one challenge noted is that not all suggestions are considered, nor are the digital solutions always able to fully meet everyone's needs.

Interactions within the authority-based hierarchical structure occur in two main ways. First, it is used to centralize decision-making regarding the prioritization of digital technology adoption projects, as well as to establish the strategic planning of DT and investment in digital resources. Technicians, nurses, and doctors request technological improvements for their processes directly through their respective departments, such as the Nursing or Medical Directorate. These directorates assess the feasibility of the requests based on factors such as process maturity and proper process mapping. If approved, the requests are forwarded to the Digital Health Core, which verifies whether they align with the broader DT scope. If validated, they are then submitted to the Hospital's Digital Governance Committee, which makes the final decision by considering investment, required resources, available infrastructure, and strategic alignment. This committee also monitors implementation and seeks to understand the outcomes of interactions between professionals and digital technologies within processes. Thus, although professionals possess the necessary knowledge to propose improvements to their routines, the

execution of these proposals depends on the approval by these governing bodies, as explained by interviewee R02: *“In the HCDT, the Digital Governance Committee is the highest authority responsible for deciding which digital technologies will be implemented in the hospital, with a focus on project prioritization. Its main role is to align initiatives with the hospital’s capacity. After approval, the committee also monitors the allocation of digital and human resources, ensuring that each project is carried out within the planned timeframe, budget, and scope.”*

Second, POs are appointed to manage the interaction between unit staff and HCDT in units with many professionals, as highlighted by interviewees from Interview I17. In this hierarchical structure, although POs gather insights, knowledge, and suggestions from frontline professionals, the direct influence of these professionals on HCDT decisions is more limited. The PO centralizes the communication of process-related issues. They define guidelines and dedicate time to meeting with teams to deeply understand professionals’ interactions within processes, as well as to identify which HMS module updates and digital technology applications should be considered by the board and later forwarded to the hospital’s digital health committee. Once a technological project or new HMS module is approved, the PO takes the lead in interacting with the IT team for development and implementation. They serve as the feedback channel and absorb knowledge regarding digital workflow changes. Due to the authority delegated to them, POs are also responsible for passing on updates to professionals through training and capacity-building sessions. A practical example of this structure was shared by nurse R09: *“Currently, we are developing the scheduling and verification module for chemotherapy prescriptions in the HMS. We have a dedicated project team, including a PO who serves as the liaison between the care areas and the IT team. In the case of chemotherapy, professionals from the involved areas validate the PO’s decisions before the final approval by the digital governance committee.”*

Another role of the PO in the interactions between healthcare professionals and the authority-based hierarchy was highlighted by R05, pharmacist and head of the pharmacy department: *“The pharmacy team is very large, with over 200 professionals. So, when there are HCDT initiatives, we divide into four large groups to ensure collaboration and access for everyone. However, there needs to be a PO to lead the implementation of the technologies, gather information from the professionals, and coordinate the dissemination and training after the implementation.”*

4.4.4. New digital operating routines in healthcare

Our findings suggest that, from a sociotechnical system perspective, the introduction of HCDDT shaped the microfoundations of operating routines, enabling the creation of digital healthcare operating routines across various areas and units of the hospital under study, both clinical and administrative. The interactions between professionals, structures, and processes in HCDDT went beyond the simple transition from manual/analog tasks to digital ones or the use of technologies for clinical treatments. They incorporated the social aspect of transformation, through the formation of multifunctional teams focused on the customized development of digital technologies, as well as their implementation journey adapted to the professional, rather than the professional adapting to the technology. New routines include, for example, the replacement of paper or manually made records with real-time digital records, ad hoc problem-solving, and the active participation of professionals in the development of customized digital technologies. These new digital healthcare routines were identified through the analysis of interview excerpts from participants across different areas of the hospital. Additionally, they were confirmed during participant observation, which allowed researchers to witness professionals carrying out these routines in practice.

The interaction among professionals was particularly intense in highly rigid processes and hierarchical structures based on consensus, where teams were assembled to solve well-defined problems. Units such as the Internal Regulation Center, Emergency Department, Outpatient Clinics, Pharmacy, Laboratory, Pathology, and Cardiology reported consistent adoption of the motto "learning from mistakes." This concept was particularly well incorporated in services such as bedside checks, exam and historical consultation requests, and patient diagnosis, areas where information was previously recorded manually on paper. With the development and implementation of new customized modules and associated digital technologies (such as QR Code readers, paging guns, tablets, and electronic wristbands), both in consensus-based and authority-based structures, a new operational routine emerged: *digital recording and tracking of patient and professional information*. Data collection began to occur in real time, with information being automatically or manually entered by professionals into the HMS, centralizing data such as consultation history, hospitalizations, and medications administered or restricted, even in cases where the patient cannot provide this information directly. This new operating routine enabled the mapping of the patient, regardless of the unit they were in, promoting communication, integration, and information exchange between professionals from different areas (interoperability). Additionally, it enabled the generation of

performance indicators, task automation, and quick access to the patient's clinical history. In a way, professionals began to gain new power through their interaction with technological change. In more flexible processes, they were able to quickly adapt to new tools to avoid delays in activities and later reported suggestions for improvement based on their practical experiences. The digital recording routine also contributed to strengthening a social structure driven by a consensus-based hierarchy and reorganized processes.

The *ad hoc problem-solving* routine emerged from observing processes and collecting feedback (Franke et al., 2024) from professionals about their experiences with digital technologies, especially in contexts with authority-based hierarchical structures and more rigid processes. In these cases, the responsible professionals — POs of the projects — periodically meet with care teams to collect reports on process failures, difficulties interacting with technologies, and functional deficiencies in digital modules, and to promote the sharing of knowledge and improvement suggestions, such as the creation of new HMS modules. This information is then forwarded to the respective departments and the hospital's HCDT management, to eliminate the identified problems. In practice, this routine manifested itself, for example, when the digital prescription module exhibited unexpected failures during the clinical activities of doctors or nurses. In these cases, the professional seeks a temporary solution to avoid interrupting care while also alerting the IT team for immediate correction. Another example mentioned refers to the excessive number of clicks required on HMS screen objects, which caused professionals to become lost and stressed during processes, and, in some cases, revert to using manual methods and paper records. Based on this, professionals made suggestions for improvements. With the updates, it was possible to reduce the number of clicks needed during patient care in the emergency department by over 45%, optimizing workflow and increasing the efficiency of the care process. This routine contributes to the development of resilient organizational skills, such as monitoring, anticipating, responding, and learning, reinforcing the adaptive capacity of professionals in the face of technological uncertainties.

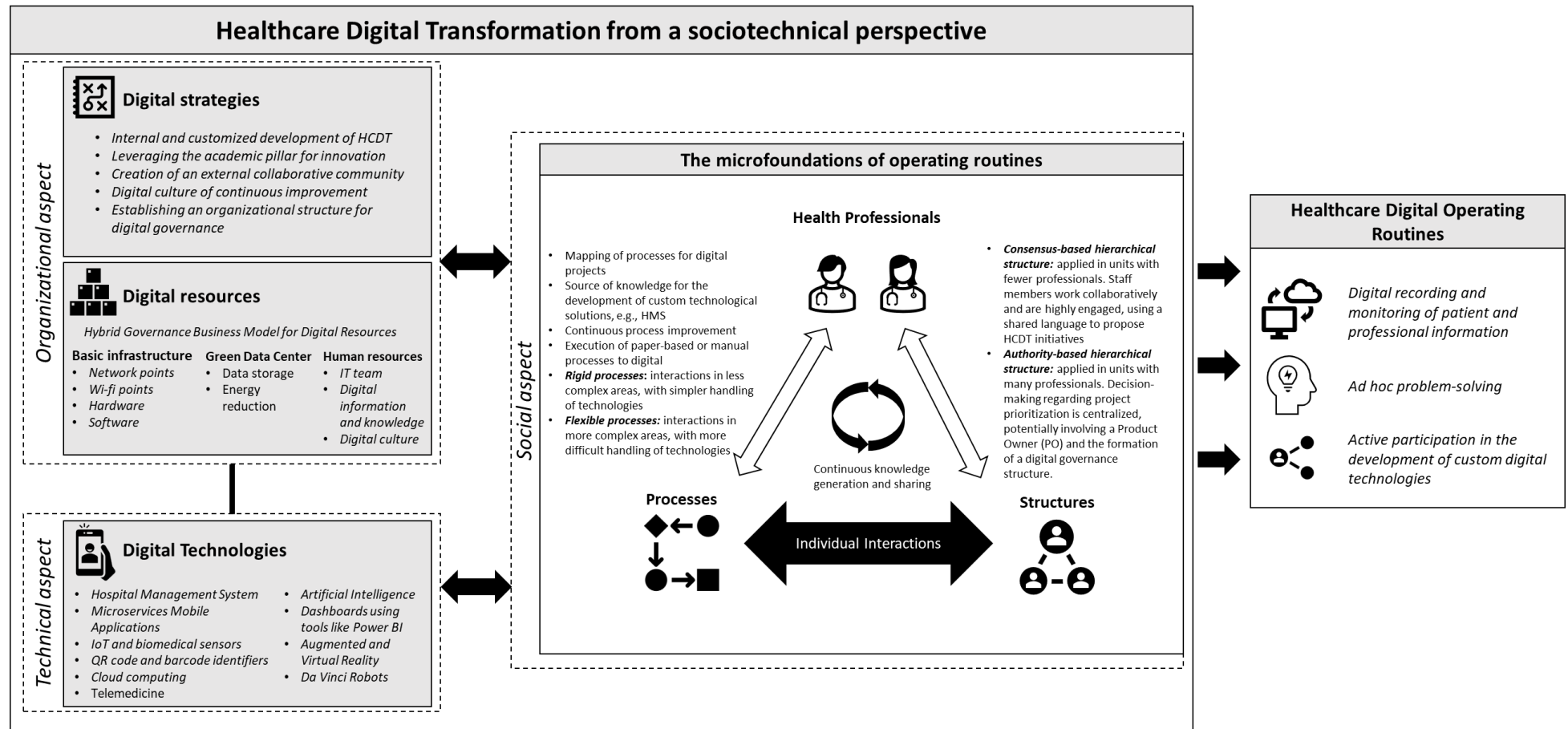
The accumulation of experience and deliberate learning by professionals led to the emergence of *active participation in the development of customized digital technologies*. This routine was especially driven by the nature of the hospital, an institution focused on care, teaching, and research, with digital strategies and resources aimed at improving operations and clinical care. This provides an environment conducive to experimentation and innovation. With the introduction of HCDT in units with fewer professionals in the hospital, such as Hematology, where activities as assessing hospitalized patients and requesting exams take place in a

predominantly consensus-based structure, healthcare professionals began to actively participate in the development of the transplant and chemotherapy module in the HMS in addition to dedicating themselves to care and administrative processes. Participation includes conducting meetings, tests, and adjustments to ensure the system's usability and alignment with the specific needs of the unit. The active involvement of professionals in HCDDT also helped mitigate common tensions in technological adoption, particularly those related to sociotechnical factors such as age, digital literacy, years of service, and the fast-paced evolution of HMS and other digital technologies.

4.5 DISCUSSIONS

Our results offer a new perspective on the study of DT in healthcare systems. They challenge existing literature by demonstrating that HCDDT should be understood as a sociotechnical system. Previous studies simplify the implementation of HCDDT to a technocentric view (e.g., Sharma et al., 2016; Wani & Malhotra, 2018; Guha & Kumar, 2018; Mukherjee & Sinha, 2020; Bavafa et al., 2021; Tushe et al., 2025), while organizational and social aspects, which involve the interactions of the microfoundations of operating routines, have not been addressed in depth. Although these microfoundational interactions have been analyzed in manufacturing contexts (Roscoe et al., 2019), our study is pioneering in the field of healthcare operations management. Figure 14 presents an empirical model of the interaction between individuals, structures, and processes shaped by HCDDT as a socio-technical system.

Figure 14. Empirical model of how HCDT as a sociotechnical system shapes the interactions among the microfoundations of operating routines



Our findings indicate that contrary to what is assumed by the model presented in Figure 10 — which was built based on the literature — the management of HCDT in healthcare systems does not necessarily occur in a strictly “top-down” manner. In other words, it does not begin by addressing organizational and technical aspects first, followed by social ones. We identified that organizational, technical, and social aspects interact differently in this context. The empirical model presented in Figure 14 suggests that, when considering HCDT as a sociotechnical system, the process for digitalization should begin from the social aspect (carries greater weight in decision-making about HCDT), with frontline healthcare professionals who demand solutions to support and shape their work. Management (e.g., the digital governance committee), in turn, also requests tools that assist in monitoring professionals in their work environments and support decision-making processes. A third stakeholder is the actual patient, which may demand digital solutions to facilitate its treatment path.

The different stakeholders create what in the IoT and Smart Manufacturing literature is called the application layer. It is the actual use of the digital solution, and the upper layer in the three-layered structure of application, core service and resource management (e.g., Qu et al., 2016). Core service and resource management are reflected in organizational and technological aspect, respectively. Both need to be aligned. To realize this alignment, it is typically assumed in the IoT and Smart Manufacturing literature that a central planner creates an integrated solution. In larger systems, such that our case hospital, it emerges organically out of multiple requests for applications. In this sense, the alignment between organizational and technological aspects (as indicated by the line connecting the organizational and technical aspects) is shaped by social interactions, according to professionals' practical needs and interactions within hierarchical structures and processes. The feedback generated by individual interactions provides the basis for structuring this alignment, as indicated by the bidirectional arrows in the model in Figure 14 that integrate the social aspects with the organizational and technical ones. This cycle is continuous: the interactions among healthcare professionals, structures, and processes enable professionals to learn through practice, including patient feedback, generate and share knowledge, and make new improvement requests. Managers then receive this information and respond by structuring the organizational and technological aspects to once again support and enhance the microfoundations.

These findings reinforce emerging themes in the field of Technology and Operations Management, which emphasize that the implementation of digital technologies (such as the

HMS studied here) should be human-centered while also promoting sustainable practices in organizational processes (Olsen et al., 2020; Heim et al., 2021; Choi et al., 2022; Angelopoulos et al., 2023; Amaya & Holweg, 2024). According to the sociotechnical perspective, organizational, technical, and social aspects need to be aligned and treated in a balanced manner to enable the adoption of technologies (Cherns, 1978; Manz & Stewart, 1997). However, this is merely a technical premise. In practice, the results revealed that there is not necessarily a balance between these aspects, but rather a continuous dynamic of interactions among different actors. It is through these interactions that operating routines are enhanced and constructed, which in turn generate demands for specific strategies, resources, and technologies. Most importantly, we enhance the existing literature by showing that in the context of digitalization, there are three layers that determine the sociotechnical system. It is not a question of balance, but the balance has to follow the architecture provided by the digital solution.

Our results further revealed that when the social aspect does not serve as the foundation for aligning with the other components of HCDDT—and instead, a top-down management approach is adopted—tensions may emerge, such as technostress, rejection, or conservative resistance to the use of technologies by professionals (Kellogg, 2022; Amaya & Holweg, 2024), which directly impacts the success of HCDDT initiatives. Although the model presented in Figure 14 is not immune to these effects, we believe it can serve as a useful approach to support decision-making in contexts that involve interactions between digital technologies and human collaboration. These results align with Kellogg's (2022) study on experimentalist governance, in which the reconfiguration of professionals' roles within local units includes the autonomy to adapt digital technologies to their specific contexts, allowing both routines and the technology itself to evolve based on lessons learned. In other words, professionals no longer serve merely as operational resources for the hospital but instead take on developmental roles within HCDDT.

As shown in Figure 14, interactions within the social aspect, when HCDDT is introduced, enable the emergence of new digital operating routines, which is particularly significant in the context of healthcare operations. The findings suggest that due to the high concentration of knowledge among healthcare professionals — stemming from their academic background and ongoing training — and their interactions within organizational structures and processes, it is possible to develop routines that enhance operations and can effectively handle unexpected situations, which are common in healthcare environments. As observed, three new routines were developed: the routine of active participation in the development of customized digital

technologies, the routine of ad hoc problem-solving, and the routine of digital health record-keeping and monitoring of patient and staff information. Focusing especially on the first two routines, they are particularly important because, when a customized digital technology fails or encounters errors, the professionals involved in its development may be able to identify the root cause of the problem and notify those responsible for fixing it — or even implementing temporary solutions. However, the most critical aspect in such situations is not just resolving technological failure but ensuring the continuity of patient care despite the absence of technology. Healthcare professionals cannot interrupt their activities, as they are dealing with human lives. In other words, the care delivery process cannot stop. Therefore, by understanding their interactions within the processes, they can implement quick solutions that allow operations to continue even without technological support.

The key however, is to transition from ad-hoc solutions to new routines. The learning from failure. For this, it is not only a question of different sources of knowledge but also of different types. In fact, knowledge has to be seen as sociotechnical. There needs to be knowledge about organizational, technical and social aspects, and their interactions, to create a solution. It needs to be known, what technical solutions can be realized given existing resources, how these resources can be leveraged given the organizational structure, and how this will impact the different stakeholders. This knowledge rarely resides in one person. Our study showed that using simple tools such as Business Process Mapping to create a joint communication, and knowledge exchange, structure over problems and their context is key to success.

The central point lies in the interactions that occur within the social aspect and how they shape the entire surrounding environment. Zooming in on this aspect, it becomes evident that HCDDT fundamentally depends on the engagement of the business areas within healthcare systems, as it is in these areas that the multidisciplinary conception and construction of digitalization take place—driven by the professionals themselves. In this context, what makes one area easier to digitalize than another is mainly related to the number of professionals involved and the complexity of the processes. Areas with a larger number of individuals and more flexible and complex processes tend to face greater challenges in digitalization. One of the reasons for this is the lack of standardization. In such cases, professionals need to deepen their understanding of the processes, applying their experience and skills to generate and share knowledge. This knowledge becomes the foundation upon which management can shape the alignment between organizational and technical aspects of digitalization. On the other hand,

areas with fewer professionals and/or more rigid processes tend to be easier to digitalize, precisely because of the clarity and standardization provided by process mapping, as well as the ease of working collectively and reaching consensus. This leads to more direct insights into how strategies, resources, and digital technologies should be configured.

Finally, as identified in the results, when interactions among microfoundations are considered the primary source of knowledge for the development of digital technologies, it becomes essential to recognize that solutions can be designed within these micro-level units to bring patients closer to healthcare systems, rather than distancing them. Technologies such as telemedicine, AI-based decision support systems, and digital monitoring tools require patients to assume a more active role, ranging from self-monitoring and data sharing to performing minor procedures with remote support, as noted by Bendoly et al. (2025). In this context, key questions emerge from the microfoundations regarding whether digitalization may create new barriers or, alternatively, foster solutions that strengthen interactions between professionals and patients (Bendoly et al., 2025). As evidenced in this study, it is precisely at the level of microfoundations that a customized HCDT can be developed—one that adapts digital technologies to professionals' tasks and patients' needs, promoting integration rather than distancing.

4.6 CONCLUSIONS

Hospitals are complex sociotechnical systems that receive much research attention. A major focus of many hospitals is improving operational routines through digitalization. But how can this HCDT be realized? In response to our research question – How does HCDT, as a sociotechnical system, shape microfoundations and generate new digital operating routines in healthcare? we found that the social aspect, which occurs in the interactions among individuals, structures, and processes defines the alignment of organizational and technical aspects to adapt to the reality of individuals (health professionals, managers, patients).

From a theoretical perspective, we present an empirically grounded model (see Figure 14) that provides a foundation for exploring the social dimension through the generation and sharing of knowledge in the interactions among the microfoundations of operating routines, as well as organizational and technical aspects. First, our analysis challenges the dominant sociotechnical perspective in literature, which tends to view social, technical, and

organizational elements as integrated and balanced. In contrast, we found that the social aspect, especially when centered on interactions between individuals, can play a more significant role, particularly in HCDT, by influencing and shaping the other aspects.

This study contributes to both the sociotechnical view and the KBV by responding to the call to open the "black box" of sociotechnical dynamics in DT (Heim et al., 2021). We contribute by showing that the sociotechnical approach can be understood differently. The relationship among these aspects is not fixed: while organizational and technical components should be aligned, it is the social dimension, when integrated into the system, that provides the knowledge and informational foundation to guide that alignment and to continuously feedback into the process for ongoing improvement. This type of relationship between the aspects can, in turn, give rise to new operating routines, transforming operations and generating benefits for the organization. In the healthcare context, as observed in this study, this may be reflected, for example, in an increased number of patient visits and a reduction in paper usage. We conclude that studies on digital transformation in organizations should consider that social aspects may play a predominant role in decision-making. This is because it is through individual interactions that DT is operated, sustained, and adapted interactions that can even lead to the emergence of new, collaborative organizational structures.

From the perspective of the KBV, this study contributes by demonstrating the ability to generate new digital operational routines based on the tacit and experiential knowledge of healthcare professionals. Given their high level of educational attainment, these professionals show strong potential to develop digital competencies through workplace interaction identifying, developing, and implementing digital solutions that enhance existing routines and enable the creation of new ones. Such contributions emerge particularly in complex contexts, where new routines may arise independently of hierarchical structures or pre-established processes. This perspective also advances the KBV by highlighting that the knowledge and experience of these professionals constitute strategic organizational resources that can be leveraged to drive operational improvements. Furthermore, another important contribution is that the accumulation of experiences by healthcare professionals gained through their daily work activities, serves as the source that enables the generation and sharing of deep knowledge about processes—whether rigid or flexible—allowing for business process mapping. This mapping is the initial step to identify processes' flaws, problems, and waste, uncover improvement opportunities, and define technological solutions that transform manual or analog

tasks into digital ones using digital technologies. This issue is particularly relevant in healthcare, where many processes are not standardized and require greater attention due to the need to handle complex situations demanding quick and precise responses (e.g., diagnoses, treatments, and patient follow-ups). Therefore, the knowledge required for mapping is often limited to frontline professionals. Finally, we contribute by presenting a new knowledge framework in which solutions for changes in the activities and interactions of healthcare professionals within hierarchical structures and clinical and administrative care processes originate from the professionals themselves, enabling them to foster a continuous improvement cycle.

From a managerial perspective, our study provides managers with a guide for implementing HCDT as a sociotechnical system, emphasizing the social aspect as particularly significant and as a key driver in shaping the relationship between organizational and technical elements to build an effective structure for HCDT. That study can also support policymakers in developing guidelines aimed at advancing the digitalization of healthcare systems in their respective regions, leveraging available resources to promote HCDT in an internalized and customized manner.

A major limitation of our study is the adoption of a single case. This was, however, necessary to have access to the in-depth insights needed to answer our research question. But this also implies that results are not directly generalizable. Future research is needed to appropriately validate and contextualize our results. Our study was further limited to examining the elements preceding HCDT and how these elements shape the microfoundations, focusing on healthcare professionals, as well as the emergence of new digital operational routines. Future research is needed to determine how healthcare digital routines, customized from the knowledge and experiences of professionals, influence the standardization of processes in highly dynamic environments, such as healthcare. We believe this is a fertile area for future research on the important topic of the intersection between HCDT, sociotechnical aspects and microfoundations.

4.7 REFERENCES

- Abeliansky, A. L., Beulmann, M., & Prettnner, K. (2024). Are they coming for us? Industrial robots and the mental health of workers. *Research Policy*, 53(3), 104956. <https://doi.org/10.1016/j.respol.2024.104956>
- Amaya, J., & Holweg, M. (2024). Using algorithms to improve knowledge work. *Journal of Operations Management*, 70(3), 482–513. <https://doi.org/10.1002/joom.1296>

- Angelopoulos, S., Bendoly, E., Fransoo, J., Hoberg, K., Ou, C., Tenhiälä, A. (2023). Digital transformation in operations management: Fundamental change through agency reversal. *Journal of Operations Management*, 69, 6, 876-889. <https://doi.org/10.1002/joom.1271>
- Armstrong, D., Gosling, A., Weinman, J., & Marteau, T. (1997). The place of inter-rater reliability in qualitative research: An empirical study. *Sociology*, 31, 597-606. <https://doi.org/10.1177/0038038597031003015>
- Atkinson, P., & Hammersley, M. (1994). Ethnography and participant observation. In N. K. Denzin & Y. S. Lincoln (Eds.), *Handbook of qualitative research* (pp. 248-261). Thousand Oaks, CA: Sage Publications.
- Bardhan, I., Kohli, R., Oborn, E., Mishra, A., Tan, C. H., Tremblay, M. C., & Sarker, S. (2025). Human-centric information systems research on the digital future of healthcare. *Information Systems Research*. Articles in Advance, pp. 1-20, <https://doi.org/10.1287/isre.2025.editorial.v36.n1>
- Barney, J. B., & Felin, T. (2013). What are microfoundations? *Academy of Management Perspectives*, 27, 138-155. <https://doi.org/10.5465/amp.2012.0107>
- Barney, J. (1991). Firm resources and sustained competitive advantage. *Journal of Management*, 17(1), 99-120. <https://doi.org/10.1177/014920639101700108>
- Bavafa, H., Savin, S., & Terwiesch, C. (2021). Customizing primary care delivery using E-visits. *Production and Operations Management*, 30(11), 4306-4327. <https://doi.org/10.1111/poms.13528>
- Bendoly, E., Li, M., Smith, J., Pavlou, P. A. (2025). Operational Process and Agency Considerations in Managing Emerging Technologies in Healthcare [Editorial]. *Journal of Operations Management*, 1-12. <https://doi.org/10.1002/joom.70029>
- Bradley, R. V., Esper, T. L., In, J., Lee, K. B., Bichescu, B. C., & Byrd, T. A. (2018). The joint use of RFID and EDI: Implications for hospital performance. *Production and Operations Management*, 27(11), 2071-2090. <https://doi.org/10.1111/poms.12955>
- Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3, 77-101. <https://doi.org/10.1191/1478088706qp063oa>
- Catchpole, K., Cohen, T., Alfred, M., Lawton, S., Kanji, F., Shouhed, D., Nemeth, L., & Anger, J. (2024). Human Factors Integration in Robotic Surgery. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 66(3), 683-700. <https://doi.org/10.1177/00187208211068946>
- Chandrasekaran, A., Senot, C., & Boyer, K. K. (2012). Process management impact on clinical and experiential quality: Managing tensions between safe and patient-centered healthcare. *Manufacturing & Service Operations Management*, 14(4), 548-566. <https://doi.org/10.1287/msom.1110.0374>
- Cherns, A. (1978). The principles of sociotechnical design. In: Pasmore, W., Sherwood, J. (Eds.), *Sociotechnical Systems: A Source Book*. University Associates, La Jolla, pp. 61-71.
- Choi, T. M., Kumar, S., Yue, X., & Chan, H. L. (2022). Disruptive technologies and operations management in the Industry 4.0 era and beyond. *Production and Operations Management*, 31(1), 9-31. <https://doi.org/10.1111/poms.13622>
- Constantiou, I. D., & Kallinikos, J. (2015). New games, new rules: Big data and the changing context of strategy. *Journal of Information Technology*, 30(1), 44-57. <https://doi.org/10.1057/jit.2014.17>
- Corbin, J., & Strauss, A. (2014). Basics of qualitative research: Techniques and procedures for developing grounded theory. Sage Publications.
- Correia, L. M. A. M., Souza, P. R., Frank, A. G. (2025). Aligning digital strategy and resources to create Healthcare 4.0 organizations: Implications for smart operations and smart working. *Technological Forecasting and Social Change*, 221, 124339. <https://doi.org/10.1016/j.techfore.2025.124339>
- Correia, L. M. A. M., Frank, A. G. (2025). Structuring resources in healthcare digital transformation: a comparison across public, private and research hospitals. *Technovation*, 147, 103320. <https://doi.org/10.1016/j.technovation.2025.103320>
- Cousins, P. D., Handfield, R. B., Lawson, B., & Petersen, K. J. (2006). Creating supply chain relational capital: The impact of formal and informal socialization processes. *Journal of Operations Management*, 24, 851-863. <https://doi.org/10.1016/j.jom.2005.08.007>
- Dai, T., & Tayur, S. (2022). Designing AI-augmented healthcare delivery systems for physician buy-in and patient acceptance. *Production and Operations Management*, 31(12), 4443-4451.

- <https://doi.org/10.1111/poms.13850>
- Delana, K., Deo, S., Ramdas, K., Subburaman, G.-B. B., & Ravilla, T. (2023). Multichannel delivery in healthcare: The impact of telemedicine centers in southern India. *Management Science*, 69(5), 2568–2586. <https://doi.org/10.1287/mnsc.2022.4488>
- Devaraj, S., Ow, T. T., & Kohli, R. (2013). Examining the impact of information technology and patient flow on healthcare performance: A Theory of Swift and Even Flow (TSEF) perspective. *Journal of Operations Management*, 31(4), 181–192. <https://doi.org/10.1016/j.jom.2013.03.001>
- Dyer, J. H., & Singh, H. (1998). The relational view: Cooperative strategy and sources of interorganizational competitive advantage. *Academy of Management Review*, 23(4), 660. <https://doi.org/10.2307/259056>
- Easton, G. S., & Rosenzweig, E. D. (2015). Team leader experience in improvement teams: A social networks perspective. *Journal of Operations Management*, 37(1), 13–30. <https://doi.org/10.1016/j.jom.2015.05.001>
- Eisenhardt, K. M. (1989). Building theories from case study research. *Academy of Management Review*, 14(4), 532–550. <https://doi.org/10.2307/258557>
- Eisenhardt, K. M., & Graebner, M. E. (2007). Theory building from cases- opportunities and challenges. *Academy of Management Journal*, 50, 25–32. <https://doi.org/10.5465/amj.2007.24160888>
- Felin, T., Foss, N. J., Heimeriks, K. H., & Madsen, T. L. (2012). Microfoundations of routines and capabilities: Individuals, processes, and structure. *Journal of Management Studies*, 49, 1351–1374. <https://doi.org/10.1111/j.1467-6486.2012.01052.x>
- Felin, T., & Hesterly, W. S. (2007). The knowledge based view, nested heterogeneity, and new value creation: Philosophical considerations on the locus of knowledge. *Academy of Management Review*, 32, 195–218. <https://doi.org/10.5465/amr.2007.23464020>
- Franco, C. W., Benitez, G. B., Sousa, P. R. de, Kliemann Neto, F. J., & Frank, A. G. (2024). Managing resources for digital transformation in supply chain integration: The role of hybrid governance structures. *International Journal of Production Economics*, 278(109428), 109428. <https://doi.org/10.1016/j.ijpe.2024.109428>
- Frank, A. G., Thürer, M., Godinho Filho, M., & Marodin, G. A. (2024). Beyond Industry 4.0 – integrating Lean, digital technologies and people. *International Journal of Operations & Production Management*, 44(6), 1109–1126. <https://doi.org/10.1108/ijopm-01-2024-0069>
- Franke, H., Kwasnitschka, D., Schmutz, J. B., & Netland, T. H. (2024). Emphasizing worker identification with skills to increase helping and productivity in production: A field experiment. *Journal of Operations Management*, 70(5), 712–732. <https://doi.org/10.1002/joom.1300>
- Gardner, J. W., Boyer, K. K., & Gray, J. V. (2015). Operational and strategic information processing: Complementing healthcare IT infrastructure. *Journal of Operations Management*, 33–34(1), 123–139. <https://doi.org/10.1016/j.jom.2014.11.003>
- Galletta, A. (2013). Mastering the semi-structured interview and beyond: From research design to analysis and publication. New York, NY: NYU Press.
- Gioia, D. A., Corley, K. G., & Hamilton, A. L. (2012). Seeking qualitative rigor in inductive research: Notes on the Gioia methodology. *Organizational Research Methods*, 16, 15–31. <https://doi.org/10.1177/1094428112452151>
- Grant, R. M. (1996). Toward a knowledge-based theory of the firm. *Strategic Management Journal*, 17, 109–122. <https://doi.org/10.1002/smj.4250171110>
- Guha, S., & Kumar, S. (2018). Emergence of big data research in operations management, information systems, and healthcare: Past contributions and future roadmap. *Production and Operations Management*, 27(9), 1724–1735. <https://doi.org/10.1111/poms.12833>
- Heiberg, J., Truffer, B., Binz, C. (2022). Assessing transitions through socio-technical configuration analysis—a methodological framework and a case study in the water sector. *Research Policy*, 51(1). <https://doi.org/10.1016/j.respol.2021.104363>
- Heim, G. R., Nair, A., & Bendoly, E. (2021). Aims and criteria for advancing technology management research at the journal of operations management. *Journal of Operations Management*, 67(8), 920–925. <https://doi.org/10.1002/joom.1164>

- Kc, D. S., Scholtes, S., & Terwiesch, C. (2020). Empirical research in healthcare operations: Past research, present understanding, and future opportunities. *Manufacturing & Service Operations Management*, 22(1), 73–83. <https://doi.org/10.1287/msom.2019.0826>
- Kellogg, K. C., Valentine, M. A., & Christin, A. (2020). Algorithms at work: The new contested terrain of control. *Academy of Management Annals*, 14(1), 366–410. <https://doi.org/10.5465/annals.2018.0174>
- Kellogg, K. C., Myers, J. E., Gainer, L., & Singer, S. J. (2021). Moving violations: Pairing an illegitimate learning hierarchy with trainee status mobility for acquiring new skills when traditional expertise erodes. *Organization Science*, 32(1), 181–209. <https://doi.org/10.1287/orsc.2020.1374>
- Kellogg, K. C. (2022). Local adaptation without work intensification: Experimentalist governance of digital technology for mutually beneficial role reconfiguration in organizations. *Organization Science*, 33(2), 571–599. <https://doi.org/10.1287/orsc.2021.1445>
- Kraus, S.; Schiavone, F.; Pluzhnikova, A.; Invernizzi, A. C. (2021). Digital transformation in healthcare: analyzing the current state-of-research. *Journal Business Research*, 123, 557–567. <https://doi.org/10.1016/j.jbusres.2020.10.030>
- Lee, T. W. (1999). Using qualitative methods in organizational research. Broadgate, London: SAGE Publications.
- Leiblein, M. (2003). The choice of organizational governance form and performance: Predictions from transaction cost, resource-based, and real options theories. *Journal of Management*, 29(6), 937–961. [https://doi.org/10.1016/s0149-2063\(03\)00085-0](https://doi.org/10.1016/s0149-2063(03)00085-0)
- Lowry, P. B., Boh, W. F., Petter, S., Leimeister, J. M., & Editors, G. (2025). Long live the metaverse: Identifying the potential for market disruption and future research. *Journal of Management Information Systems*, 42(1), 3–38. <https://doi.org/10.1080/07421222.2025.2455770>
- Malhotra, A., Majchrzak, A., & Lyytinen, K. (2021). Socio-technical affordances for large-scale collaborations: Introduction to a virtual special issue. *Organization Science*, 32(5), 1371–1390. <https://doi.org/10.1287/orsc.2021.1457>
- Malik, M., Andargoli, A., Chavez Clavijo, R., & Mikalef, P. (2024). A relational view of how social capital contributes to effective digital transformation outcomes. *Journal of Strategic Information Systems*, 33(2), 101837. <https://doi.org/10.1016/j.jsis.2024.101837>
- Manz, C., Stewart, G. (1997). Attaining flexible stability by integrating total quality management and socio-technical systems theory. *Organization Science*, 8(1), 59–70. <http://www.jstor.org/stable/2635228>
- Mithas, S., Chen, Z.-L., Saldanha, T. J. V., & De Oliveira Silveira, A. (2022). How will artificial intelligence and Industry 4.0 emerging technologies transform operations management?. *Production and Operations Management*, 31(12), 4475–4487. <https://doi.org/10.1111/poms.13864>
- Mukherjee, U. K., & Sinha, K. K. (2020). Robot-assisted surgical care delivery at a hospital: Policies for maximizing clinical outcome benefits and minimizing costs. *Journal of Operations Management*, 66(1–2), 227–256. <https://doi.org/10.1002/joom.1058>
- Nenova, Z., & Shang, J. (2022). Chronic disease progression prediction: Leveraging case-based reasoning and big data analytics. *Production and Operations Management*, 31(1), 259–280. <https://doi.org/10.1111/poms.13532>
- Nickerson, J. A., & Zenger, T. R. (2004). A knowledge-based theory of the firm—The problem-solving perspective. *Organization Science*, 15, 617–632. <https://doi.org/10.1287/orsc.1040.0093>
- Nonaka, I. (1994). A dynamic theory of organizational knowledge creation. *Organization Science*, 5, 14–37.
- Nwafor, O., Ma, X., Johnson, N. A., Singh, R., & Aron, R. (2023). Differential impacts of technology-network structures on cost efficiency: Knowledge spillovers in healthcare. *Journal of Management Information Systems*, 40(3), 840–882. <https://doi.org/10.1080/07421222.2023.2229126>
- Qu, T., Lei, S. P., Wang, Z. Z., Nie, D. X., Chen, X., & Huang, G. Q. (2016). IoT-based real-time production logistics synchronization system under smart cloud manufacturing. *The International Journal of Advanced Manufacturing Technology*, 84, 147–164. <https://doi.org/10.1007/s00170-015-7220-1>
- Queenan, C., Cameron, K., Snell, A., Smalley, J., & Joglekar, N. (2019). Patient heal thyself: Reducing hospital readmissions with technology-enabled continuity of care and patient activation. *Production and Operations Management*, 28(11), 2841–2853. <https://doi.org/10.1111/poms.13080>

- Olsen, T. L., & Tomlin, B. (2020). Industry 4.0: Opportunities and challenges for operations management. *Manufacturing & Service Operations Management*, 22(1), 113–122. <https://doi.org/10.1287/msom.2019.0796>
- Ransbotham, S., Overby, E. M., & Jernigan, M. C. (2021). Electronic trace data and legal outcomes: The effect of electronic medical records on malpractice claim resolution time. *Management Science*, 67(7), 4341–4361. <https://doi.org/10.1287/mnsc.2020.3725>
- Roscoe, S., Cousins, P. D., & Handfield, R. (2019). The microfoundations of an operational capability in digital manufacturing. *Journal of Operations Management*, 65(8), 774–793. <https://doi.org/10.1002/joom.1044>
- Rousseau D. M. 1977. Technological differences in job characteristics, employee satisfaction, and motivation: A synthesis of job design research and sociotechnical systems theory. *Organizational Behavior and Human Performance*, 19 (1): 18–42. [https://doi.org/10.1016/0030-5073\(77\)90052-6](https://doi.org/10.1016/0030-5073(77)90052-6)
- Sharma, L., Chandrasekaran, A., Boyer, K. K., & McDermott, C. M. (2016). The impact of Health Information Technology bundles on Hospital performance: An econometric study. *Journal of Operations Management*, 41(1), 25–41. <https://doi.org/10.1016/j.jom.2015.10.001>
- Silva, M. E., Santos, K. A., Pereira, S. C. F., & Hendry, L. C. (2025). Switching the telescope lens: A sociomaterial perspective of sustainable agricultural (proto)practices transfer in an agrifood supply chain. *Journal of Operations Management*. <https://doi.org/10.1002/joom.1369>
- Spradley, J.P. (2016). *Participant Observation*. Waveland Press.
- Stark, A., Ferm, K., Hanson, R., Johansson, M., Khajavi, S., Medbo, L., Öhman, M., & Holmström, J. (2023). Hybrid digital manufacturing: Capturing the value of digitalization. *Journal of Operations Management*, 69(6), 890–910. <https://doi.org/10.1002/joom.1231>
- Stevens, M., & Schaik, J. (2020). Implementing new technologies for complex care: The role of embeddedness factors in team learning. *Journal of Operations Management*, 66(1–2), 112–134. <https://doi.org/10.1002/joom.1034>
- Sykes, T. A., & Aljafari, R. (2022). We are all in this together, or are we? Job strain and coping in the context of an E-healthcare system implementation. *Journal of Management Information Systems*, 39(4), 1215–1247. <https://doi.org/10.1080/07421222.2022.2127450>
- Tushe, S., Kc, D. S., Ding, H., & Yeung, H. (2025). Multichannel healthcare: Impact of asynchronous telemedicine adoption on patient flow. *Manufacturing & Service Operations Management*, 27(1), 59–74. <https://doi.org/10.1287/msom.2022.0235>
- van Marrewijk, A., & Dessing, N. (2019). Negotiating reciprocal relationships: Practices of engaged scholarship in project studies. *International Journal of Project Management*, 37(7), 884–895. <https://doi.org/10.1016/j.ijproman.2019.07.001>
- Wani, D., & Malhotra, M. (2018). Does the meaningful use of electronic health records improve patient outcomes? *Journal of Operations Management*, 60(1), 1–18. <https://doi.org/10.1016/j.jom.2018.06.003>
- Webb, E. J., Campbell, D. T., Schwartz, D., & Sechrest, L. (1999). *Unobtrusive Measures*. Thousand Oaks, CA: Sage Publications.
- Wowak, K. D., Lalor, J. P., Somanchi, S., & Angst, C. M. (2023). Business analytics in healthcare: Past, present, and future trends. *Manufacturing & Service Operations Management*, 25(3), 975–995. <https://doi.org/10.1287/msom.2023.1192>
- Yin, R. K. (2014). *Case study research: Design and methods* (5th ed.). Thousand Oaks, CA: Sage Publications.
- Zhang, L., Shao, Z., & Benitez, J. (2025). How to leverage digital platforms in preventing industrial accidents: Evidence from a multi-level empirical study. *Production and Operations Management*. <https://doi.org/10.1177/10591478241305263>
- Zhang, X., Denicol, J., Chan, P. W., & Le, Y. (2024). Designing the transition to operations in large inter-organizational projects: Strategy, structure, process, and people. *Journal of Operations Management*, 70(1), 107–136. <https://doi.org/10.1002/joom.1275>

APPENDIX A

Leaders responsible for the hospital's HCDT						
Area	Interview	Time	Interviewee	Professional Education	Role (Type)	Experience in the hospital
Digital Health Unit – round 1	I01	02:15	R01	Doctor	Coordinator	6 years
Digital Health Unit – round 2	I02	01:10	R01	Doctor	Coordinator	6 years
Digital Governance Committee	I03	00:44	R02	Doctor	Coordinator	8 years
Information and Communication Technology Management Coordination – round 1	I04	02:12	R03	Systems Analyst	Coordinator	17 years
Information and Communication Technology Management Coordination – round 2	I05	00:36	R03	Systems Analyst	Coordinator	17 years
Professionals from different departments of the hospital						
Area	Interview	Time	Interviewee	Professional Education	Role (Type)	Experience in the hospital
Surgical Block – round 1	I06	00:43	R04	Doctor	Head – Head of the Surgical Block	24 years
			R05	Pharmacist	Head of Medication Management and Logistics Section	14 years
Surgical Block – round 2	I07	00:45	R06	Administrator	Manager	29 years
			R07	Nurse	Head of Surgical Nursing Service	22 years
			R08	Nurse	Technical Nurse	15 years
Nursing staff – round 1	I09	00:54	R09	Nurse	Manager	14 years
			R10	Nurse	Quality and Patient Safety Manager	14 years
Nursing staff – round 2	I09	00:36	R11	Nurse	Neonatal ICU Nurse	10 years
			R12	Nurse	Nurse of Education and Nursing Services	12 years
Pediatric Intensive Care Unit (ICU) and Nursing Process Committee	I10	00:35	R13	Nurse	Member of the Nursing Process	2 years
Rapid Response Team	I11	00:38	R14	Doctor	Supervisor	5 years
Emergency	I12	00:46	R15	Doctor	Head of Emergency Service	16 years
Adult Intensive Care Unit (ICU)	I13	00:43	R16	Doctor	Head of ICU Safety and Quality	4 years
			R17	Doctor	Head of Critical Patient Management	12 years

Nutrition	I14	00:42	R18	Nutritionist	Quality Control Supervisor	11 years
Laboratory Testing – round 1	I15	00:55	R19	Doctor	Oncologist of Laboratory Diagnostic Service	4 years
Laboratory Testing – round 2	I16	00:45	R20	Biomedical Scientist	Supervisor of Laboratory Informatics	7 years
Pharmacy	I17	01:02	R05	Pharmacist	Head of Medication Management and Logistics Section	14 years
			R21	Pharmacist	Supervisor of Management and Logistics Section	7 years
			R22	Pharmacist	Medication and Logistics Manager	12 years
Hemodialysis	I18	00:44	R23	Nurse	Head of Hemodialysis Unit	25 years
			R24	Doctor	Head of Nephrology Service	14 years
Outpatient Clinic	I19	01:02	R25	Doctor	Supervisor of SUS Hospital Billing Administration	16 years
			R26	Doctor	Head of Outpatient Clinics	10 years
			R27	Administrator	Manager of Scheduling and Consultation System Rules	17 years
Medical Imaging	I20	01:22	R28	Physicist	Head of Medical Physics and Radiation Protection Service	18 years
			R29	Physicist	Imaging Physicist of Medical Physics and Radiation Protection Service	14 years
			R30	Physicist	Internal Auditor of Diagnostic Imaging Accreditation Program	12 years
			R31	Physicist	Medical Physicist	6 years
Pathology	I21	00:40	R32	Doctor	Pathologist	10 years
Cardiology	I22	00:41	R33	Doctor	Cardiologist of Heart Failure and Cardiac Transplantation	9 years
Information Technology	I23	01:30	R03	Systems Analyst	Coordinator	17 years
			R34	Systems Analyst	Head of Business Management Service	29 years
			R35	Computer Scientist	Head of Technology Management Department	12 years
			R36	Systems Analyst	Head of Support and Relationship Service	18 years
Internal Regulation Core	I24	01:15	R37	Nurse	Operations Nurse	13 years
			R38	Doctor	Executive Coordinator	5 years
Hematology	I25	01:02	R39	Doctor	Hematologist	21 years
			R40	Doctor	Head of Hematology and Bone Marrow Transplantation Service	12 years

APPENDIX B: INTERVIEW GUIDELINE

1 – Information on the professional background and operational context.

2 – Questions about the interaction effects of the microfoundations of operating routines with the introduction of HCDT.

- a) Could you describe the main processes in which you are involved? How do you interact with the functionalities of the HMS and/or other digital technologies in these processes?*
- b) Do you remember how the processes were before the implementation of the HMS and/or associated digital technologies? What were the main changes observed after this implementation?*
- c) With the implementation of HMS and/or digital technologies in the processes, what were the main benefits and drawbacks observed for professionals?*
- d) How are the requests for improvements in operational routines through the HMS and/or new digital technologies made?*
- e) Do you seek to stay updated on trends and innovations related to the digitalization of healthcare processes? If so, how do you do this? Is the acquired knowledge applied or shared in any way within the hospital?*
- f) Have you and/or your colleagues participated in the development and implementation of any technological project? If so, how did you and your colleagues contribute to this process?*
- g) How is feedback management and the continuous improvement process for technology implementation conducted? Are there specific criteria used to evaluate the success or failure of technological initiatives from the professionals' perspective?*
- h) Have new operational routines been created or discontinued following the implementation of the HMS and/or other digital technologies? If so, could you highlight which ones?*

5. CONCLUSIONS

This study comprises three articles, each corresponding to a specific objective of this PhD dissertation. Figure 15 illustrates the relationship between the three articles. While Article 1 addresses the management of digital resources for HCDT by considering the technological, organizational, and environmental contexts across different hospital management strategy models, Article 2 seeks to validate the alignment between strategy and digital resources, supported by network embeddedness, for the implementation of Healthcare 4.0 technologies, considering two main dimensions: smart operations and smart working. In turn, Article 3 demonstrates how the alignment among resources, strategies, and digital technologies shapes the microfoundations of operating routines — comprising individuals, hierarchical structures, and processes — which represent the level at which digital transformation is effectively operationalized in practice and enables the creation of new digital healthcare operating routines.

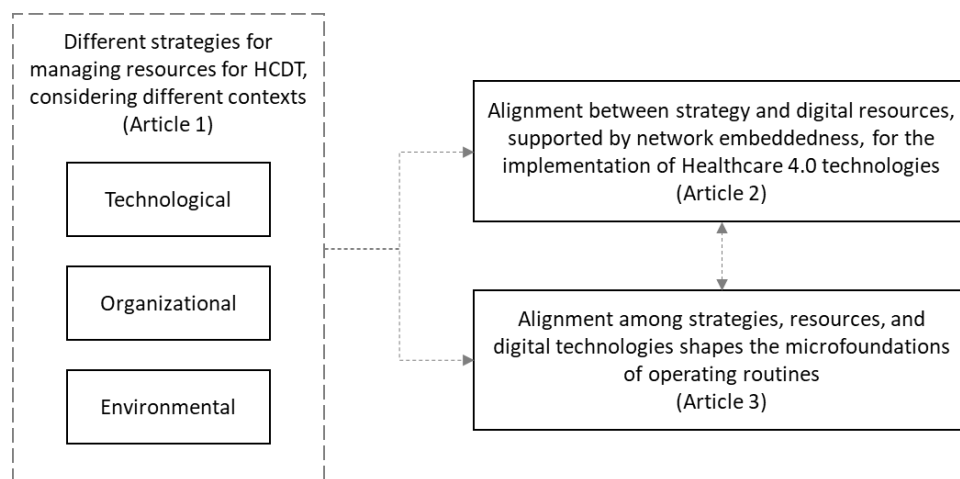


Figure 15. Articles relationship

The three articles that comprise this dissertation aim to build a theoretical framework to understand how healthcare organizations can organizationally structure healthcare digital transformation and promote the digitalization of their operations. Figure 16 presents an organizational structure that synthesizes the main findings and guides the organizational structuring of digital transformation in healthcare organizations.

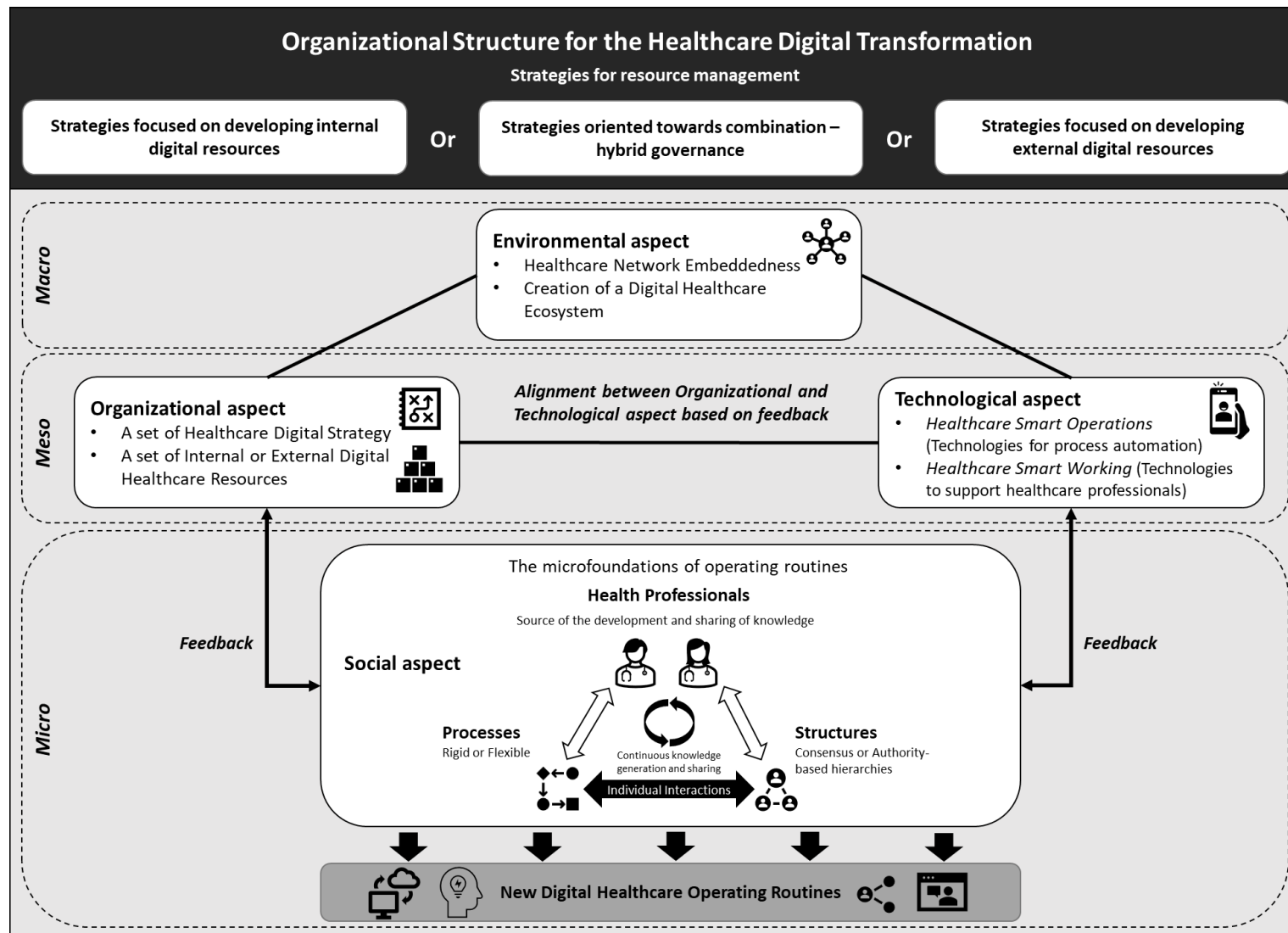


Figure 16. Organizational Structure for the Healthcare Digital Transformation

Observing Figure 16, it can be noted that it is hierarchically structured from top to bottom and organized into four main levels. At the first level, three types of strategies for digital resource management are presented: strategies focused on developing internal digital resources, strategies oriented toward combination–hybrid governance, and strategies focused on developing external digital resources. Managers may choose and adopt one of these strategies according to the capacity and specific characteristics of each type of hospital—public, private, or teaching and research institutions—considering their specificities, constraints, and capital availability., as discussed in Article 1 of Chapter 2.

Subsequently, the three organizational levels that should be considered in the organizational structuring of HCDT are presented: Macro, Meso, and Micro, reflected in the integration of the Environmental, Organizational, Technological, and Social Aspects. At the Macro level, the Environmental Aspect is identified, which corresponds to an organizational level external to the healthcare system. At this level, the healthcare system should consider Healthcare Network Embeddedness and/or the creation of a Digital Healthcare Ecosystem, with the aim of supporting and accessing external digital resources from partners that may collaborate with HCDT. The boundaries of the macro level encompass the institutional environment in which the healthcare organization is embedded, including public actors, private organizations, regulatory bodies, policymakers, and other stakeholders that exert direct or indirect influence on HCDT. This level corresponds to the sphere of governance and institutional pressures that shape external norms, expectations, and directives. The macro level is delineated at the point where decision-making shifts from being predominantly conditioned by external forces to being internalized within the organization's governance structure, where it is translated into strategic directives by managers, decision-makers, directors, and executive leaders.

The Meso level corresponds to the internal organizational sphere of governance and management within the healthcare organization. It encompasses the presidency, executive boards, directories, and departments responsible for strategic formulation and decision-making regarding HCDT. At this level, directives stemming from the supplier and institutional environment (Macro level) are interpreted and translated into organizational digital strategies, policies, digital resources and action plans. Simultaneously, the Meso level analyzes and consolidates demands emerging from the micro level by assessing needs, priorities, and the feasibility of implementation. Thus, the Meso level functions as a mediating and alignment

mechanism, connecting external institutional pressures with internal operational dynamics, ensuring strategic coherence and organizational coordination in the HCDDT process.

At the Meso level, positioned at an intermediate layer, the alignment between the Organizational and Technological Aspects is established. This alignment is constructed and continuously reinforced by feedback originating from the Micro level, indicating that HCDDT does not necessarily need to originate from top management, but can instead be managed based on insights emerging from the Micro level, while simultaneously being supported by the Environmental Aspect at the Macro level. More specifically, the Organizational Aspect comprises a set of Digital Healthcare strategies aimed at digitalization, as well as a set of Internal and External Digital Resources that enable the operationalization of these strategies and support the implementation of digital technologies. Additionally, this aspect should consider the establishment of digital governance committees to support the management of HCDDT within healthcare systems.

Still at the meso level, the Technological Aspect represents the practical manifestation of HCDDT, corresponding to the dimensions of digital technology applications to be implemented. These dimensions may focus on Healthcare Smart Operations, which consists of a set of digital technologies aimed at process automation, and Healthcare Smart Working, which encompasses digital technologies designed to support healthcare professionals in their activities. Both the technological dimensions and the types of digital resources and strategies are presented in Article 2, in Chapter 3.

Finally, the micro level corresponds to the operational sphere of the healthcare organization, where clinical and administrative activities are carried out. It encompasses healthcare professionals, digital technologies used in daily practice, work processes, hierarchical structures, and the operating routines that structure care delivery. At this level, HCDDT materializes through interactions among the microfoundations of operating routines — namely, professionals, processes (rigid or flexible), and hierarchical arrangements (based on consensus or authority). These interactions shape how digital technologies are operationalized in practice and how professional knowledge is generated and shared. The experiences and adaptations emerging from this level provide feedback to the Meso level, informing strategic decisions, organizational adjustments, and the definition of resources and strategies necessary to support implementation. Such feedback may involve improving existing digital applications or identifying opportunities for digitalizing previously manual or analog activities. As a result,

operating routines may be reconfigured, and new digitally enabled routines may emerge. The interaction patterns among professionals, processes, and hierarchical structures, as well as the routines that arise from them, are further examined in Article 3, presented in Chapter 4.

5.1 CONTRIBUTIONS

This study contributes to HCDT literature by offering a new organizational structure model that highlights the interconnections among digital strategies, digital resources, digital technologies, and the microfoundations of operating routines, as illustrated in Figure 16. The results indicate that these interconnections require the integration of macro-, meso-, and micro-level organizational dimensions, reflected in environmental, organizational, technological, and social aspects, respectively, as suggested by Ahmadi et al. (2018), Mauro et al. (2024), Mao et al. (2024) and Duus et al. (2026). When combined, these aspects support throughout the digital transformation journey and must be considered jointly, since digital technologies simultaneously affect all organizational levels. Addressing these aspects in isolation, or neglecting any of them, may compromise the effectiveness of the transformation process. In this way, the study helps address a gap in literature by demonstrating that HCDT cannot be understood from an exclusively technocentric perspective, as implied by the models presented by Tortorella et al. (2020a), Tortorella et al. (2022a), and Bendoly et al. (2025). Instead, it incorporates additional dimensions into an organizational structure, showing that HCDT can be also conceptualized as a sociotechnical system (Marcon et al., 2022; 2025) in which technical, social, organizational, and environmental aspects are treated with equal importance.

The proposed organizational structure (resulting from the organizational structuring) also contributes by addressing gaps identified in established models and structures in the literature. Although it was developed for the HCDT context — which involves sector-specific mechanisms, such as public hospitals that rely on procurement processes, for example — the structure can be adapted to other contexts, such as manufacturing, or even serve as a basis for developing new variants. Furthermore, the structure complements the consolidated RBV model proposed by Barney (1991) by extending the discussion beyond resource ownership as a source of competitive advantage. The proposed structure demonstrates that organizations may simultaneously possess strategic resources and rely on a network of external actors to access and integrate additional digital resources necessary for digital transformation. It also clarifies how such resources can be orchestrated (Sirmon et al., 2007). The model further complements

the organizational structure proposed by Verhoef et al. (2021) by illustrating how transformation-related resources and capabilities can be managed, as well as which governance mechanisms — internal, external, or hybrid — may be adopted to implement digital technologies. Furthermore, complement this by considering different organizational levels—macro, meso, and micro—for digital transformation.

The organizational structure proposed in this study also incorporates elements found in established HCDDT-oriented models in the literature, such as the structure developed by Kraus et al. (2021), which considers interconnected dimensions to be addressed in the HCDDT implementation process. However, the model developed in this study goes further by integrating environmental aspects, such as the creation of digital health ecosystems and the importance of being embedded in a network of actors to enable access to resources and support the implementation of digital technologies. In addition, it highlights the social aspect by demonstrating how the workforce and its interactions with other microfoundations of operating routines can contribute to the implementation of a customized HCDDT. Although microfoundations have previously been examined in the manufacturing context (Roscoe et al., 2019), it remained unclear how they could be shaped and mobilized within the HCDDT context, thus representing an additional contribution and novel insight into the HCDDT literature.

The proposed structure also advances beyond organizational structure models focused on the implementation of a single technology, such as the DTEM model for telehealth by Mishra and Sharma (2022), the framework by Chaudhuri et al. (2023) for the co-creation process in implementing 3D printing services in healthcare, and the model by Duus et al. (2026) for the implementation of Artificial Intelligence in healthcare. Unlike these approaches, the structure proposed in this thesis is not restricted to isolated technologies but addresses bundles of digital technologies, which may serve as entry points to digitalization and feedback mechanisms, while simultaneously guiding strategic definition and access to digital resources for multiple technologies. Furthermore, it presents a continuous HCDDT flow originating from the organization's micro level.

Additionally, the results show that the organizational structure of HCDDT must account for changes in the microfoundations of operating routines, which are considered here as a social aspect, as they involve people. It is at this micro level of the organization that digital technologies are effectively operationalized, making it essential to understand how these microfoundations are shaped by HCDDT and how new healthcare digital operating routines emerge from this process. The findings reveal that, through the interaction of these three

elements, professionals mobilize their knowledge, practical experience, and learning derived from trial-and-error processes to support the alignment between digital strategies, resources, and technologies. This process enables HCDT to be tailored to the needs of professionals, who are directly responsible for operationalizing digital technologies in patient care activities. In turn, this provides hospital managers with insights into the strategies, resources and technologies that need to be considered according to user requirements.

The study also contributes by demonstrating that the healthcare sector is highly dependent on professionals and their interactions with processes, hierarchical structures, and regulatory frameworks. Although regulatory, normative, and legal aspects were not directly addressed in this study, they are inherently part of organizational structuring and therefore represent a limitation in this study. Unlike manufacturing, for example, where processes can often be fully automated, healthcare involves clinical care also characterized by high levels of uncertainty for both patients and professionals. These uncertainties may pose risks to patient safety, particularly in contexts involving professionals–machine interactions and potential conflicts between technological systems and professional judgment.

Considering the proposed organizational structure, the central role of professionals and their interactions with processes and hierarchical structures at the micro level of the organization is emphasized. These interactions connect to the meso level, where project owners and HCDT leaders bridge operational activities and strategic decision-making. At this level, decision-makers define which digital strategies and resources are required. In turn, the meso level interacts with the macro level—external to the organization—through contracts and partnerships that enable access to and acquisition of digital resources and technologies, as well as insights from the broader ecosystem to support internal digitalization efforts. Therefore, the healthcare context plays a particularly significant role in digital transformation, making it different from other areas since it involves lives. Proposing an organizational structure that explicitly accounts for these multilevel interactions is essential for effectively supporting and advancing HCDT.

Nevertheless, the results can support managers and policymakers in defining digital transformation strategies, particularly in the healthcare context, which is characterized as a complex system due to the high level of process uncertainty. From a practical perspective, to support the organizational structuring of HCDT, healthcare systems—such as hospitals—may begin by defining a digital resource management model that integrally considers environmental, organizational, technical, and social aspects. In this regard, healthcare systems should develop

a digital transformation strategy that assesses which management model is most suitable for governing the digital resources required for technology implementation, for instance, both for process automation and for supporting healthcare professionals. This approach requires the prior structuring of critical organizational elements to ensure successful implementation. Subsequently, it is essential to encourage healthcare professionals to provide continuous feedback on the implementation process and the use of digital technologies, as they interact directly with patients and are therefore well-positioned to indicate how these interactions can be improved, ultimately generating greater benefits for task performance and clinical care.

Another practical contribution, based on the three articles developed, a digital maturity assessment model for healthcare systems was constructed. Although this model is not described in detail throughout this thesis, it constitutes a direct outcome of the research and was applied at HCPA. This application enabled the assessment of the institution's level of HCDT maturity, as well as the generation of insights to support decision-making and advance its organizational structuring. Overall, the model is structured around three main dimensions. The first is the technological dimension, which evaluates the degree of customization, scalability, expansion, integration, and traceability of digital technologies. The second is the organizational dimension, encompassing project management, infrastructure, and available digital resources. Finally, the social dimension addresses changes in work routines and knowledge management practices. This tool shows strong potential for use in future digital maturity assessments and analyses across other healthcare systems.

The thesis may also significantly contribute to strengthening Brazil's Unified Health System (Sistema Único de Saúde - SUS), particularly within the context of the SUS Digital Program, established by Ordinance GM/MS No. 3,232/2024, which aims to modernize the Brazilian healthcare system through digital transformation. By proposing a structured approach to HCDT, the thesis provides theoretical and practical support to assist the SUS in integrating data through the National Health Data Network (Rede Nacional de Dados em Saúde - RNDS), expanding citizens' access to health information via the Meu SUS Digital application, and improving care coordination across healthcare networks. Furthermore, the insights generated are also applicable to private healthcare systems, which face similar challenges related to interoperability, digital strategy development, technological resource allocation, and the reconfiguration of operating routines. By addressing social, organizational, technological, and environmental aspects, the thesis contributes to the development of a more integrated digital health ecosystem (particularly in Brazil, an emerging economy undergoing a progressive

transition toward higher levels of digitalization) capable of expanding access, strengthening clinical care networks, enhancing interoperability, and maximizing resource efficiency in both public and private healthcare contexts.

5.2 LIMITATIONS AND FUTURE RESEARCH

The thesis also presents some limitations. The study focuses on the practical context of Brazil, an emerging economy. In developing countries, digitalizing healthcare systems often presents challenges, especially in public healthcare environments, where most financial resources are allocated almost entirely to healthcare professionals' salaries and rely on public procurement processes to acquire external resources. As a result, HCDT in the public sector of emerging economies may progress at a slower pace. In contrast, developed countries, with greater capital to invest in digital transformation, are likely to be more technologically advanced and have more initiatives to achieve HCDT. Therefore, the territorial context is considered a limitation of this study. Future studies examining healthcare systems in other emerging or developed economies, as well as different organizational structuring models, are needed to enhance the generalizability of the findings.

Another limitation of this thesis is related to the size of the hospitals analyzed. This study focused on large hospitals, as these institutions are more likely to be at more advanced stages of HCDT compared to small hospitals. However, the organizational structuring required for HCDT in smaller healthcare systems may differ substantially due to constraints in human, financial, and infrastructural resources (e.g., Primary Health Care Units – UBS's). In this regard, future studies are needed to investigate the organizational structure of HCDT in small hospitals or UBS's, considering, for example, the implementation of specific digital technologies—such as telemedicine or artificial intelligence—as entry points for the gradual adoption of other digital solutions. Moreover, literature remains limited in this context, highlighting relevant opportunities for further research in this area.

The organizational structure proposed in this thesis focuses on analyzing key organizational elements — strategy, resources, digital technologies, and operating routines — that are critical to the successful implementation of HCDT. However, each of these elements encompasses a broader set of components that extend beyond the scope of this study. For instance, within the digital strategy dimension, factors such as institutional pressures, regulatory and legislative frameworks, normative guidelines, self-regulation mechanisms, interoperability

requirements, and policies addressing data duplication across health information systems — particularly those established by the Brazilian Ministry of Health — play a significant role. Nevertheless, these aspects are not explored in depth. Addressing them would require highly detailed analysis, including access to confidential data and the examination of multiple stakeholders' roles and interactions. Therefore, this study was delimited to a broader analytical scope, aiming to understand the organizational structure of HCDT. However, this delimitation opens avenues for future research about advances in organizational structure, given that healthcare is a highly regulated environment that becomes even more complex with the adoption of digital technologies. Understanding regulatory approvals, compliance mechanisms, and institutional alignments represents a fertile field for further investigation.

Another limitation of this study concerns the lack of a deeper examination of healthcare professionals' digital literacy (also an element of the social aspect) within the organizational structure of HCDT. Although it is well recognized that this behavioral dimension is highly relevant—since different levels of knowledge regarding digitalization and digital technologies, as well as professionals' age, may influence the speed and effectiveness of HCDT adoption—this perspective was not analyzed in depth. Accordingly, future research is encouraged to investigate these effects and related strategies, exploring differences across levels of digital literacy, particularly within the context of interactions among the microfoundations of operating routines. Furthermore, considering studies on professionals' resistance, work overload and continuing education, as well as strategies to address this issue, represents an opportunity for structural advancement.

Additionally, another limitation relates to the fact that the organizational structure proposed for HCDT does not focus on the implementation of a single specific technology—such as artificial intelligence, which is currently a prominent topic—but rather on the managerial organization required for the implementation of digital technology packages. In this context, the study also does not examine in depth the security mechanisms associated with these technologies, especially in the case of artificial intelligence and other emerging digital solutions. The scope of the thesis is centered on the organizational and managerial structuring necessary to enable the adoption of such technology packages, rather than on a detailed technical analysis of how these technologies function or their security mechanisms. Consequently, this delimitation opens avenues for future research to explore these technical aspects, thereby further enhancing the organizational structuring of HCDT.

Finally, it is important to highlight that the organizational structure model developed in this thesis has not been implemented in practice, which currently represents a limitation. Therefore, it opens opportunities for implementation and, consequently, for analysis of its application. Furthermore, considering the response time associated with organizational structuring, strategies, resources, and technologies allocated to HCDT in healthcare systems, future research may investigate the returns on investment and the associated costs of different management models. Such an approach may help explain the nuances, as well as the advantages and disadvantages, of different organizational structuring models for HCDT.

Statement on the use of artificial intelligence

Artificial intelligence tools were used to support the review of English grammar.

The principal author takes full responsibility for the content of the articles.